

The block below carries no module guard, so docstrip copies it into EVERY generated file. That is what we want for the TeX files and wrong for the latexmkrc, which is Perl and does not understand lines opening with two percent signs – hence the negated guard around it.

The guard lines themselves are hidden behind a false conditional. docstrip reads a guard line either way, but ltxdoc, which typesets this file, would otherwise set the literal text of the guard as a stray paragraph in the body of the manual.

Nothing in this area may name a TeX conditional, not even in prose: outside the meta-comment at the top of the file, every documentation line here IS LaTeX source and is executed, and a conditional named in passing opens or closes a real one. That is not hypothetical – an earlier draft of this very comment swallowed the rest of the manual.

Three conditional primitives are deliberately absent from the lists below: the ones that close or branch a conditional. `\DoNotIndex` scans its argument with TeX’s own conditional machinery live, so a bare one of them closes the wrong conditional. The manual ended its run with “Too many }’s” and exited with status 1, which failed the docs step of the harness even though the PDF had been written. Leaving them out of the lists means they get indexed, which is harmless. Note that they cannot be named here either, not even inside . . . shorthand split across a line.

The coppe document class

Version 4.0

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a WhatsApp group where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Name}{Surname}{Degree}{Institution}
\advisor{Second Advisor's Name}{Surname}{Degree}{Institution}
\advisor{Third Advisor's Name}{Surname}{Degree}{Institution}
```

`\coadvisor` takes exactly the same four arguments.

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently – given name and surname go in one argument:

```
\examiner{First Examiner's Name Surname}{Degree}{Institution}
\examiner{Second Examiner's Name Surname}{Degree}{Institution}
...
\examiner{N-th Examiner's Name Surname}{Degree}{Institution}
```

The argument order changed in v4.1. The treatment (“Prof.”, “Prof.^a”) used to be the first mandatory argument and the institution an optional one; now the treatment is the optional argument and is EMPTY by default, and the institution is the last mandatory argument. 3.1.2.1.3(e) asks for name, titulacao and institution, and asks for no treatment at all, so that is what the folha de aprovacao now prints. The institution may be left empty – `\advisor{Ana}{Lima}{D.Sc.}{}{}` compiles and simply prints nothing in its place.

To print a treatment anyway, pass it in the optional argument:

```
\advisor[Prof.]{Ana}{Lima}{D.Sc.}{UFRJ}
\examiner[Prof.\textsuperscript{a}]{Bia Sousa}{Ph.D.}{UFF}
```

A document written for v4.0 or earlier has to be rewritten. See `MIGRATION_v3_to_v4.md` for the mechanical rule.

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information

is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Folha adicional: Coleta CAPES

Since August 2026 every thesis and dissertation must carry a *folha adicional* immediately after the folha de rosto, holding the fields the CAPES/MEC collection requires plus the ficha catalografica (SiBI manual, 9th rev. ed. 2026, 3.1.2.1.2 and Annex H). The sheet is filled in by the graduate programme, and is neither counted nor numbered.

The macros below feed that sheet. They are institutional data, so the sheet is always typeset in Portuguese regardless of the main language of the work, as the COPPE norm requires of institutional identity.

Since v4.1 the sheet is drawn as Annex H shows it: a CLOSED FRAME the width of the text block, one cell per field, cells separated by a horizontal rule, and **no rules to write on** – the space to fill in is the blank space of the cell. A field left empty simply leaves its cell blank. The ficha catalografica sits centred below the frame, on the same sheet.

`\areaconcentracao` The area of concentration, e.g. `\areaconcentracao{Engenharia de Sistemas e Computacao}`. It appears on the folha de rosto, on the folha de aprovacao and on the folha adicional.

`\linhapesquisa` The research line, required by the CAPES model of the folha de rosto, where it is printed at the left margin just above the advisors. Its label follows the MAIN LANGUAGE of the work (“Linha de pesquisa”, “Research line”, “Línea de investigación”, “Axe de recherche”, “Linea di ricerca”). Leave it out, or give it an empty argument, and the folha de rosto comes out without the line and without any error.

`\tipoproducao` The kind of intellectual output, one of `bibliografica`, `artistica`, `tecnologica` or `tecnica`. Anything else raises a warning.

`\projetovinculado` Whether the work belongs to a research project (`sim` or `nao`) and, if so, its `\nomeprojeto` name.

`\agenciafomento` A funding agency, given as full name and acronym; repeat once per agency:

```
\agenciafomento{Conselho Nacional de Desenvolvimento
                  Cientifico e Tecnologico}{CNPq}
\agenciafomento{Fundacao Carlos Chagas Filho de Amparo a
                  Pesquisa do Estado do Rio de Janeiro}{FAPERJ}
```

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call

`\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.

`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.

`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is provided, it will be used as sort key. This was later, in 2024, implemented also for `\printloabbreviations` `\abbrev`, otherwise `Symbol`, or `Abreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

Illustrations: caption on top, source below

`\source` 2.10 of the Manual puts the CAPTION ABOVE every illustration – figure, table, `\fonte` quadro, code listing, algorithm – with the designating word, the number in arabic figures, an em dash and the title; and it makes the SOURCE mandatory, BELOW `\newcoppefloat`

the illustration, in the smaller size. The class does that for you: the caption goes on top wherever you write `\caption`, and `\source{...}` (or `\fonte{...}`, the same command) sets the source line. The source advances no counter and appears in no list.

```
\begin{figure}[ht]
  \centering
  \caption{0 que a figura mostra}    % em cima
  \label{fig:algo}
  \includegraphics[width=0.5\textwidth]{arquivo}
  \fonte{Elaboracao propria.}        % embaixo
\end{figure}
```

If another package has already defined `\source` or `\fonte`, use `\cpsource` and `\cpfonte`, which are always available.

A **quadro** is an illustration whose data is bordered on all sides, unlike a *tabela*, which is ruled only at the top and the bottom. The **quadro** environment behaves like **figure** and **table** and has its own list, `\listofquadros`.

For an illustration the Manual does not name – a map, a photograph, a score – declare your own float in the preamble:

```
\newcoppefloat{mapa}{Mapa}{Lista de Mapas}
```

You get the **mapa** environment, numbered within the chapter like every other illustration, the caption on top, `\source` below, `\autoref` support, and a `\listofmapas` command to put with the other pre-textual lists. The example does exactly this.

Long quotations, footnotes, captions and the folio all come out in the smaller uniform size 2.2(b) prescribes; you do not set that yourself.

References

COPPE_{TEX} produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_{TEX} `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_{TEX} styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography               % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book (@livro)`; `@article (@artigo)`; `@artigodejornal` (newspaper); `@incollection (@partedelivro)`; `@inbook (@emlivro)`; `@inreference (@verbete)`; `@proceedings (@anais)`; `@inproceedings (@trabalhodeevento)`; `@thesis (@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@tesephD`, `@tcc`); `@report (@relatorio)`; `@periodical (@periodico)`; `@manual`; `@misc (@diverso)`; `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent (@patente)`; `@legislation (@legislacao/@atnorma` `@jurisdiction (@jurisprudencia`, with a `relator` field); `@legal (@documentojuridico)`; `@standard (@norma)`; `@music (@partitura)`; `@map (@mapa)`; `@image (@iconografico)`; `@artwork (@obradearte)`; `@video (@audiovisual)`; `@movie (@filme)`; `@letter (@correspondencia)`; `@dataset (@conjuntodedados)`.

Games and TV `@game (@jogo)`; `@boardgame (@jogodetabuleiro)`; `@videogame (@jogoeletronico)`; `@tvshow (@programadetv)`.

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals. A periodical is the exception: 4.2.3 sets its title in capitals throughout.

Fields the ABNT categories need

Beyond the usual `biblatex` fields, each with an unaccented Portuguese synonym:

Events `eventtitle` (`evento`) — the name of the event, set in capitals ahead of the title of the proceedings, as 4.2.4 requires; `number` (`numero`) the edition; `eventdate` (`dataevento`) the year it was held; `venue` (`localevento`) the city. The title (`@anais`) or `booktitle` (`@trabalhodeevento`) then carries only “Anais [...]”, which is the element in bold.

Patents `filingdate` (`deposito`) and `grantdate` (`concessao`), printed as “Depósito: Concessão:” (4.2.5).

Theses `course` (`curso`), `institution` (`instituicao`), `pagetotal` — rendered as “1997. 203 f. Dissertação (Mestrado em ...) — Instituição, Local, 1997.” (4.2.1.1).

Legal documents `relator`, plus volume, number and pages of the vehicle in which the act was published (4.2.6).

Anything electronic `url` and `urldate` (`dataacesso`), closing the entry as “Disponível em: Acesso em:”

Every one of these is exercised, with the manual’s own example data, by `adversativa/referencias-manual.bib` — one entry per category of 4.2 — and checked against the manual’s printed form by `tools/conferir-referencias.py`. Neither is distributed: they prove the styles, they are not part of them.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard L^AT_EX command used before all appendices. COPPET_{EX} introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Latin Modern* (`lmodern`), which is what the class loads. Redefine if you loaded `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPETeX class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsceexam, msceexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msceexam`, and `dsceexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is `brazilian`.

- `brazilian` and `english` are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- `spanish`, `french`, `italian` are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

pdfa Produce the deposited file as **PDF/A-2b**, which 2.2(d) of the manual requires of the digital version. The class loads `pdfx` at level `a-2b` and writes the XMP metadata from the document’s own `\title`, `\author`, `\keyword` and `\department`. Level `a-2b` rather than `a-1b`: `a-1b` forbids transparency, which `tcolorbox` and several graphics packages produce. The metadata

reaches the PDF on the SECOND run, because `pdfx` reads it from a file the class writes at the end of the first one – which is no burden, since three runs are needed for cross-references anyway.

assinaturas Gone in v4.1, and now a no-op that only warns. It used to draw a signature rule above each board member. The deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand; the folha de aprovacao now always uses the compact list. The option is still accepted so a document written before v4.1 still compiles.

coorientador List the coadvisors on the three abstract pages, under the advisors, in the same shape the folha de rosto uses. Off by default: the abstract page listing only the advisors is a COPPE tradition, and 3.1.2.1.4 does not require either form.

rascunhoficha While writing, fill the folha adicional with a ficha catalografica typeset by the class itself, so the page is not an empty frame. **Never valid for deposit:** the real ficha comes from the SiBI generator or the Programa's library, and goes in with `\fichacatalografica`.

twoside, oneside The 2026 edition dropped the verso margins and the deposit is digital, so the class sets one-sided pages with a 3 cm left margin on every folha. **twoside** restores the mirrored layout for anyone who is going to PRINT and bind the work; **oneside** is the default and is there to be explicit.

semmorewrites, morewrites TeX has sixteen output streams and a work that asks for every list runs out of them; the failure, “No room for a new `\write`”, lands on whichever list came last and explains nothing. Under pdfTeX the class therefore loads the **morewrites** package by itself when it is installed, which removes the ceiling. **semmorewrites** turns that off, for the rare case where the package disagrees with something else in the document – it buffers what is written until shipout, so a package that writes a file and reads it back in the same run can misbehave. **morewrites** means “I insist”: a missing package becomes an error instead of a warning. Under LuaLaTeX, which has 128 streams, none of this is loaded and none of it is needed.

comserifa, semserifa Since v4.1 the class sets the whole document in the sans-serif family by default; **comserifa** goes back to the serif one. **Neither the manual nor the COPPE norm prescribes a typeface:** 2.2(b) fixes the colour, the body size 12 and the smaller uniform size of the four items it lists, and says nothing about the family. The choice is the author's, and the SiBI manual that carries the rule is itself set in Arial – which is why sans is now the default. The class loads **lmodern**, which brings both families as vector fonts, so either way the document is PDF/A material. **semserifa** is still accepted and now does nothing, so a document written before v4.1 still compiles. For a DIFFERENT sans face – Helvetica, say – load its package in the preamble and redefine `\familydefault` there; the default is for the common case of an author who does not want to choose a font at all. Redefine `\coppefinalfont` if you do load another face, so the colophon does not announce Latin Modern.

listasnosumario Put the pre-textual lists – figures, tables, quadros, programas, abreviaturas, simbolos – back into the sumario. **Contrary to 3.1.2.1.6,**

which tells the author to follow the manual’s own sumario, and that one opens at “1 INTRODUCAO”: the lists are pre-textual elements and come BEFORE the sumario, so listing them inside it is listing what the reader has already passed. The option exists because every CoppeTeX document written before this release had them there, and a work in its final weeks should not have its sumario change under it.

numeraisromanos Number the pre-textual folhas in lowercase roman numerals.

Contrary to 2.7, which has the count running from the folha de rosto with nothing printed until the textual part. Kept only so that documents written under the older norm still compile; do not use it for a new work.

5.1 Changing document identification

\freeconfig The user could *optionally* use the command **freeconfig** to modify the parameters that print the document identification. The command **freeconfig** needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the **abstract** environment. Picked by the class option (**brazilian** by default; one of **english**, **spanish**, **french**, **italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the **foreignabstract** environment. **english** by default for every main language; switches automatically to **brazilian** when **main = english**.

third optional. Loaded on demand with **\usecoppelanguage{<lang>}** in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

```
\titlein
brazilianabstract (env.)
\usecoppelanguage
```

Two pieces of user-facing API come with the multilingual model:

- **\titlein{<lang>}{<text>}** records the title in any language. **\title{...}** still writes the Brazilian-Portuguese title and **\foreigntitle{...}** still writes the English one (back-compat); for a main language outside the **brazilian/english** pair, also call **\titlein{<lang>}{...}**. The cover and folha de rosto print whichever title matches **\coppe@mainlang**.
- The **brazilianabstract** environment is the optional third abstract used when the main language is neither **brazilian** nor **english** – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in **\begin{otherlanguage}{brazilian}** and uses the Portuguese template strings. A document with **main = brazilian** can simply keep using **abstract**; **brazilianabstract** is redundant then.

- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthesis`, `phdthesis`, `tcc`, `coppemscdeg`, `coppedscdeg`, `coppephddeg`, `coppetccdeg`, `coppefiling`, `coppegrant`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\'i tulo em portugu\'es} % Brazilian abstract slot
  \foreigntitle{English title} % English abstract slot
  \titlein{spanish}{T\'i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\'erez}
```

```

\advisor{Mar\'}{i a}{Garc\'}{i a}{D.Sc.}{UFRJ}
\department{PESC}\date{01}{2026}
\keyword{multilenguaje}
\maketitle
\frontmatter
\begin{abstract}          Resumen principal en espa\~nol.      \end{abstract}
\begin{foreignabstract} English foreign abstract.              \end{foreignabstract}
\begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
\tableofcontents
\mainmatter
\chapter{Introducci\'}{on} Cuerpo en espa\~nol\dots
\end{document}

```

6 Files in your project folder

In a normal install where CoppeTeX is part of your TEXINPUTS path (e.g. inside the local texmf tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```

\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}

```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard `book` class (`12pt`, `a4paper`, `oneside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdf $\text{\TeX}$ and the Unicode engines (Lua $\text{\TeX}$ /Xe $\text{\TeX}$).

`inputenc (utf8)` UTF-8 input; loaded *only under pdf $\text{\TeX}$* (the Unicode engines read UTF-8 natively).

`fontenc (T1)` T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdf $\text{\TeX}$* . Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 one-sided margins (3/3/2/2 cm) and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms; the natural-language option tracks the main language.

Bibliography, quotations and language

babel multilingual typesetting; the language options are computed from the main and foreign language slots.

csquotes context-sensitive quotation marks (the recommended companion of **biblatex**).

biblatex (**backend=biber**) the bibliography engine and the ABNT **coppe** styles (**coppe** or, with the **numbers** option, **coppe-numeric**). The backend is **biber**, not **bibTeX**.

Hyperlinks

hyperref clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

etoolbox e-TeX programming toolkit.

ltxcmds low-level command helpers (`\ltx@ifpackageloaded, ...`).

ifthen the `\ifthenelse` conditional.

hyphenat hyphenation control (`\nohyphens`, used on the cover).

lastpage `\pageref{LastPage}` for the cataloging page count.

hologo TeX-family logos (`\hologo{LaTeX}, ...`).

xcolor colour support (used by **listings** and **hyperref**).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_latex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the **lmodern** package. To use Times, it is recommended to load the package **mathptmx**, rather than the deprecated **times**. There is also an enhanced Times version available with the **tgtermes** package. You can still use the Arial font face with the package **uarial**.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the **hyperref** package, which is loaded by the **coppe** class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under `pdfLaTeX`: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (`LuaLaTeX`, `XeLaTeX`). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation A quotation longer than three lines is INDENTED BY 4 cm from the left margin – not set with a 4 cm left margin, which is a different and much narrower thing. 2.2(b) of the manual says “recuo de 4 cm da margem esquerda”, so on an A4 page with a 3 cm margin the quoted text starts 7 cm from the edge of the paper. It carries no quotation marks and is set in the smaller font, single spaced. The `coppe` class provides the `longquote` environment, which does all of that; the indentation is `\recuolongquote`, should a Programa ever need to change it.

9 A simple example

```
1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating} % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz} % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 %\usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
```

```

19 \addbibresource{example.bib}% base de referências (bibtex/biber)
20
21 \makelossymbols
22 \makeloabbreviations
23 % Flutuante criado pelo autor: mesma regra dos demais (legenda acima, fonte
24 % abaixo) e uma lista própria, \listofmapas.
25 \newcoppfloat{mapa}{Mapa}{Lista de Mapas}
26 % Fonte: a norma fixa o corpo 12, e não a família. A classe compõe sem serifa
27 % por padrão; para compor com serifa, passe a opção de classe 'comserifa' lá
28 % em cima -- NÃO redefine \familydefault aqui, porque isso venceria a opção.
29 % Para uma sem serifa diferente da Latin Modern, carregue o pacote da fonte e
30 % troque a família:
31 %\usepackage{helvet}\renewcommand{\familydefault}{\sfdefault}
32
33
34 \begin{document}
35 \title{Titulo da Tese}
36 \foreigntitle{Thesis Title}
37 \author{Nome do Autor}{Sobrenome}
38 % Orientador: nome, sobrenome, titulacao e instituicao. O tratamento
39 % ("Prof.") e o argumento OPCIONAL entre colchetes e vem vazio por padrao --
40 % a norma nao pede tratamento nenhum. A instituicao e obrigatoria, mas pode
41 % ficar vazia: \advisor{Ana}{Lima}{D.Sc.}{} compila sem erro.
42 \advisor{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}{UFRJ}
43 % O segundo orientador vai com tratamento, so para mostrar o argumento
44 % opcional funcionando; a norma nao pede tratamento nenhum.
45 \advisor[Prof.]{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}{UFRJ}
46 % Coorientador: mesmos quatro argumentos de \advisor. Aparece na folha de
47 % rosto e na folha de aprovacao; para ve-lo tambem nas paginas de resumo,
48 % passe a opcao de classe 'coorientador'.
49 \coadvisor{Nome do Coorientador}{Sobrenome}{D.Sc.}{UFF}
50 % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
51 % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
52 % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
53 % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
54
55 \examiner{Nome do Primeiro Examinador Sobrenome}{D.Sc.}{UFRJ}
56 \examiner{Nome do Segundo Examinador Sobrenome}{Ph.D.}{UFRJ}
57 \examiner{Nome do Terceiro Examinador Sobrenome}{D.Sc.}{UFF}
58 \examiner{Nome do Quarto Examinador Sobrenome}{Ph.D.}{UNICAMP}
59 % Instituicao vazia: o argumento e obrigatorio, mas aceita ficar em branco e
60 % entao nada e impresso no lugar dela.
61 \examiner{Nome do Quinto Examinador Sobrenome}{Ph.D.}{}
62 \department{PESC}
63 \date{06}{2026}
64
65 \keyword{Primeira palavra-chave}
66 \keyword{Segunda palavra-chave}
67 \keyword{Terceira palavra-chave}
68
69 % Dados da folha adicional (Coleta CAPES), obrigatoria desde agosto de 2026.
70 % A folha e preenchida pelo Programa; estes campos apenas a compoem.
71 \areaconcentracao{Engenharia de Sistemas e Computa\c c\~ao}
72 \linhapesquisa{Engenharia de Dados e Conhecimento}

```

```

73 \tipoproducao{bibliografica} % bibliografica | artistica | tecnologica | tecnica
74 \projeto{vinculado{sim}} % sim | nao
75 \nomeprojeto{Nome do Projeto de Pesquisa ao qual o trabalho se vincula}
76 \agenciafomento{Conselho Nacional de Desenvolvimento Cientifico e
77 Tecnologico}{CNPq}
78 \agenciafomento{Fundacao Carlos Chagas Filho de Amparo a Pesquisa do
79 Estado do Rio de Janeiro}{FAPERJ}
80 % A ficha oficial vem de fichacatalografica.sibi.ufrj.br; tendo o arquivo,
81 % descomente a linha abaixo e a moldura vazia da folha adicional some.
82 % \fichacatalografica{ficha.pdf}
83
84 \maketitle
85
86 \frontmatter
87 \dedication{A alguém cujo valor é digno desta dedicatória.}
88
89 \chapter*{Agradecimentos}
90
91 Gostaria de agradecer a todos os que contribuíram para este trabalho:
92 orientadores, colegas, familiares e às agências de fomento. Lorem ipsum
93 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
94 ut labore et dolore magna aliqua.
95
96 \begin{abstract}
97
98 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
99 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
100 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
101 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
102 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
103 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
104 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
105 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
106 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
107 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
108 aspernatur aut odit aut fugit.
109
110 \end{abstract}
111
112 \begin{foreignabstract}
113
114 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
115 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
116 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
117 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
118 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
119 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
120 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
121 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
122 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
123 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
124 aspernatur aut odit aut fugit.
125
126 \end{foreignabstract}

```

```

127
128 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
129 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
130 \listoffigures
131 \listoftables
132 \listofquadros
133 \listofmapas
134 \listofprogramas
135 \listofalgorithms
136 \printloabbreviations
137 \printlosymbols
138 \tableofcontents
139
140 \mainmatter
141 \chapter{Introdução}
142
143 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos do
144
145 A classe \verb|coppe| foi criada originalmente por Vicente Helano e George Ainsworth, poré
146 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M
147
148 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação
149
150 A versão mais atual dessa classe é mantida no GitHub, no repositório \url{https://github.c
151
152 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po
153
154 Há um grupo de WhatsApp para perguntas e discussões do estilo: \href{https://github.com/CO
155
156 Bugs devem ser reportados na \href{https://github.com/COPPE-UFRJ/CoppeTeX/issues}{página d
157
158 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p
159
160 É importante de notar que essa classe não foi construída sobre a classe \LaTeX \ para a A
161
162 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma
163
164 Este documento não substitui, mas complementa, o documento que descreve a classe.
165
166
167
168 \chapter{Configurações Iniciais}
169
170 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c
171
172 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra
173
174 Recomendo ler o documento “The \verb*|coppe| document class” para entender melhor todas
175
176 \section{Linguagem principal do texto}
177
178 Essa classe considera que o texto principal está em português e algumas partes específicas
179 \begin{itemize}
180 \item A opção \verb|english| deve ser usada no comando \verb|\documentclass|.

```

```

181 \item A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p
182 \end{itemize}
183
184 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe \CopeT
185
186
187 \section{Por que usar o \LaTeX}
188
189 Há uma grande discussão entre usuário de Word e \LaTeX, principalmente, quanto ao uso dess
190
191 Nós escolhemos o \LaTeX por alguns motivos: grande facilidade de seguir um estilo sem se p
192
193 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob
194
195 \section{Como e onde usar o \LaTeX}
196
197 Existem muitos tutoriais de \LaTeX, mas basicamente, em 2024, ele é usado em dois ambiente
198 \begin{enumerate}
199 \item Na sua máquina, instalando uma versão completa como o \hologo{MiKTeX}, típico do Win
200 \item Usar um ambiente na rede, como o Overleaf.
201 \end{enumerate}
202
203 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o b
204
205
206 \chapter{Algumas Regras da COPPE}
207
208 Todas abreviaturas e símbolos devem ser definida antes de utilizada. Isso é facilmente fei
209
210 É imprescindível definir os símbolos, tal como o
211 conjunto dos números reais  $\mathbb{R}$  e o conjunto vazio  $\emptyset$ .
212 \syml{\mathbb{R}}{Conjunto dos números reais}
213 \syml{\emptyset}{Conjunto vazio}. Usamos esse exemplo aqui justamente para mostrar como
214
215 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o \ve
216
217 Como as listas de símbolos e de abreviaturas usamo o mesmo comando usado para criar índice
218
219 \section{Citações}
220 \label{sec:citacoes}
221
222 Citações curtas podem ser feitas \enquote{o comando quote} ou direto com ‘‘duas crases e do
223
224 \begin{longquote}
225 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
226 feita com o ambiente \verb*|longquote| The primary objective of this
227 investigation was to determine the feasibility of detecting corrosion in
228 aluminum Naval aircraft components with neutron radiographic interrogation
229 and the use of standard corrosion penetrameters. Secondary objectives
230 included the determination of the effect of object thickness on image quality,
231 the defining of minimum levels of detectability and a preliminary investigation
232 of a means whereby the degree of corrosion could be quantified with neutron
233 radiographic data. \citep{article-example}
234 \end{longquote}

```

```

235
236 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote \verb*|na
237
238 Em todo caso, \textbf{deve se tomar enorme atenção com as citações, para evitar ocorrer em
239
240 Quando se cita uma obra por meio de outra (\textit{apud}), pela ABNT entra na lista de ref
241
242 \chapter{Floats}
243
244 Grande parte dos problemas de iniciantes, e veteranos, em \LaTeX é da localização dos \texti
245
246 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesm
247
248 \textbf{Pela norma atual (manual2024), a legenda} \verb|\caption| \textbf{de toda ilustração}
249
250 Quadros são ilustrações cujos dados são delimitados por linhas em \emph{todas} as bordas (ao
251
252
253 \section{Tabelas e Figuras Padrão}
254
255 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
256
257 \begin{table}[ht]
258 \centering % Centraliza a tabela
259 \caption{Exemplo de Tabela de Números}
260 \label{tab:exemplo_numeros}
261 \begin{tabular}{ccc} % Define a quantidade de colunas
262 \hline % Linha superior
263 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabeçalhos
264 \hline % Linha média
265 1 & 2 & 3 \\ % Primeira linha de dados
266 \hline
267 4 & 5 & 6 \\ % Segunda linha de dados
268 \hline
269 7 & 8 & 9 \\ % Terceira linha de dados
270 \hline
271 10 & 11 & 12 \\ % Quarta linha de dados
272 \hline % Linha inferior
273 \end{tabular}
274 \fonte{Elabora\c{c}\~ao pr'opria.} % fonte obrigatoria, abaixo do float
275 \end{table}
276
277
278
279 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
280
281 \begin{figure}[ht]
282 \centering % Centraliza a figura
283 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
284 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
285 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
286 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
287 \end{figure}
288

```

```

289 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
290
291 \begin{quadro}[ht]
292 \centering
293 \caption{Exemplo de Quadro}
294 \label{qua:exemplo}
295 \begin{tabular}{|l|l|}
296 \hline
297 \textbf{Ilustração} & \textbf{Delimitação} \\
298 \hline
299 Tabela & Linhas apenas no topo e na base. \\
300 \hline
301 Quadro & Linhas em todas as bordas. \\
302 \hline
303 \end{tabular}
304 \fonte{Elabora\c{c}\~ao pr\'opria.}
305 \end{quadro}
306
307 Um segundo quadro, no \autoref{qua:segundo}, prova que o contador anda e que a lista de quad
308
309 \begin{quadro}[ht]
310 \centering
311 \caption{Segundo Quadro, para provar a numeração}
312 \label{qua:segundo}
313 \begin{tabular}{|l|l|}
314 \hline
315 \textbf{Elemento} & \textbf{Onde fica} \\
316 \hline
317 Legenda & Acima da ilustração. \\
318 \hline
319 Fonte & Abaixo da ilustração. \\
320 \hline
321 \end{tabular}
322 \fonte{Adaptado do manual do SiBI, 9.\textsuperscript{a} ed. rev., 2026.}
323 \end{quadro}
324
325 \section{Criando um float novo}
326
327 \verb|\newcoppefloat{nome}{Nome}{Titulo da Lista}| cria um flutuante que segue a mesma regra
328
329 \begin{mapa}[ht]
330 \centering
331 \caption{Primeiro mapa, num float criado pelo autor}
332 \label{mapa:primeiro}
333 \begin{tikzpicture}[scale=0.8]
334 \draw[thick] (0,0) rectangle (4,2.4);
335 \draw[fill=gray!20] (0.4,0.4) rectangle (1.8,2);
336 \draw[fill=gray!40] (2.2,0.4) rectangle (3.6,1.2);
337 \node at (2,-0.4) {\footnotesize esquema de ocupação};
338 \end{tikzpicture}
339 \fonte{Elabora\c{c}\~ao pr\'opria.}
340 \end{mapa}
341
342 \begin{mapa}[ht]

```



```

343 \centering
344 \caption{Segundo mapa, para provar o contador e a lista}
345 \label{mapa:segundo}
346 \begin{tikzpicture}[scale=0.8]
347   \draw[thick] (0,0) circle (1.2);
348   \draw[->] (-1.6,0) -- (1.6,0);
349   \draw[->] (0,-1.6) -- (0,1.6);
350   \node at (0,-2) {\footnotesize esquema de alcance};
351 \end{tikzpicture}
352 \fonte{Elabora\c{c}\~ao pr\'opria.}
353 \end{mapa}
354
355
356 \section{Tabelas mais elegantes}
357
358 Atualmente a tend\^encia \e usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
359
360 \begin{table}[ht]
361 \centering % Centraliza a tabela
362 \caption{Exemplo de Tabela de N\^umeros mais elegantes}
363 \label{tab:exemplo_numerosbom}
364 \begin{tabular}{ccc} % Define a quantidade de colunas
365 \toprule % Linha superior
366 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ % Cabe\c{c}alhos
367 \midrule % Linha m\^edia
368 1 & 2 & 3 \\ % Primeira linha de dados
369 4 & 5 & 6 \\ % Segunda linha de dados
370 7 & 8 & 9 \\ % Terceira linha de dados
371 10 & 11 & 12 \\ % Quarta linha de dados
372 \bottomrule % Linha inferior
373 \end{tabular}
374 \fonte{Elabora\c{c}\~ao pr\'opria.}
375 \end{table}
376
377 \section{Tabelas Longas ou Largas}
378
379 Se sua tabela \e muito longa ou larga, existem v\arias op\c{c}oes.
380 \begin{itemize}
381   \item alterar o tamanho da letra
382   \item Usar o longtable
383   \item rodar a tabela, fazendo ela em \textit{landscape}
384   \item fazer a tabela dentro de um minibox
385 \end{itemize}
386
387
388 \subsection{Tabelas largas demais}
389
390 \E comum em teses que as tabelas sejam largas demais. H\aa v\arias formas de resolver isso.
391
392 A \autoref{tab:tabela_largafns} \e larga demais, e nela isso \e resolvido diminuindo a fonte p
393
394 \begin{table}[ht]
395 \centering % Centraliza a tabela
396 \caption{Exemplo de Tabela Larga com Fonte Menor}

```

```

397 \label{tab:tabela_largafns}
398 \footnotesize % Aplica uma fonte menor para a tabela
399 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
400 \toprule
401 \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} & \textbf{Col. 4} & \textbf{Col. 5} & \textbf{Col. 6} & \textbf{Col. 7} \\
402 \midrule
403 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\
404 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
405 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
406 \bottomrule
407 \end{tabular}
408 \fonte{Elabora\c{c}\~ao pr\'opria.}
409 \end{table}
410
411 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer conteúdo.
412
413 \begin{table}[ht]
414 \centering
415 \caption{Exemplo de Tabela Redimensionada}
416 \label{tab:examplerb}
417 \resizebox{\textwidth}{!}{%
418 \begin{tabular}{llll}
419 \toprule
420 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
421 \midrule
422 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
423 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
424 \bottomrule
425 \end{tabular}%
426 }
427 \fonte{Elabora\c{c}\~ao pr\'opria.}
428 \end{table}
429
430
431 Para rodar uma tabela muito larga em 90 graus no LaTeX, você pode usar o pacote \verb*|rotatble|.
432
433 Aqui está um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela.
434
435 \begin{sidewaystable}
436 \centering
437 \caption{Sua Legenda Aqui}
438 \label{tab:sua_tabela}
439 \begin{tabular}{llll}
440 \toprule
441 Coluna 1 & Coluna 2 & Coluna 3 \\
442 \midrule
443 Item 1 & Item 2 & Item 3 \\
444 Item 4 & Item 5 & Item 6 \\
445 \bottomrule
446 \end{tabular}
447 \fonte{Elabora\c{c}\~ao pr\'opria.}
448 \end{sidewaystable}
449
450 Se a tabela for muito longa, o ambiente \verb|longtable| é o ideal. Ele fornece comandos para

```

```

451
452 % Exemplo de tabela longa que se estende por várias páginas
453 \begin{longtable}{|c|c|c|}
454 % primeiro cabeçalho (é o caption)
455 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
456 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
457 \endfirsthead
458 % cabeçalho normal
459 \multicolumn{3}{c}%
460 {\tablelongname\ \thetable} -- continuação da página anterior} \\
461 \hline \textbf{Col. 1} & \textbf{Col. 2} & \textbf{Col. 3} \\ \hline
462 \endhead
463 % pé normal
464 \hline \multicolumn{3}{|r|}{Continua na próxima página} \\ \hline
465 \endfoot
466 \hline
467 % último pé
468 \multicolumn{3}{|r|}{Continua na próxima página}} \\
469 \hline \hline
470 \endlastfoot
471
472 % Conteúdo da tabela
473 1 & 2 & 3 \\
474 4 & 5 & 6 \\
475 1 & 2 & 3 \\
476 4 & 5 & 6 \\
477 1 & 2 & 3 \\
478 4 & 5 & 6 \\
479 1 & 2 & 3 \\
480 4 & 5 & 6 \\
481 1 & 2 & 3 \\
482 4 & 5 & 6 \\
483 1 & 2 & 3 \\
484 4 & 5 & 6 \\
485 1 & 2 & 3 \\
486 4 & 5 & 6 \\
487 1 & 2 & 3 \\
488 4 & 5 & 6 \\
489 1 & 2 & 3 \\
490 1 & 2 & 3 \\
491 4 & 5 & 6 \\
492 1 & 2 & 3 \\
493 4 & 5 & 6 \\
494 1 & 2 & 3 \\
495 4 & 5 & 6 \\
496 1 & 2 & 3 \\
497 4 & 5 & 6 \\
498 1 & 2 & 3 \\
499 4 & 5 & 6 \\
500 1 & 2 & 3 \\
501 4 & 5 & 6 \\
502 1 & 2 & 3 \\
503 4 & 5 & 6 \\
504 1 & 2 & 3

```

505 4 & 5 & 6 \\
 506 1 & 2 & 3 \\
 507 4 & 5 & 6 \\
 508 1 & 2 & 3 \\
 509 4 & 5 & 6 \\
 510 1 & 2 & 3 \\
 511 4 & 5 & 6 \\
 512 1 & 2 & 3 \\
 513 4 & 5 & 6 \\1 & 2 & 3 \\
 514 4 & 5 & 6 \\
 515 1 & 2 & 3 \\
 516 4 & 5 & 6 \\
 517 1 & 2 & 3 \\
 518 1 & 2 & 3 \\
 519 4 & 5 & 6 \\
 520 1 & 2 & 3 \\
 521 4 & 5 & 6 \\
 522 1 & 2 & 3 \\
 523 4 & 5 & 6 \\
 524 1 & 2 & 3 \\
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 547 4 & 5 & 6 \\
 548 1 & 2 & 3 \\
 549 4 & 5 & 6 \\
 550 1 & 2 & 3 \\
 551 4 & 5 & 6 \\
 552 1 & 2 & 3 \\
 553 4 & 5 & 6 \\
 554 1 & 2 & 3 \\
 555 4 & 5 & 6 \\
 556 1 & 2 & 3 \\
 557 4 & 5 & 6 \\
 558 1 & 2 & 3 \\

```

559 4 & 5 & 6 \\
560 1 & 2 & 3 \\
561 4 & 5 & 6 \\
562 1 & 2 & 3 \\
563 4 & 5 & 6 \\
564 1 & 2 & 3 \\
565 4 & 5 & 6 \\
566 1 & 2 & 3 \\
567 4 & 5 & 6 \\
568 1 & 2 & 3 \\
569 4 & 5 & 6 \\
570 1 & 2 & 3 \\
571 4 & 5 & 6 \\
572 1 & 2 & 3 \\
573 4 & 5 & 6 \\
574 1 & 2 & 3 \\
575 4 & 5 & 6 \\
576 1 & 2 & 3 \\
577 4 & 5 & 6 \\
578 1 & 2 & 3 \\
579 4 & 5 & 6 \\
580
581 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
582 \end{longtable}
583 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
584
585 \section{Subfiguras (subcaption)}
586
587 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
588
589 \begin{figure}[ht]
590   \centering
591   \begin{subfigure}[b]{0.42\textwidth}
592     \centering\includegraphics[width=\linewidth]{example-image-a}
593     \caption{Primeira vista}\label{fig:sub-a}
594   \end{subfigure}
595   \hfill
596   \begin{subfigure}[b]{0.42\textwidth}
597     \centering\includegraphics[width=\linewidth]{example-image-b}
598     \caption{Segunda vista}\label{fig:sub-b}
599   \end{subfigure}
600   \caption{Figura com duas subfiguras}
601   \label{fig:subs}
602   \fonte{Elabora\c{c}\~ao pr\'opria.}
603 \end{figure}
604
605 Referencie a figura inteira com \verb|\autoref{fig:subs}|. Cada parte tem
606 rótulo próprio: \verb|\autoref{fig:sub-a}| leva à \autoref{fig:sub-a}.
607
608 \section{Imagens (graphicx)}
609
610 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
611 \begin{itemize}
612   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;

```

```

613 \item \verb|[angle=90]| --- rotação;
614 \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, dire
615 \end{itemize}
616 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt
617
618 \section{Introdução ao tikz}
619
620 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio
621
622 \begin{figure}[ht]
623 \centering
624 \begin{tikzpicture}
625 \node[draw, rounded corners] (a) at (0,0) {Início};
626 \node[draw, rounded corners] (b) at (4,0) {Fim};
627 \draw[->, thick] (a) -- (b) node[midway, above] {processo};
628 \end{tikzpicture}
629 \caption{Diagrama simples em tikz}
630 \label{fig:tikz}
631 \fonte{Elabora\c{c}\~ao pr\'opria.}
632 \end{figure}
633
634 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|ti
635
636 \section{Float ou direto no texto?}
637
638 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \L
639
640 \chapter{Using BibLaTeX}
641 \label{cap:biblatex}
642
643 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
644
645 \textbf{É obrigatório compilar com \texttt{biber}} (e não com \hologo{BibTeX}). A sequênci
646
647 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
648
649 \begin{table}[h]
650 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
651 \texttt{\textbackslash cite} do LaTeX\ e
652 o comando \texttt{\textbackslash citep},
653 disponível pela opção \texttt{natbib} do BibLaTeX.}
654 \label{tab:citation}
655 \centering
656 {\footnotesize
657 \begin{tabular}{|c|c|c|}
658 \hline
659 Tipo da Publicação & \verb|\cite| & \verb|\citep|\\
660 \hline
661 Livro & \cite{book-example} & \citep{book-example}\\
662 Artigo & \cite{article-example} & \citep{article-example}\\
663 Relatório & \cite{techreport-example} & \citep{techreport-example}\\
664 Relatório & \cite{techreport-exampleIn} & \citep{techreport-exampleIn}\\
665 Anais de Congresso & \cite{inproceedings-example} &
666 \citep{inproceedings-example}\\

```

```

667      Séries & \cite{incollection-example} & \citep{incollection-example}\\
668      Em Livro & \cite{inbook-example} & \citep{inbook-example}\\
669      Dissertação de mestrado & \cite{mastersthesis-example} &
670      \citep{mastersthesis-example}\\
671      Tese de doutorado & \cite{phdthesis-example} & \citep{phdthesis-example}\\
672      \hline
673 \end{tabular}}
674 \fonte{Elabora\c{c}\~ao pr\'opria.}
675 \end{table}
676
677 \begin{table}[h]
678 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao
679 \texttt{\textbackslash cite} do \LaTeX\ e
680 o comando \texttt{\textbackslash citet},
681 disponível pela opção \texttt{\textbackslash natbib} do BibLaTeX. Além disso, usando o booktabs.}
682 \label{tab:citation1}
683 \centering
684 {\footnotesize
685 \begin{tabular}{ccc}
686 \toprule
687 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
688 \midrule
689 Livro & \cite{book-example} & \citet{book-example}\\
690 Artigo & \cite{article-example} & \citet{article-example}\\
691 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
692 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
693 Anais de Congresso & \cite{inproceedings-example} &
694 \citet{inproceedings-example}\\
695 Séries & \cite{incollection-example} & \citet{incollection-example}\\
696 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
697 Dissertação de mestrado & \cite{mastersthesis-example} &
698 \citet{mastersthesis-example}\\
699 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
700 \bottomrule
701 \end{tabular}}
702 \fonte{Elabora\c{c}\~ao pr\'opria.}
703 \end{table}
704
705 \section{Comandos de citação}
706
707 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
708
709 \subsection{Parâmetro opcional: página e “tradução nossa”}
710
711 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
712
713 \subsection{Citação de citação (\emph{apud})}
714
715 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
716
717 \subsection{Múltiplas bibliografias}
718
719 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|\refsection|
720

```

```

721 \section{Tipos de entrada bibliográfica}
722
723 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinônimo
724
725 \subsection{Tipos acadêmicos principais}
726 \begin{itemize}
727   \item Livro / monografia --- \verb|@book| (\verb|@livro|)
728   \item Artigo de periódico --- \verb|@article| (\verb|@artigo|); matéria de jornal --- \verb|@journal|
729   \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
730   \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento --- \verb|@inproceedings|
731   \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo --- \verb|@phdthesis|
732   \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); periódico --- \verb|@article|
733   \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/site --- \verb|@online|
734 \end{itemize}
735
736 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
737 \begin{itemize}
738   \item Patente --- \verb|@patent| (\verb|@patente|)
739   \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato|)
740   \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\verb|@musica|)
741   \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
742   \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb|@filme|)
743 \end{itemize}
744
745 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
746 \begin{itemize}
747   \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@jogo|)
748   \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadettv|, \verb|@serietv|)
749 \end{itemize}
750
751 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
752
753 \chapter{Alguns outros exemplo úteis}
754
755 \begin{tcolorbox}[title=Meu Textbox]
756 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
757 \end{tcolorbox}
758
759 \begin{tcolorbox}
760 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem como
761 \end{tcolorbox}
762
763 \begin{figure}[ht]
764   \centering
765   \begin{tikzpicture}
766     \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\textwidth]{image}}
767     \begin{scope}[x={(image.south east)},y={(image.north west)}]
768       % Definindo o textbox dentro da figura
769       \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text=
770         \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
771           Este textbox fala sobre como inserir um textbox dentro de uma figura usando
772         \end{tcolorbox}
773       ];
774     \end{scope}
775   \end{tikzpicture}

```



```

775 \end{tikzpicture}
776 \caption{Figura com Textbox}
777 \label{fig:figura_com_textbox1}
778 \fonte{Elabora\c{c}\~ao pr\'opria.}
779 \end{figure}
780
781
782 \begin{figure}[ht]
783 \centering
784 \begin{tcolorbox}
785 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
786 \end{tcolorbox}
787 \caption{Figura com Textbox simples}
788 \label{fig:figura_com_textbox}
789 \fonte{Elabora\c{c}\~ao pr\'opria.}
790 \end{figure}
791
792 \section{Listagens de código e algoritmos}
793
794 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
795
796 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
797 def saudacao(nome):
798     # imprime uma saudação acentuada
799     print(f"Olá, {nome}!")
800
801 saudacao("COPPE")
802 \end{python}
803 \source{Elaboração própria.}
804
805 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
806
807 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
808 Olá, COPPE!
809 \end{Verbatim}
810
811 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|. O \auto
812
813 \begin{java}[caption={Exemplo de programa em Java},label=prog:java]
814 public class Saudacao {
815     public static void main(String[] args) {
816         System.out.println("Olá, COPPE!");
817     }
818 }
819 \end{java}
820 \fonte{Elaboração própria.}
821
822 Há também \verb|\pythoninline| e \verb|\javainline| para código no meio do
823 texto, como em \javainline{new Saudacao()}, e o ambiente \verb|Verbatim| (com as opções acim
824
825 \begin{algorithm}[H]
826 \caption{Soma dos elementos de um vetor}
827 \label{alg:soma}
828 \Dados{vetor  $v$  com  $n$  elementos}

```

```

829 \Resultado{soma $$ dos elementos de $v$}
830 $s \leftarrow 0$;
831 \Para{$i \leftarrow 1$ \Ate{} $n$}{
832   $s \leftarrow s + v[i]$;
833 }
834 \Retorna $$;
835 \end{algorithm}
836
837 0 \autoref{alg:busca} é o segundo algoritmo, e usa \verb|\Enquanto|. Como no caso das listag
838
839 \begin{algorithm}[H]
840 \caption{Busca sequencial em um vetor}
841 \label{alg:busca}
842 \Dados{vetor $v$ com $n$ elementos e uma chave $k$}
843 \Resultado{posição de $k$ em $v$, ou $0$ se $k$ não estiver em $v$}
844 $i \leftarrow 1$;
845 \Enquanto{$i \leq n$ \textbf{e} $v[i] \neq k$}{
846   $i \leftarrow i + 1$;
847 }
848 \Se{$i > n$}{\Retorna $0$;}
849 \Retorna $i$;
850 \end{algorithm}
851
852 \chapter{Escrevendo matemática}
853
854 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
855
856 \section{Alinhamento (align)}
857 0 ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
858 \begin{align}
859   f(x) &= (x+1)^2 \\
860         &= x^2 + 2x + 1 .
861 \end{align}
862
863 \section{Matrizes (matrix)}
864 0 \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
865 \begin{equation}
866   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
867   \quad
868   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
869 \end{equation}
870
871 \section{Subequações (subequations)}
872 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
873 \begin{subequations}
874 \begin{align}
875   a &= b + c , \\
876   d &= e - f .
877 \end{align}
878 \end{subequations}
879
880 \section{Estruturas alinhadas (array)}
881 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
882 \begin{equation}

```

```

883     |x| = \left\{ \begin{array}{l}
884         x, \quad \& \text{se } x \geq 0, \\
885         -x, \quad \& \text{caso contrário.}
886     \end{array} \right.
887 \end{equation}
888
889 \section{Operadores (\textbackslash DeclareMathOperator)}
890 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
891 \begin{verbatim}
892 \usepackage{amsmath}
893 \DeclareMathOperator{\RMSE}{RMSE}
894 \DeclareMathOperator{\diag}{diag}
895 \end{verbatim}
896 Assim:
897 \begin{itemize}
898     \item \verb|\RMSE(y,\hat{y})| produz  $\text{\RMSE}(y,\hat{y})$ ;
899     \item \verb|\diag(\lambda_1,\lambda_2)| produz
900          $\text{\diag}(\lambda_1,\lambda_2)$ .
901 \end{itemize}
902
903 \section{Unicode e LuaLaTeX (unicode-math)}
904 \label{sec:unicode-math}
905 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
906 \begin{verbatim}
907 \usepackage{amsmath}           % a classe já carrega
908 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
909 \setmathfont{Latin Modern Math} % ou STIX Two Math
910 \end{verbatim}
911 O código clássico continua valendo ( $\mathbb{R}$ ,  $\int_a^b f(x)dx$ ,  $\alpha$ ) e você ai
912
913 \chapter{Referências cruzadas, notas e índices}
914 \label{cap:refs}
915
916 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
917
918 \section{Rótulos e referências cruzadas}
919
920 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
921
922 \section{Notas de rodapé}
923
924 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
925
926 \section{hyperref e marcadores}
927
928 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref
929
930 \section{URLs}
931
932 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
933 use \verb|\href{https://exemplo.org}{texto}|. Evite digitar URLs como texto comum --- elas n
934
935 \section{Índice remissivo}
936

```

```

937 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
938
939 \section{Traduções (NBR 10520)}
940
941 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
942
943 \chapter{Truques de \LaTeX}
944
945 \section{Comentários mágicos (\texttt{\% !TeX})}
946 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
947 \begin{verbatim}
948 % !TeX program = lualatex
949 % !TeX encoding = UTF-8
950 % !TeX spellcheck = pt_BR
951 % !BIB program = biber
952 \end{verbatim}
953 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
954
955 \section{Espaço engolido}
956 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como “\LaTeX é
957
958 \section{Escopo (grupos)}
959 Chaves \verb|\{...}| (ou \verb|\begingroup|\dots\verb|\endgroup|) criam um escopo: mudanças
960
961 \section{Descarregar figuras: \texttt{\textbackslash clearpage} * \texttt{\textbackslash m
962 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
963
964 \section{\texttt{latexmkrc}, perl e Overleaf}
965 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
966
967 \chapter{Método Proposto}
968 \chapter{Resultados e Discussões}
969
970 \section{Algumas Demonstrações}
971
972 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
973 \syml[beta]{Beta}{A palavra Beta more e corrigida}
974 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
975 \syml{beta}{A palavra beta}
976 \syml{alpha}{A palavra alpha}
977 \syml[alpha]{Alpha}{A palavra Alpha}
978 \syml[zzalpha]{\alpha}{A letra $\alpha$ corrigida}
979 \syml[marco]{Marco}{A palavra Marco corrigida}
980
981 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exempl
982 \abbrev{GoT}{Game of Thrones}
983 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
984 \abbrev[iot]{IoT}{IoT ordenado como iot}
985 \abbrev[IOT]{IoT}{IoT ordenado como IoT}
986 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
987 \abbrev{IoT}{IoT com ordenação default}
988 \abbrev[ITU]{ITU}{ITU mesmo}
989
990 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}

```

```

991 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
992
993 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas auto
994
995
996
997
998 \chapter{Conclusões}
999
1000 \backmatter
1001
1002 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
1003 % partitura --- declarados via sinônimos em português em example.bib.
1004 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
1005 \nocite{jurisprudencia-example}
1006 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
1007 \nocite{jornalcaderno-example,jornal-example}
1008 \printbibliography
1009
1010
1011
1012 \appendix
1013
1014 \chapter{Um apêndice}
1015
1016 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1017
1018 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo
1019
1020
1021 \chapter{Outro apêndice}
1022
1023 \annex
1024
1025
1026 \chapter{Um Anexo}
1027 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização
1028
1029
1030
1031 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que
1032
1033 No modelo \CoppeTeX os anexos devem obrigatoriamente vir depois dos apêndices e usam o comand
1034
1035
1036 \chapter{Outro Anexo}
1037
1038
1039 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
1040
1041 \end{document}
1042
1043 \example

```

10 Implementation

10.1 The ‘coppe.cls’ file

```
1044 <*class>
1045 \def\filename{coppe.dtx}
1046 \def\fileversion{v4.1}
1047 \def\filedate{2026/09/10}
1048 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
1049 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
1050 % Save the class version and date into private macros immediately, before any
1051 % required package can redefine the very common \fileversion / \filedate
1052 % global names (LaTeX has no convention reserving them for the host class).
1053 % \coppe@version / \coppe@date are what user-facing macros (e.g.
1054 % \coppetexfinalpage) should reference.
1055 \let\coppe@version\fileversion
1056 \let\coppe@date\filedate
1057 % Document date stored in private slots so \date can set them without ever
1058 % touching TeX's \month/\year primitives (which would break \today). They
1059 % default to the current system month/year, so a document without \date
1060 % still renders sensibly on the cover and folha de rosto.
1061 \edef\coppe@docmonth{\the\month}
1062 \edef\coppe@docyear{\the\year}
1063 \LoadClass[12pt,a4paper,oneside]{book}
1064 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
1065 % the END of the class (after every class hook is defined), in that order:
1066 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
1067 % it patches (TOC, refs, lists) are already defined, and biblatex must come
1068 % after hyperref so its citation links and back-references work.
1069 \RequirePackage{hyphenat}
1070 \RequirePackage{lastpage}
1071 \RequirePackage{ifthen}
1072 \RequirePackage{graphicx}
1073 \RequirePackage{tabularx}
1074 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
1075 \RequirePackage{longtable} % tabelas que quebram entre páginas
1076 \RequirePackage{etoolbox}
1077 \RequirePackage{ltxcmds}
1078 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
1079 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
1080 \def\TeX{\hologo{TeX}}
1081 \def\LaTeX{\hologo{LaTeX}}
1082 \def\LaTeXe{\hologo{LaTeXe}}
1083 \def\BibTeX{\hologo{BibTeX}}
1084 \def\pdfLaTeX{\hologo{pdfLaTeX}}
1085 \def\pdfTeX{\hologo{pdfTeX}}
1086 \def\LuaLaTeX{\hologo{LuaLaTeX}}
1087 \def\XeLaTeX{\hologo{XeLaTeX}}
1088 \def\MiKTeX{\hologo{MiKTeX}}
1089 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
1090 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
1091 % files may use accented Latin characters (âéíóú) directly. Under pdfLaTeX we also
1092 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
1093 \RequirePackage{iftex}
```

```

1094 \ifPDFTeX
1095   \RequirePackage[utf8]{inputenc}
1096   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
1097     Although slower, you should try compiling with LuaLaTeX}}
1098 \fi
1099 % lmodern MUST come with the T1 encoding. Without it, T1 falls back to the EC
1100 % fonts, which exist only as METAFONT bitmaps in most installations, and the
1101 % whole document is typeset in Type3 bitmap fonts. Measured before this change:
1102 % example.pdf carried 87 Type3 fonts against 6 Type1, and coppe.pdf 133 against
1103 % 2. The consequences are not cosmetic:
1104 %
1105 %   * the text renders blurred at any zoom and prints poorly;
1106 %   * copying and searching text in the PDF is unreliable;
1107 %   * PDF/A -- required for deposit by 2.2(d) of the 2026 manual -- is not
1108 %     attainable with bitmap text.
1109 %
1110 % Latin Modern is the vector successor of Computer Modern, metrically and
1111 % visually near-identical, and ships with every current distribution, so the
1112 % page layout does not move.
1113 \RequirePackage{lmodern}
1114 \RequirePackage[T1]{fontenc}
1115 % Math support. amsmath is universal and works on every engine, so the class
1116 % brings it in unconditionally (it must precede hyperref, which is loaded at
1117 % the end of the class). amssymb is the legacy AMS symbol package, built
1118 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
1119 % XeLaTeX) it clashes with unicode-math, which already provides all of its
1120 % symbols through Unicode math codes. So amssymb is loaded ONLY under
1121 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
1122 % unicode-math (the class does not auto-load unicode-math because choosing
1123 % an OpenType math font is a typographic decision that belongs to the author).
1124 \RequirePackage{amsmath}
1125 \ifPDFTeX
1126   \RequirePackage{amssymb}
1127 \else
1128   % Unicode engine: defer the check to end of document. Only warn if the
1129   % user did NOT add either unicode-math (the modern OpenType-math route)
1130   % or amssymb (the legacy route, which they would have loaded manually).
1131   % This avoids nagging authors who already followed the documented
1132   % \usepackage{unicode-math} workflow.
1133   \AtEndDocument{%
1134     \@ifpackageloaded{unicode-math}{\}%
1135     \@ifpackageloaded{amssymb}{\}%
1136     \ClassWarningNoLine{coppe}{%
1137       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1138       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1139       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1140       Add to your preamble:\MessageBreak
1141       \space\space\string\usepackage{unicode-math}\MessageBreak
1142       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1143       Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1144       with the OpenType math font setup}}}%
1145 \fi
1146 % Margins per the SiBI manual, 9th rev. ed. (2026), 2.3: left 3cm, top 3cm,
1147 % right 2cm, bottom 2cm -- ONE-SIDED. The 2025 edition also specified verso

```

1148 % margins (right 3cm / left 2cm) and recommended printing the body on both
1149 % sides; the 2026 revision DELETED the verso margins entirely, following CEPG
1150 % Resolution 246/2023, which ended the printed deposit. A digital-only document
1151 % has no gutter, so the mirrored layout (previously inner=outer=2cm plus a 1cm
1152 % bindingoffset) no longer has any normative basis and is now opt-in via the
1153 % \texttt{twoside} class option, handled after \ProcessOptions below.
1154 %
1155 % headheight=15pt is given here (not patched later) so geometry's own page
1156 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header
1157 % height to host the running head, and the standard 12pt default triggers
1158 % its "headheight is too small" warning. Letting geometry know up front
1159 % keeps \topmargin a derived value -- we never touch it by hand.
1160 % (geometry has no \texttt{oneside} key -- only \texttt{twoside}; with the key
1161 % absent it follows the class, which \LoadClass set to one-sided above.)
1162 % headsep is given explicitly to place the folio. 2.7 measures it the way it
1163 % measures the right margin -- to the digits themselves: "a 2cm da borda
1164 % superior, ficando o ultimo algarismo a 2cm da borda direita". The right edge
1165 % already landed at 2.02cm; the top of the digits sat at 1.87cm, too high.
1166 % geometry derives \topmargin from top - 1in - headheight - headsep, so
1167 % shortening headsep lowers the header and leaves the body top untouched:
1168 % topmargin + headheight + headsep is 13.088pt either way, and 3cm is where the
1169 % body still starts. The value is CALIBRATED, not computed -- the folio sits a
1170 % little lower than the arithmetic alone predicts, because fancyhdr's box does
1171 % not put the digits flush with the top of \headheight. Two measured points,
1172 % taken from the rendered page rather than from the PDF's font metrics:
1173 % headsep 18.07pt -> digits at 1.8669cm headsep 14.29pt -> 2.0616cm
1174 % which put 2.0000cm at 15.49pt while the folio was still set at the body size.
1175 % It is \footnotesize now (2.2(b) puts pagination in the smaller font), the
1176 % digits are shorter, and their top therefore sits lower for the same baseline:
1177 % re-measured, 15.49pt gave 2.0659cm and the value moved to 16.87pt. That is
1178 % precisely why this is a measured constant and not a computed one -- change
1179 % the folio's size or headheight and it has to be measured again, not scaled.
1180 % Verified on the rendered page: the top of the digits and their right edge come
1181 % out at the same 2.019cm, and the right margin is 2cm by construction, so that
1182 % shared 0.019cm is the ruler's own bias and not a residual error.
1183 \RequirePackage[a4paper,headheight=15pt,headsep=16.87pt,%
1184 top=3cm,bottom=2cm,left=3cm,right=2cm]{geometry}
1185 % The first paragraph of every section must be indented like any other. LaTeX's
1186 % default is the Anglo-Saxon convention -- no indent after a heading -- which is
1187 % not what Brazilian typography (nor the SiBI manual's own examples) uses.
1188 % indentfirst was mentioned only as a commented suggestion in the example
1189 % preamble, so every document built with this class came out with the first
1190 % paragraph of each section flush left. Loading it in the class fixes it for
1191 % everybody instead of asking each author to remember.
1192 %
1193 % Loaded BEFORE titlesec so titlesec's \@afterindent handling sees the patched
1194 % \@afterindentfalse.
1195 % Long unbreakable tokens -- \verb|...| in running prose, command names, URLs --
1196 % cannot be hyphenated, so TeX overfills the line rather than stretching the
1197 % spaces. example.tex alone produced 17 overfull boxes this way, the worst 86pt,
1198 % which is the "text does not always respect the margins" a recent review
1199 % reported. \emergencystretch gives TeX a final pass in which it may stretch
1200 % interword spaces instead of running into the margin. It cannot save a single
1201 % token longer than the measure, but it clears the ordinary cases.


```

1202 \setlength\emergencystretch{3em}
1203 \RequirePackage{indentfirst}
1204 \RequirePackage{titlesec}
1205 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1206 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1207 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1208 % keep "N Title". \appendix and \annex raise the flag (see below).
1209 \newif\if@coppeappendix \@coppeappendixfalse
1210 \titleformat{\chapter}[hang]
1211   {\normalfont\Large\bfseries}
1212   {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1213   {1ex}\{MakeUppercase}
1214 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1215 % 2.6 says the numeric indicative of a section precedes its title flush left,
1216 % and that titles WITHOUT a numeric indicative are centred -- and then lists
1217 % them, "apendice(s)" and "anexo(s)" among them. A letter is not a numeric
1218 % indicative, and the manual sets its own Annex A centred. The chapter format is
1219 % therefore re-issued when \appendix or \annex is called: same label, same
1220 % travessao, but a centred block instead of a flush-left hang. Numbered chapters
1221 % are untouched -- they DO carry a numeric indicative and stay flush left.
1222 \newcommand\coppe@appendixheadings{%
1223   \titleformat{\chapter}[block]
1224     {\normalfont\Large\bfseries\filcenter}
1225     {\MakeUppercase{\appendixname}\ \thechapter\ \textendash}
1226     {1ex}\{MakeUppercase}%
1227   \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}}
1228 \let\coppe@orig@appendix\appendix
1229 \renewcommand\appendix{%
1230   \coppe@orig@appendix\@coppeappendixtrue\coppe@appendixheadings}
1231 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1232 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1233 % which macro is written (so it is correct at .toc-render time even though the
1234 % appendixname has changed back); the chapter letter is baked in at write time.
1235 \newif\if@coppeannex \@coppeannexfalse
1236 \newcommand\coppe@tocapp[1]{%
1237   \MakeUppercase{\coppestring{appendix}}%
1238   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1239 \newcommand\coppe@tocanx[1]{%
1240   \MakeUppercase{\coppestring{annex}}%
1241   \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1242 \def\@chapter[#1]#2{%
1243   \ifnum \c@secnumdepth >\m@ne
1244     \if@mainmatter
1245       \refstepcounter{chapter}%
1246       \typeout{\@chapapp\space\thechapter.}%
1247       \if@coppeappendix
1248         \if@coppeannex
1249           \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1250         \else
1251           \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1252         \fi
1253       \else
1254         \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1255       \fi

```

```

1256     \else
1257         \addcontentsline{toc}{chapter}{#1}%
1258     \fi
1259 \else
1260     \addcontentsline{toc}{chapter}{#1}%
1261 \fi
1262 \chaptermark{#1}%
1263 \addtocontents{lof}{\protect\advspace{10\p@}}%
1264 \addtocontents{lot}{\protect\advspace{10\p@}}%
1265 \if@twocolumn
1266     \@topnewpage[\@makechapterhead{#2}]%
1267 \else
1268     \@makechapterhead{#2}%
1269     \@afterheading
1270 \fi}
1271 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1272 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1273 \titleformat{name=\chapter,numberless}[hang]
1274     {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1275 \titlespacing*{name=\chapter,numberless}{Opt}{10pt}{1.5\baselineskip}
1276 % section: ALL CAPS, normal weight -- the \bfseries here contradicted this very
1277 % comment until v4.1, and made the secondary level as heavy as the primary one,
1278 % defeating the "destaque gradativo" of 2.6. The gradation is now
1279 % chapter = ALL CAPS bold, section = ALL CAPS regular, subsection = Title Case
1280 % bold, subsubsection = Title Case bold italic, as read from the manual.
1281 \titleformat{\section}
1282     {\normalfont\large}{\thesection}{1ex}{\MakeUppercase}
1283 % subsection: bold (author writes Title Case).
1284 \titleformat{\subsection}
1285     {\normalfont\bfseries}{\thesubsection}{1ex}{\bfseries}
1286 % subsubsection: bold italic.
1287 \titleformat{\subsubsection}
1288     {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\bfseries}
1289 % paragraph = quinary section: italic, the author capitalising only the first
1290 % word. 2.6 recommends going as deep as the quinary, and the annotation on the
1291 % 2025 manual (specs/ANOTACOES.md) describes exactly this shape.
1292 %
1293 % book makes \paragraph a RUN-IN heading, so the text continued on the same
1294 % line -- 2.6 requires "o texto deve ser iniciado em outra linha". The positive
1295 % "after" spacing in \titlespacing turns it into a displayed heading.
1296 \titleformat{\paragraph}[hang]
1297     {\normalfont\itshape}{\theparagraph}{1ex}{\bfseries}
1298 \titlespacing*{\paragraph}{Opt}{2.5ex plus 1ex minus .2ex}{1.5ex plus .2ex}
1299 % book defaults secnumdepth to 2, so \subsubsection and \paragraph came out
1300 % UNNUMBERED however carefully \titleformat named \thesubsubsection. 2.6 has
1301 % the numbering following the whole chain down to the quinary, and the manual's
1302 % own sumario is four levels deep, so tocdepth follows.
1303 \setcounter{secnumdepth}{4}
1304 \setcounter{tocdepth}{4}
1305 % --- 0 sumario (2.6 e 3.1.2.1.6) -----
1306 %
1307 % Duas exigencias o governam, e a segunda tem um exemplo obrigatorio.
1308 %
1309 % A 2.6: "no sumario, as secoes devem ser grafadas conforme apresentadas no

```

```

1310 % corpo do trabalho". A gradacao do corpo -- primaria em CAIXA ALTA negrito,
1311 % secundaria em CAIXA ALTA, terciaria em negrito, quaternaria em negrito
1312 % italico, quinario em italico -- tem de reaparecer aqui. So as duas primeiras
1313 % reapareciam: da terciaria em diante o sumario saia em redondo, contradizendo
1314 % o corpo.
1315 %
1316 % A 3.1.2.1.6: os titulos "devem ser alinhados a esquerda... pela margem do
1317 % titulo do indicativo mais extenso", e "utilize como exemplo o sumario deste
1318 % trabalho". Medido no PDF do manual: o indicativo comeca em 3,00 cm nos cinco
1319 % niveis -- a margem -- e o titulo em 4,75 cm nos cinco. Uma coluna so.
1320 %
1321 % 0 \@dottedtocline do LaTeX faz o contrario. Recua progressivamente (3,00 /
1322 % 3,62 / 4,57 / 5,89 / 7,13 cm nesta classe) e da a cada nivel uma caixa de
1323 % indicativo de largura fixa. Isso empurrava o titulo quinario para 9,20 cm --
1324 % 6,2 dos 16 cm de mancha -- e, pior, a caixa fixa ESTOURA: com o indicativo
1325 % "10.10.10.10.10", que uma tese de dez capitulos alcanca, o numero entrava
1326 % 4,6 mm POR CIMA do titulo. A 2.6 quer o titulo "separado por um espaco" do
1327 % indicativo; ali nao havia espaco nenhum.
1328 %
1329 % A coluna abaixo e medida, nao arbitrada: \coppe@tocnumberline mede cada
1330 % indicativo enquanto compoe o sumario e guarda o maior; o valor vai para o
1331 % .aux e governa a passada seguinte. E literalmente "a margem do titulo do
1332 % indicativo mais extenso". Custa uma passada -- a mesma que as referencias
1333 % cruzadas ja custam.
1334 \newlength\coppe@tocnumwidth
1335 \newlength\coppe@tocnummax
1336 \newlength\coppe@tocnumsep
1337 % 0 espaco entre o indicativo e o titulo (2.6, "separado por um espaco").
1338 \setlength\coppe@tocnumsep{0.5em}
1339 % Piso: a coluna do proprio manual. Pela regra literal, um trabalho so com
1340 % capitulos teria uma coluna de meio centimetro e o titulo quase colado na
1341 % margem; o piso evita isso. A coluna nunca fica MENOR que o indicativo mais
1342 % extenso mais o espaco -- o piso so a impede de ficar estreita demais.
1343 \setlength\coppe@tocnumwidth{1.75cm}
1344 \setlength\coppe@tocnummax{\z@}
1345 \newcommand\coppe@tocnumberline[1]{%
1346   \settowidth\@tempdima{#1}%
1347   \ifdim\@tempdima>\coppe@tocnummax \global\coppe@tocnummax=\@tempdima \fi
1348   \hb@xt@\coppe@tocnumwidth{#1\hfil}}
1349 % A largura medida na passada anterior chega pelo .aux, que o \document le
1350 % antes dos ganchos de \AtBeginDocument.
1351 \AtBeginDocument{%
1352   \@ifundefined{coppe@tocnumsaved}{\fi}%
1353   \setlength\@tempdima{\coppe@tocnumsaved}%
1354   \addtolength\@tempdima\coppe@tocnumsep
1355   \ifdim\@tempdima>\coppe@tocnumwidth
1356     \setlength\coppe@tocnumwidth\@tempdima
1357   \fi}}
1358 \AtEndDocument{%
1359   \ifdim\coppe@tocnummax>\z@
1360     \if@filesw
1361       \immediate\write\@auxout{\string\gdef\string\coppe@tocnumsaved
1362         {\the\coppe@tocnummax}}%
1363     \fi

```

```

1364 \setlength\@tempdima\coppe@tocnummax
1365 \addtolength\@tempdima\coppe@tocnumsep
1366 \ifdim\@tempdima>\coppe@tocnumwidth
1367 \ClassWarning{coppe}{A coluna do sumario cresceu; rode LaTeX de novo}%
1368 \fi
1369 \fi}
1370 % Uma linha do sumario. #1 nivel, #2 fonte, #3 entrada, #4 folha. A entrada
1371 % traz \numberline{N} seguido do titulo; o \hskip-\leftskip puxa a primeira
1372 % linha de volta a margem, e o \leftskip deixa as linhas seguintes de um
1373 % titulo longo alinhadas pela coluna do titulo -- nunca debaixo do numero.
1374 % A folha herda a fonte do nivel, como no sumario do manual.
1375 \newcommand\coppe@tocline[4]{%
1376 \ifnum #1>\c@tocdepth\else
1377 \vskip\z@ \@plus.2\p@
1378 {\leftskip\z@ \rightskip\@tocrmarg \parfillskip-\rightskip
1379 \parindent\z@ \interlinepenalty\@M
1380 \leavevmode
1381 #2%
1382 \advance\leftskip\coppe@tocnumwidth
1383 \null\nobreak\hskip-\leftskip
1384 #3\nobreak
1385 \leaders\hbox{$\m@th\mkern\@dotsep mu\hbox{.}\mkern\@dotsep mu$}%
1386 \hfill\nobreak
1387 \hb@xt@\@pnumwidth{\hss #4}\par}%
1388 \fi}
1389 % O pontilhado do manual e mais fechado que o do book; \@dotsep e a distancia
1390 % entre os pontos, em unidades matematicas.
1391 \renewcommand\@dotsep{2}
1392 % Os cinco niveis. O \MakeUppercase vai so nos dois primeiros porque so eles
1393 % saem em caixa alta no corpo; ele atravessa os algarismos sem toca-los e
1394 % preserva comandos robustos, inclusive os do hyperref.
1395 %
1396 % O nivel 1 nao ganha mais o \addvspace{1.0em} do book: o sumario do manual
1397 % tem espacamento uniforme, sem folga extra antes da secao primaria. As
1398 % penalidades ficam, para que uma entrada de capitulo nao seja abandonada
1399 % sozinha no pe da folha.
1400 \renewcommand*\l@chapter[2]{%
1401 \addpenalty{-\@highpenalty}%
1402 \coppe@tocline{0}{\bfseries}{\MakeUppercase{#1}}{#2}%
1403 \penalty\@highpenalty}
1404 \renewcommand*\l@section[2]{%
1405 \coppe@tocline{1}{\normalfont}{\MakeUppercase{#1}}{#2}}
1406 \renewcommand*\l@subsection[2]{%
1407 \coppe@tocline{2}{\normalfont\bfseries}{#1}{#2}}
1408 \renewcommand*\l@subsubsection[2]{%
1409 \coppe@tocline{3}{\normalfont\bfseries\itshape}{#1}{#2}}
1410 \renewcommand*\l@paragraph[2]{%
1411 \coppe@tocline{4}{\normalfont\itshape}{#1}{#2}}
1412 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1413 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1414 \RequirePackage{caption}
1415 % justification=centering makes MULTI-LINE captions centre like the one-line
1416 % ones. Without it, caption's singlelinecheck centres a caption that fits on one
1417 % line and justifies one that wraps, so in a document with mixed caption lengths

```

```

1418 % a couple of tables looked left-aligned while the rest looked centred -- which
1419 % is what a recent review noticed about tables 5.1 and 5.2. 2.10 does not
1420 % prescribe the alignment; it does not tolerate two alignments either.
1421 \captionsetup{font=footnotesize,labelsep=endash,justification=centering}
1422 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1423 % caption so it picks up the ABNT captionsetup above.
1424 \RequirePackage{subcaption}
1425 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1426 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1427 % at the bottom is mandatory). The float package's plaintop style forces the
1428 % caption above the body for figure, table and the new quadro float. -----
1429 \RequirePackage{float}
1430 \floatstyle{plaintop}
1431 \restylefloat{figure}
1432 \restylefloat{table}
1433 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1434 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1435 % unlike a tabela (rules on top and bottom only).
1436 \providecommand{\quadroname}{\coppestring{frame}}
1437 \providecommand{\listquadroname}{\coppestring{listframe}}
1438 \newfloat{quadro}{htbp}{loq}[chapter]
1439 \floatname{quadro}{\quadroname}
1440 \providecommand{\quadroautorefname}{\quadroname}
1441 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1442 % (float's own \l@quadro lays the number/title out incorrectly).
1443 \let\l@quadro\l@figure
1444 \newcommand\listofquadros{%
1445   \chapter*{\listquadroname}%
1446   \coppe@listtoc{\listquadroname}%
1447   \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1448   \@starttoc{loq}}
1449 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1450 % float as a smaller, single-spaced, centred line; it does NOT advance any
1451 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1452 % the label is "Fonte" (pt) / "Source" (en).
1453 \newcommand{\cpsourcename}{\coppestring{source}}
1454 \newcommand{\cpsource}[1]{%
1455   \par\addvspace{\smallskipamount}%
1456   {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
1457 \let\cpfonte\cpsource
1458 \@ifundefined{source}{\let\source\cpsource}{}%
1459 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1460 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1461 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1462 % \floatstyle is NOT wrapped in a group here: \newfloat defines the environment
1463 % locally, so a group around it threw the new environment away as soon as it
1464 % closed and the document then failed with "Environment undefined". plaintop is
1465 % the class-wide style anyway, set once above and never changed, so leaving it
1466 % selected costs nothing.
1467 % The new float is numbered WITHIN THE CHAPTER, like figure, table and quadro,
1468 % and its list entries are laid out by \l@figure -- float's own \l@<env> runs
1469 % the number into the title and drops the leaders, which is what the Lista de
1470 % Mapas of the example showed the first time.
1471 \newcommand{\newcoppefloat}[3]{%

```

```

1472 \floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}[chapter]%
1473 \floatname{#1}{#2}%
1474 \expandafter\let\csname l@#1\endcsname\l@figure
1475 \expandafter\providecommand\csname #1autorefname\endcsname{#2}%
1476 \expandafter\newcommand\csname listof#1s\endcsname{%
1477   \chapter*{#3}\coppe@listtoc{#3}%
1478   \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1479   \@starttoc{lo#1}}
1480 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1481 %   (filete) from the left margin; footnotesize is the book default. -----
1482 \let\coppe@orig@footnotetext\@footnotetext
1483 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1484 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1485 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1486 %   (manual2024 2.6 / proposta P24). -----
1487 \RequirePackage{enumitem}
1488 \newlist{alineas}{enumerate}{2}
1489 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1490 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1491 % --- Page numbering: arabic in the top OUTER corner on textual pages
1492 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1493 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1494 %   class option). -----
1495 % Every list goes through \@starttoc, which allocates a stream the first time
1496 % it sees an extension. That is where the ceiling is hit in practice, so that
1497 % is where the class checks -- one \ifnum before the allocation, and an error
1498 % that names the three ways out instead of the kernel's bare "No room".
1499 %
1500 % \count17 is the write allocation counter, and 15 is the last usable stream.
1501 % Under LuaTeX there are 128 and the counter is managed differently, so the
1502 % check is skipped there; the same is true when morewrites is in charge.
1503 % The wrap waits for \begin{document} on purpose: hyperref redefines
1504 % \@starttoc as well, and it is loaded at the very end of this class, so a
1505 % \let taken here would be thrown away.
1506 \newcommand\coppe@checkwrites[1]{%
1507   \ifdefined\directlua\else
1508     \if@coppemorewrites\else
1509       \ifundefined{tf@#1}{%
1510         \ifnum\count17>14
1511           \ClassError{coppe}{Acabaram os 16 fluxos de escrita do TeX}{%
1512             Este trabalho pede mais listas do que o pdfTeX consegue manter
1513             abertas ao mesmo tempo.\MessageBreak
1514             Tres saidas, nesta ordem de preferencia:\MessageBreak
1515             (1) compile com lualatex, que tem 128 fluxos;\MessageBreak
1516             (2) passe a opcao 'morewrites' a classe coppe;\MessageBreak
1517             (3) remova uma lista que o trabalho nao usa.\MessageBreak
1518             A lista que faltou agora foi a de extensao '#1', mas o problema
1519             nao e dela: e a soma de todas.}%
1520           \fi}{}%
1521         \fi
1522       }
1523 % The wrapping itself lives in a macro of its own rather than inside the hook:
1524 % \AtBeginDocument stores its argument by expansion, and a parameter written
1525 % there does not survive it -- the document died with "Parameters must be

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1526 % numbered consecutively" at \begin{document}. One level of nesting, one
1527 % doubling, and the hook only has to call it.
1528 \newcommand\coppe@wrapstarttoc{%
1529   \let\coppe@orig@starttoc\@starttoc
1530 % \numberline mede o indicativo (ver \coppe@tocnumberline) apenas no sumario:
1531 % a lista de figuras e a de tabelas numeram ilustracoes, nao secoes, e nao tem
1532 % por que compartilhar a coluna.
1533 \def\@starttoc##1{%
1534   \coppe@checkwrites{##1}%
1535   \begingroup
1536     \def\coppe@@ext{##1}\def\coppe@@toc{toc}%
1537     \ifx\coppe@@ext\coppe@@toc \let\numberline\coppe@tocnumberline \fi
1538     \coppe@orig@starttoc{##1}%
1539   \endgroup}}
1540 \AtBeginDocument{\coppe@wrapstarttoc}
1541 \RequirePackage{fancyhdr}
1542 % \headheight is set to 15pt by geometry up above (see the geometry call),
1543 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1544 % 2.2(b) lists what is set in "fonte menor e uniforme": long quotations,
1545 % footnotes, PAGINATION, and the captions of illustrations and tables. The
1546 % folio was the one item on that list still coming out at the body size of 12pt.
1547 % It is \footnotesize now, which is the same smaller size the other three
1548 % already use -- "uniforme" is part of the requirement, not just "menor".
1549 % Pre-textual lists reach the sumario only under the listasnosumario option
1550 % (see the option's declaration for the reason). Post-textual material calls
1551 % \addcontentsline directly and is unaffected.
1552 \newcommand\coppe@listtoc[1]{%
1553   \if@coppe@listastoc\addcontentsline{toc}{chapter}{#1}\fi
1554 % \coppe@glosspre marks the two PRE-textual uses of the glossary environment --
1555 % the list of abbreviations and the list of symbols -- apart from the genuine
1556 % post-textual Glossario, which shares the same environment but must appear in
1557 % the sumario.
1558 \newif\if@coppe@glosspre \@coppe@glossprefalse
1559 \newcommand\coppe@folio{{\footnotesize\thepage}}
1560 \fancypagestyle{coppe}{{fancyhf}{} \fancyhead[R0,LE]{\coppe@folio}%
1561   \renewcommand{\headrulewidth}{0pt}}
1562 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1563 % number may leak onto them (e.g. the verso right after the cover).
1564 \renewcommand{\cleardoublepage}{%
1565   \clearpage
1566   \if@twoside
1567     \ifodd\c@page\else
1568       \hbox{}\thispagestyle{empty}\newpage
1569     \if@twocolumn\hbox{}\newpage\fi
1570   \fi
1571   \fi}
1572 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1573   p\kern-.025em e}}\kern-.08em
1574   T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1575 \newboolean{maledoc}
1576 \setboolean{maledoc}{false}
1577 % maledoc diz apenas que o substantivo do tipo de documento e masculino
1578 % ("Exame de Qualificacao ... apresentado", "SUBMETIDO"): serve a concordancia,
1579 % e nada mais. Quem NAO vai a deposito -- e portanto nao tem ficha

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```

1580 % catalografica nem folha da Coleta CAPES -- e o exame de qualificacao, que e
1581 % outra coisa. As duas condicoes coincidem hoje por acaso, e por isso ganham
1582 % cada uma o seu booleano.
1583 \newif\if@coppeexame \@coppeexamefalse
1584 %
1585 % Class options.
1586 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1587 % fixed language slots:
1588 % * main      -- the language of the body, captions, and the main abstract.
1589 %              Set by a class option: brazilian (default), english,
1590 %              spanish, french, italian, or any user-supplied language
1591 %              that ships a coppe-lang-<name>.def file.
1592 % * foreign   -- the language used by the foreignabstract environment and
1593 %              by \foreign@... registers. Defaults to english (or to
1594 %              brazilian when main=english) for back-compat.
1595 % * third     -- optional, loaded on demand with \usecoppelanguage{<name>}
1596 %              in the preamble (for citations or quotations in a third
1597 %              language). Has no class option of its own.
1598 % Note: the legacy \if@english boolean (defined and toggled here in
1599 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1600 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.ltx)
1601 % tests it. Code that needs to branch on the main language should test
1602 % \ifx\coppe@mainlang\coppe@englishstr instead.
1603 \def\coppe@mainlang{brazilian}
1604 \def\coppe@foreignlang{english}
1605 \DeclareOption{brazilian}{%
1606   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1607 \DeclareOption{english}{%
1608   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1609 \DeclareOption{spanish}{%
1610   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1611 \DeclareOption{french}{%
1612   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1613 \DeclareOption{italian}{%
1614   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1615 \DeclareOption{msc}{%
1616   \newcommand{\@degree}{M.Sc.}
1617   \newcommand{\@degreename}{Mestrado}
1618   \newcommand{\local@degname}{Mestre}
1619   \newcommand{\foreign@degname}{Master}
1620   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1621   \newcommand\foreign@doctype{Dissertation}
1622 }
1623 \DeclareOption{dsccexam}{%
1624   \newcommand{\@degree}{D.Sc.}
1625   \newcommand{\@degreename}{Doutorado}
1626   \newcommand{\local@degname}{Doutor}
1627   \newcommand{\foreign@degname}{Doctor}
1628   \setboolean{maledoc}{true}\@coppeexamtrue
1629   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1630   \newcommand\foreign@doctype{Qualifying Exam}
1631 }
1632 \DeclareOption{mscexam}{%
1633   \newcommand{\@degree}{M.Sc.}

```



```

1634 \newcommand{\@degreename}{Mestrado}
1635 \newcommand{\local@degname}{Mestre}
1636 \newcommand{\foreign@degname}{Master}
1637 \setboolean{maledoc}{true}\@coppeexamtrue
1638 \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1639 \newcommand\foreign@doctype{Qualifying Exam}
1640 }
1641 \DeclareOption{dsc}{%
1642 \newcommand{\@degree}{D.Sc.}
1643 \newcommand{\@degreename}{Doutorado}
1644 \newcommand{\local@degname}{Doutor}
1645 \newcommand{\foreign@degname}{Doctor}
1646 \newcommand\local@doctype{Tese}
1647 \newcommand\foreign@doctype{Thesis}
1648 }
1649 \newif\if@coppenumeric\@coppenumericfalse
1650 \DeclareOption{numbers}{\@coppenumerictrue}
1651 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1652 % pre-textual pages are counted but not numbered (manual2024 2.7).
1653 \newif\if@copperoman\@copperomanfalse
1654 \DeclareOption{numeraisromanos}{\@copperomantrue}

```

Default line spacing is one-and-a-half; the `doublespacing` option switches to double spacing. Since `setspace` is loaded at the end of the class (after `hyperref`), the actual `\onehalfspacing`/`\doublespacing` call is deferred: the option only flips a flag, which the late-load block reads.

```

1655 \newif\if@coppedoublespace\@coppedoublespacefalse
1656 \DeclareOption{doublespacing}{\@coppedoublespacetrue}
1657 % Two-sided layout is opt-in since v4.1: the 2026 manual dropped the verso
1658 % margins and the deposit is digital (CEPG 246/2023). Given \texttt{twoside},
1659 % the mirrored 3cm inner / 2cm outer layout is restored after \ProcessOptions.
1660 % With no \fichacatalografica file given, the folha adicional leaves a framed
1661 % blank where the ficha goes. The rascunhoficha option fills that blank with the
1662 % class's own LaTeX-composed ficha instead -- useful while writing, never valid
1663 % for deposit: the official ficha comes from the SiBI generator.
1664 % Annex D of the manual draws one signature rule per board member, with the
1665 % identification underneath. The COPPE folha de aprovacao has listed the names
1666 % compactly, without rules, for years. The compact list stays the default and
1667 % the rules are opt-in, so existing theses do not change shape overnight.
1668 % PDF/A output. Opt-in for now: 2.2(d) of the 2026 manual requires the deposited
1669 % file to be PDF/A, but the constraints (embedded fonts, colour profile, no
1670 % encryption, restricted transparency) can trip up a document that pulls in
1671 % arbitrary images or packages, and breaking existing theses to enforce it would
1672 % be worse than letting the author switch it on when depositing.
1673 \newif\if@coppepdfa \@coppepdfafalse
1674 \DeclareOption{pdfa}{\@coppepdfatrue}
1675 % A opcao 'assinaturas' punha uma regua para assinatura a mao acima do nome de
1676 % cada membro da banca. Com a entrega so digital (Resolucao CEPG n. 246/2023)
1677 % nao ha o que assinar a mao, e a v4.1 tirou as reguas. A opcao continua aceita,
1678 % sem efeito, para nao quebrar documento escrito antes disso.
1679 \DeclareOption{assinaturas}{\ClassWarningNoLine{coppe}{A opcao 'assinaturas'
1680   nao faz mais nada: a folha de aprovacao nao tem mais linhas de assinatura,
1681   porque a entrega e digital desde a Resolucao CEPG n. 246/2023}}
1682 % As paginas de resumo listam os ORIENTADORES abaixo do titulo. O coorientador

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1683 % nao aparece ali, e isso e tradicao da COPPE, nao exigencia do manual: a
1684 % 3.1.2.1.4 descreve o resumo e nao lista quem assina a orientacao. Como ha
1685 % Programas que preferem ve-lo, a opcao acrescenta os coorientadores nas tres
1686 % paginas de resumo, no mesmo formato da folha de rosto. Sem a opcao nada muda.
1687 \newif\if@coppecoadvresumo \@coppecoadvresumofalse
1688 \DeclareOption{coorientador}{\@coppecoadvresumotrue}
1689 % 3.1.2.1.6 says the sumario enumerates the divisions "na mesma ordem e grafia
1690 % em que a materia nele se sucede" and adds "inclusive os elementos
1691 % pos-textuais" -- and then settles the question by pointing at its own sumario
1692 % as the model. That sumario opens at "1 INTRODUCAO": the lists of figures, of
1693 % tables, of abbreviations and of symbols come BEFORE the sumario and are not
1694 % listed in it, which is also the rule of NBR 6027 for pre-textual elements.
1695 % The class was adding all of them, so every thesis carried six entries the
1696 % manual's own example does not have. They come out. Post-textual elements --
1697 % Referencias, Glossario, Apendices, Anexos, Indice -- stay, as required.
1698 %
1699 % Programas that want the old behaviour ask for it by name.
1700 \newif\if@coppelistastoc \@coppelistastocfalse
1701 \DeclareOption{listasnosumario}{\@coppelistastoctrue}
1702 \newif\if@copperascunho \@copperascunhofalse
1703 \DeclareOption{rascunhoficha}{\@copperascunhotrue}
1704 \newif\if@coppetwoside \@coppetwosidefalse
1705 \DeclareOption{twoside}{\@coppetwosidetrue}
1706 \DeclareOption{oneside}{\@coppetwosidefalse}
1707 % --- Fluxos de escrita -----
1708 % TeX has SIXTEEN \write streams and no more. Every list costs one for the
1709 % whole run: the .toc, the .lof, the .lot, and one for each list this class
1710 % adds (.loq for quadros, .lol for programas, .loa for algoritmos), plus the
1711 % .abx and .syx behind the lists of abbreviations and of symbols, the .idx of
1712 % the index, hyperref's .out, biblalex's .bcf and the kernel's .aux and friends.
1713 % A work that asks for all of them sits exactly at the ceiling.
1714 %
1715 % The kernel's failure in that situation -- "No room for a new \write" -- names
1716 % neither the cause nor the way out, and lands on whichever list happens to be
1717 % requested last, which is usually \tableofcontents and has nothing to do with
1718 % it. The class therefore checks before the allocation and says what is
1719 % happening (see \coppe@starttoc below).
1720 %
1721 % There are three ways out and the class chooses none of them for the author:
1722 %
1723 % * compile with LuaLaTeX. It has 128 streams; nothing else has to change,
1724 % and the class supports it.
1725 % * let the class load 'morewrites', which multiplexes the streams so the
1726 % ceiling disappears under pdfLaTeX.
1727 % * drop a list the work does not use. Each one returns a stream.
1728 %
1729 % The DEFAULT is the second one, and it is on. An error telling the author to
1730 % pass an option is advice for someone who already knows: a student two days
1731 % from the deposit reads "No room for a new \write" and concludes the class is
1732 % broken. So under pdfTeX the class loads morewrites by itself.
1733 %
1734 % Three things keep that from becoming a burden:
1735 %
1736 % * it is loaded ONLY IF INSTALLED. Missing, the class carries on without it

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1737 %      and warns; morewrites does not become a hard dependency of a work that
1738 %      would never have reached the ceiling anyway.
1739 %      * 'semmorewrites' turns it off, for whoever hits a conflict. morewrites
1740 %      patches \newwrite, \openout and \write and buffers what is written until
1741 %      shipout, so a package that writes a file and reads it back in the SAME run
1742 %      can misbehave. That is the real cost of this default, and it is why the
1743 %      escape exists.
1744 %      * 'morewrites' still exists and now means "I insist": with it, a missing
1745 %      package is an error rather than a warning.
1746 %
1747 % Under LuaTeX none of this applies and nothing is loaded.
1748 \newif\if@coppemorewrites \@coppemorewritestrue
1749 \newif\if@coppemwforced \@coppemwforcedfalse
1750 \DeclareOption{morewrites}{\@coppemorewritestrue\@coppemwforcedtrue}
1751 \DeclareOption{semmorewrites}{\@coppemorewritestruefalse}
1752 % A 2.2(b) fixa a cor, o corpo 12 e o corpo menor uniforme dos quatro itens
1753 % listados nela, e nao diz uma palavra sobre familia de fonte -- nem a Norma da
1754 % COPPE. A escolha e do autor. Desde a v4.1 o padrao da classe e a SEM SERIFA,
1755 % que e como o proprio manual do SiBI, que fixa a regra, esta composto;
1756 % 'comserifa' volta para a serifada, que e a do LaTeX. 'semserifa' continua
1757 % aceita, agora sem efeito, para nao quebrar documento escrito antes da v4.1.
1758 \newif\if@coppesansserif \@coppesansseriftrue
1759 \DeclareOption{semserifa}{\@coppesansseriftrue}
1760 \DeclareOption{comserifa}{\@coppesansseriffalse}
1761 \ProcessOptions\relax
1762 % A troca da familia padrao vai aqui porque so agora se sabe se a opcao foi
1763 % pedida; o lmodern ja esta carregado la em cima e traz as duas familias, a
1764 % serifada e a sem serifa, ambas vetoriais -- que e o que o PDF/A exige.
1765 % Quem quiser uma sem serifa DIFERENTE (Helvetica, por exemplo) carrega o
1766 % pacote da fonte no preambulo e refaz \familydefault ali; o padrao existe
1767 % para o caso comum, em que ninguem quer escolher fonte.
1768 \if@coppesansserif
1769   \renewcommand{\familydefault}{\sfdefault}%
1770 \fi
1771 % morewrites has to come before anything else allocates, so it goes first
1772 % thing after the options are read.
1773 \if@coppemorewrites
1774   \ifdefined\directlua
1775 % Under LuaTeX there is nothing to solve: 128 streams already. The flag is
1776 % cleared too, so the \@starttoc guard further down knows the ceiling is not
1777 % in play here either.
1778   \ClassInfo{coppe}{morewrites dispensado: o LuaTeX ja tem 128 fluxos}%
1779   \@coppemorewritestrue
1780 \else
1781   \IfFileExists{morewrites.sty}
1782   {\RequirePackage{morewrites}}
1783   {\if@coppemwforced
1784     \ClassError{coppe}{Opcao 'morewrites' pedida, mas morewrites.sty nao
1785      esta instalado}{Instale o pacote morewrites, ou compile com lualatex,
1786      que nao precisa dele.}%
1787   \else
1788     \ClassWarningNoLine{coppe}{O pacote morewrites nao esta instalado.\MessageBreak
1789      Sem ele o pdfTeX so aguenta 16 fluxos de escrita, e um trabalho\MessageBreak
1790      com todas as listas passa disso. Se a compilacao parar por falta\MessageBreak

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1791         de fluxo, instale morewrites ou compile com lualatex}%
1792         \@coppemorewritesfalse
1793     \fi}%
1794 \fi
1795 \fi
1796 % \LoadClass and geometry ran before \ProcessOptions (see above), so the
1797 % two-sided layout can only be applied here. geometry reads left/right as
1798 % inner/outer once twoside is on, so the 3cm/2cm pair set above becomes the
1799 % mirrored gutter with no further arithmetic. \@twosidetrue is what book's own
1800 % twoside option sets, and is what \cleardoublepage and fancyhdr test.
1801 % In one-sided mode fancyhdr treats every page as odd, so the \fancyhead[R0,LE]
1802 % declarations elsewhere already put the folio in the top right corner of every
1803 % folha, as 2.7 requires.
1804 \if@coppetwoside
1805     \@twosidetrue
1806     \geometry{twoside}%
1807 \fi
1808 % Built-in language names (used by the loader and the dispatcher below).
1809 \def\coppe@@brazilianstr{brazilian}
1810 \def\coppe@@englishstr{english}
1811 % Load Babel with the foreign language first and the main language last
1812 % (Babel treats the LAST option in the list as the main language). With
1813 % pt-main this produces [english,brazilian]{babel} and with en-main
1814 % [brazilian,english] -- identical to the historical \if@english calls.
1815 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1816 % Some Babel modules change LaTeX beyond translating captions, and one of
1817 % those changes breaks the norm. With Spanish as the MAIN language,
1818 % babel-spanish redefines \numberline to put a period after the section
1819 % number, so the sumario reads "2.1. SECCION SECUNDARIA" while the section
1820 % heading itself -- which the class formats -- reads "2.1 SECCION SECUNDARIA".
1821 % The 2.6 separates the indicative from the title by a space (NBR 6024) and
1822 % the 3.1.2.1.6 wants the sumario to reproduce the text; the period fails
1823 % both. 'es-nosectiondot' turns off that single adjustment and leaves the rest
1824 % of babel-spanish (quoting, shorthands, roman numerals, captions) alone.
1825 % \coppe@babelextra@<lang> is the general hook: it applies to the MAIN
1826 % language only, which is where these layout modules act, and it is kept out
1827 % of \coppe@babelopts because the language-pack loader below iterates that
1828 % list and would look for coppe-lang-es-nosectiondot.def.
1829 \@namedef{coppe@babelextra@spanish}{,es-nosectiondot}
1830 \let\coppe@babelload\coppe@babelopts
1831 \@ifundefined{coppe@babelextra@\coppe@mainlang}{\%
1832     \edef\coppe@babelload{\coppe@babelload\@nameuse{coppe@babelextra@\coppe@mainlang}}
1833 \expandafter\RequirePackage\expandafter[\coppe@babelload]{babel}
1834 % --- Multilingual string dispatcher -----
1835 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1836 % language <lang>. The three retrievers below look strings up either at the
1837 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1838 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1839 \providecommand{\copperdefstring}[3]{%
1840     \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1841 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@\language @#1}}
1842 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1843 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1844 % Load an extra language pack at any point in the preamble (third-language

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1845 % slot for citations or quotations beyond the main+foreign pair).
1846 \providecommand{\usecoppelanguage}[1]{%
1847   \def\coppe@@cur{#1}%
1848   \ifx\coppe@@cur\coppe@@brazilianstr\else
1849   \ifx\coppe@@cur\coppe@@englishstr\else
1850     \InputIfFileExists{coppe-lang-#1.def}{}
1851     {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1852   \fi\fi}
1853 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1854 % consulted by the cover/folha renderers once they are migrated in step 2.
1855 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1856 % still drive every visible site, so behavior for pt/en is unchanged.
1857 \providecommand{\titlein}[2]{%
1858   \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1859 % --- Built-in string tables: brazilian + english. The values mirror the
1860 % literals currently scattered through the class; step-1 only populates the
1861 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1862 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1863 \copperdefstring{english}{appendix}{Appendix}
1864 \copperdefstring{brazilian}{annex}{Anexo}
1865 \copperdefstring{english}{annex}{Annex}
1866 \copperdefstring{brazilian}{frame}{Quadro}
1867 \copperdefstring{english}{frame}{Frame}
1868 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1869 \copperdefstring{english}{listframe}{List of Frames}
1870 \copperdefstring{brazilian}{source}{Fonte}
1871 \copperdefstring{english}{source}{Source}
1872 \copperdefstring{brazilian}{program}{Programa}
1873 \copperdefstring{english}{program}{Program}
1874 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1875 \copperdefstring{english}{listprogram}{List of Programs}
1876 \copperdefstring{brazilian}{references}{Refer\~encias}
1877 \copperdefstring{english}{references}{References}
1878 \copperdefstring{brazilian}{in:}{em}
1879 \copperdefstring{english}{in:}{in}
1880 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1881 \copperdefstring{english}{art}{Art}
1882 \copperdefstring{brazilian}{platform}{Plataforma}
1883 \copperdefstring{english}{platform}{Platform}
1884 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao de}
1885 \copperdefstring{english}{directedby}{Directed by}
1886 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao de}
1887 \copperdefstring{english}{producedby}{Produced by}
1888 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1889 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1890 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~i}mbolos}
1891 \copperdefstring{english}{listsymbol}{List of Symbols}
1892 \copperdefstring{brazilian}{glossary}{Gloss{\~a}rio}
1893 \copperdefstring{english}{glossary}{Glossary}
1894 \copperdefstring{brazilian}{advisor}{Orientador}
1895 \copperdefstring{english}{advisor}{Advisor}
1896 \copperdefstring{brazilian}{advisors}{Orientadores}
1897 \copperdefstring{english}{advisors}{Advisors}
1898 \copperdefstring{brazilian}{dept}{Programa}

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1899 \copperdefstring{english} {dept} {Department}
1900 \copperdefstring{brazilian}{approvedmale} {Aprovado por}
1901 \copperdefstring{brazilian}{approvedfemale}{Aprovada por}
1902 \copperdefstring{brazilian}{approvedonmale} {Aprovado em}
1903 \copperdefstring{brazilian}{approvedonfemale}{Aprovada em}
1904 \copperdefstring{brazilian}{keywordsname} {Palavras-chave}
1905 \copperdefstring{brazilian}{volumename} {Volume}
1906 \copperdefstring{brazilian}{coadvisor} {Coorientador}
1907 \copperdefstring{brazilian}{coadvisors} {Coorientadores}
1908 \copperdefstring{brazilian}{ofname} {de}
1909 \copperdefstring{brazilian}{researchline}{Linha de pesquisa}
1910 \copperdefstring{english} {approvedmale} {Approved by}
1911 \copperdefstring{english} {approvedfemale}{Approved by}
1912 \copperdefstring{english} {approvedonmale} {Approved on}
1913 \copperdefstring{english} {approvedonfemale}{Approved on}
1914 \copperdefstring{english} {keywordsname} {Keywords}
1915 \copperdefstring{english} {volumename} {Volume}
1916 \copperdefstring{english} {coadvisor} {Co-advisor}
1917 \copperdefstring{english} {coadvisors} {Co-advisors}
1918 \copperdefstring{english} {ofname} {of}
1919 \copperdefstring{english} {researchline}{Research line}
1920 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1921 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1922 \copperdefstring{english} {universityname}{Universidade Federal do Rio de Janeiro}
1923 \copperdefstring{brazilian}{cityname} {Rio de Janeiro}
1924 \copperdefstring{english} {cityname} {Rio de Janeiro}
1925 \copperdefstring{brazilian}{statename} {RJ}
1926 \copperdefstring{english} {statename} {RJ}
1927 \copperdefstring{brazilian}{countryname} {Brasil}
1928 \copperdefstring{english} {countryname} {Brasil}
1929 % Month names.
1930 \copperdefstring{brazilian}{monthname1} {Janeiro}
1931 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1932 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1933 \copperdefstring{brazilian}{monthname4} {Abril}
1934 \copperdefstring{brazilian}{monthname5} {Maio}
1935 \copperdefstring{brazilian}{monthname6} {Junho}
1936 \copperdefstring{brazilian}{monthname7} {Julho}
1937 \copperdefstring{brazilian}{monthname8} {Agosto}
1938 \copperdefstring{brazilian}{monthname9} {Setembro}
1939 \copperdefstring{brazilian}{monthname10}{Outubro}
1940 \copperdefstring{brazilian}{monthname11}{Novembro}
1941 \copperdefstring{brazilian}{monthname12}{Dezembro}
1942 \copperdefstring{english}{monthname1} {January}
1943 \copperdefstring{english}{monthname2} {February}
1944 \copperdefstring{english}{monthname3} {March}
1945 \copperdefstring{english}{monthname4} {April}
1946 \copperdefstring{english}{monthname5} {May}
1947 \copperdefstring{english}{monthname6} {June}
1948 \copperdefstring{english}{monthname7} {July}
1949 \copperdefstring{english}{monthname8} {August}
1950 \copperdefstring{english}{monthname9} {September}
1951 \copperdefstring{english}{monthname10}{October}
1952 \copperdefstring{english}{monthname11}{November}

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1953 \copperdefstring{english}{monthname12}{December}
1954 % algorithm2e package option for this language (empty -> english default).
1955 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1956 \copperdefstring{english} {algorithm2eopt}{}
1957 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1958 \copperdefstring{brazilian}{babelname}{brazilian}
1959 \copperdefstring{english} {babelname}{english}
1960 % Auto-load language packs for any non-built-in language listed in babelopts.
1961 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1962 % (defined just above) and need no extra file; others (spanish/french/italian
1963 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1964 % must already be defined when these run, so the loader sits AFTER the tables.
1965 \@for\coppe@@lang:=\coppe@babelopts\do{%
1966   \edef\coppe@@cur{\coppe@@lang}%
1967   \ifx\coppe@@cur\coppe@@brazilianstr\else
1968     \ifx\coppe@@cur\coppe@@englishstr\else
1969       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1970       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1971       {\ClassError{coppe}
1972         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1973         {Install it next to coppe.cls, or pick a built-in language.}}%
1974     \fi\fi}
1975 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1976 % Portuguese alias for short, inline quotations (proposta P3).
1977 \RequirePackage{csquotes}
1978 \newcommand{\citacao}[1]{\enquote{#1}}
1979 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1980 % package gives the "Programa" listing (caption on top, ABNT R46) with
1981 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and
1982 % the language styles below are based on the user's listagens.tex. -----
1983 \RequirePackage{xcolor}
1984 \RequirePackage{listings}
1985 \RequirePackage{fancyvrb}
1986 \DeclareFixedFont{\ttb}{T1}{txxtt}{bx}{n}{10}
1987 \DeclareFixedFont{\ttm}{T1}{txxtt}{m}{n}{10}
1988 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1989   literate=%
1990   {\~{\~{a}}}{\~{a}}1 {\~{\~{a}}}{\~{a}}1 {\~{\~{a}}}{\~{a}}1 {\~{\~{a}}}{\~{a}}1 {\~{\~{e}}}{\~{e}}1%
1991   {\~{\~{e}}}{\~{e}}1 {\~{\~{e}}}{\~{e}}1 {\~{\~{i}}}{\~{i}}1 {\~{\~{i}}}{\~{i}}1 {\~{\~{o}}}{\~{o}}1%
1992   {\~{\~{o}}}{\~{o}}1 {\~{\~{o}}}{\~{o}}1 {\~{\~{u}}}{\~{u}}1 {\~{\~{u}}}{\~{u}}1 {\~{\~{c}}}{\~{c}}1%
1993   {\~{\~{C}}}{\~{C}}1 {\~{\~{A}}}{\~{A}}1 {\~{\~{A}}}{\~{A}}1 {\~{\~{A}}}{\~{A}}1 {\~{\~{E}}}{\~{E}}1%
1994   {\~{\~{E}}}{\~{E}}1 {\~{\~{I}}}{\~{I}}1 {\~{\~{O}}}{\~{O}}1 {\~{\~{O}}}{\~{O}}1 {\~{\~{O}}}{\~{O}}1%
1995   {\~{\~{U}}}{\~{U}}1%
1996 }
1997 % Babel-aware listing names, set after listings' own babel setup.
1998 \AtBeginDocument{%
1999   \renewcommand{\lstlistingname}{\coppestring{program}}%
2000   \renewcommand{\lstlistlistingname}{\coppestring{listprogram}}%
2001 % \listofprogramas in the coppe list style (heading + ToC + running head).
2002 \newcommand{\listofprogramas}%
2003   \chapter*{\lstlistlistingname}%
2004   \coppe@listtoc{\lstlistlistingname}%
2005   \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
2006   \@starttoc{lol}}

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2007 % Per-language styles + environments (python, java, xml, html, prolog).
2008 % The red hook that marks a wrapped line is built ONCE, into a box, and the
2009 % styles below insert the box. Writing the colour inline -- postbreak=\mbox{%
2010 % \textcolor{red}{\hookrightarrow}\space} -- worked for years and broke the
2011 % moment the pdfa option was switched on: pdfx puts xcolor into a converting
2012 % mode (\selectcolormodel plus \convertcolorsU/D), and a colour command run
2013 % from inside listings' line-breaking machinery then fails with "Missing
2014 % number, treated as zero" at the first wrapped line. Every listing in the
2015 % document dies, so the class's own example could not be compiled as PDF/A at
2016 % all. A box is immune: it is built here, in ordinary horizontal mode where
2017 % xcolor works normally, and listings only ever copies it.
2018 %
2019 % The \sbox waits for \begin{document} because xcolor's colours are not
2020 % defined while the class is still loading.
2021 \newsavebox\coppe@lstarrow
2022 \AtBeginDocument{\sbox\coppe@lstarrow{\textcolor{red}{\hookrightarrow}\space}}
2023 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
2024   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2025   morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
2026   emph={MyClass,__init__},emphstyle=\ttb\color[rgb]{0.6,0,0},%
2027   stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
2028   extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
2029   escapechar=',postbreak=\usebox\coppe@lstarrow}}
2030 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}%
2031   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
2032 \newcommand\pythoninline[1]{\pythonstyle\lstinline!#1!}
2033 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}
2034 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
2035   numberstyle=\small\ttfamily,numbersep=0.5em,%
2036   morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
2037   keywordstyle=\ttb\color[rgb]{0,0,0.5},%
2038   emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
2039   emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2040   showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
2041   postbreak=\usebox\coppe@lstarrow}}
2042 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}]{
2043 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
2044   numberstyle=\small\ttfamily,numbersep=0.5em,%
2045   keywordstyle=\ttb\color[rgb]{0,0,0.5},emphstyle=\ttb\color[rgb]{0.6,0,0},%
2046   stringstyle=\color[rgb]{0,0.5,0},frame=tb,showstringspaces=false,%
2047   extendedchars=true,inputencoding=utf8,breaklines=true,%
2048   postbreak=\usebox\coppe@lstarrow}}
2049 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}]{
2050 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%
2051   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
2052   keywordstyle=\ttb\color[rgb]{0,0,0.5},emph={MyClass,__init__},%
2053   emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2054   showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2055   breaklines=true,escapechar=',%
2056   postbreak=\usebox\coppe@lstarrow}}
2057 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
2058   \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
2059 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
2060   numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%

```



```

2061 morekeywords={self,as},keywordstyle=\ttb\color[rgb]{0,0,0.5},%
2062 emphstyle=\ttb\color[rgb]{0.6,0,0},stringstyle=\color[rgb]{0,0.5,0},frame=tb,%
2063 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
2064 breaklines=true,escapechar=',%
2065 postbreak=\usebox\coppe@lstarrow}}
2066 \lstnewenvironment{java}[1][\{\javastyle\lstset{#1}%
2067 \csname\@lst @SetFirstNumber\endcsname\}{\csname\@lst @SaveFirstNumber\endcsname}
2068 \newcommand\javainline[1]{\{\javastyle\lstinline!#1!}}
2069 \newcommand\javaexternal[2][\{\javastyle\lstinputlisting[1]{#2}}
2070 % fancyvrb's Verbatim environment is available for program output; the
2071 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
2072 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
2073 % example,
2074 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
2075 % ...
2076 % \end{Verbatim}
2077 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
2078 \renewcommand{\theFancyVerbLine}{\small\ttfamily\arabic{FancyVerbLine}}
2079 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
2080 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
2081 % language (English) -- needed because algorithm2e errors on an empty option.
2082 \edef\coppe@@algotlang{\coppemainstring{algorithm2eopt}}
2083 \ifx\coppe@@algotlang\@empty
2084 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
2085 \else
2086 \expandafter\RequirePackage\expandafter[\coppe@@algotlang,linesnumbered,lined,algochapter,n
2087 \fi
2088 % algorithm2e composes its own caption and therefore ignores the class-wide
2089 % \captionsetup{font=footnotesize,labelsep=endash}. Left alone it printed
2090 % "Algoritmo 6.1:Titulo" in bold at body size, while every other illustration
2091 % printed "Figura 6.1 -- Titulo" in footnotesize -- 2.10 asks for the
2092 % designating word, the number, a TRAVESSAO and the title, uniformly.
2093 %
2094 % The separator and the caption fonts are therefore set to match. The 'ruled'
2095 % layout (rules above and below) is kept: 2.10 prescribes where the caption and
2096 % the source go, not whether the float is delimited, and the rules read here
2097 % like the top-and-bottom delimitation a table carries.
2098 \SetAlgoCaptionSeparator{\textendash}
2099 \SetAlCapFnt{\footnotesize}
2100 \SetAlCapNameFnt{\footnotesize}
2101 \SetAlCapSty{textnormal}
2102 \SetAlCapNameSty{textnormal}
2103 \SetKwFor{Para}{para}{faça}{fim do para}
2104 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
2105 % setspace + hyperref + biblatex are loaded at the very end of the class
2106 % (see the block just before %</class>), so their hooks see every macro the
2107 % class defines, and the load order setspace -> hyperref -> biblatex is
2108 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

2109 \newcommand\coppe@deptcode{}
2110 \newcommand\meta@deptname{}

```

```

2111 \newcommand\department[1]{%
2112 % The programme CODE is kept alongside the resolved name, and each name is
2113 % declared TWICE. \local@deptname keeps the TeX accent escapes with braces
2114 % (Computa{\c c}{\~ a}o) that the cover and the folha de rosto typeset;
2115 % \meta@deptname is the same name in literal UTF-8, and is the one that goes
2116 % into the XMP. pdfx copies whatever it is given straight through, so feeding it
2117 % the escaped form produced "Computa{c}o" in dc:description -- which is why the
2118 % subject line used to fall back to the bare programme code. Both forms are
2119 % written out here rather than derived from one another because there is no
2120 % reliable way to strip TeX accent syntax at the point where the .xmldata is
2121 % written, and a table of thirteen entries is cheaper than that machinery.
2122 \gdef\coppe@deptcode{#1}%
2123 \ifthenelse{\equal{#1}{PEB}}{
2124   {\global\def\local@deptname{Engenharia Biom{\` e}dica}
2125    \global\def\meta@deptname{Engenharia Biomédica}
2126    \global\def\foreign@deptname{Biomedical Engineering}}{}
2127 \ifthenelse{\equal{#1}{PEC}}{
2128   {\global\def\local@deptname{Engenharia Civil}
2129    \global\def\meta@deptname{Engenharia Civil}
2130    \global\def\foreign@deptname{Civil Engineering}}{}
2131 \ifthenelse{\equal{#1}{PEE}}{
2132   {\global\def\local@deptname{Engenharia El{\` e}trica}
2133    \global\def\meta@deptname{Engenharia Elétrica}
2134    \global\def\foreign@deptname{Electrical Engineering}}{}
2135 \ifthenelse{\equal{#1}{PEM}}{
2136   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
2137    \global\def\meta@deptname{Engenharia Mecânica}
2138    \global\def\foreign@deptname{Mechanical Engineering}}{}
2139 \ifthenelse{\equal{#1}{PEMM}}{
2140   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
2141    \global\def\meta@deptname{Engenharia Metalúrgica e de Materiais}
2142 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
2143 \ifthenelse{\equal{#1}{PEN}}{
2144   {\global\def\local@deptname{Engenharia Nuclear}
2145    \global\def\meta@deptname{Engenharia Nuclear}
2146    \global\def\foreign@deptname{Nuclear Engineering}}{}
2147 \ifthenelse{\equal{#1}{PEN0}}{
2148   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
2149    \global\def\meta@deptname{Engenharia Oceânica}
2150    \global\def\foreign@deptname{Ocean Engineering}}{}
2151 \ifthenelse{\equal{#1}{PPE}}{
2152   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
2153    \global\def\meta@deptname{Planejamento Energético}
2154    \global\def\foreign@deptname{Energy Planning}}{}
2155 \ifthenelse{\equal{#1}{PEP}}{
2156   {\global\def\local@deptname{Engenharia de Produ{\c c}{\~ a}o}
2157    \global\def\meta@deptname{Engenharia de Produção}
2158    \global\def\foreign@deptname{Production Engineering}}{}
2159 \ifthenelse{\equal{#1}{PEQ}}{
2160   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
2161    \global\def\meta@deptname{Engenharia Química}
2162    \global\def\foreign@deptname{Chemical Engineering}}{}
2163 \ifthenelse{\equal{#1}{PESC}}{
2164   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\~ a}o}

```

```

2165 \global\def\meta@deptname{Engenharia de Sistemas e Computação}
2166 \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
2167 \ifthenelse{\equal{#1}{PET}}
2168 {\global\def\local@deptname{Engenharia de Transportes}
2169 \global\def\meta@deptname{Engenharia de Transportes}
2170 \global\def\foreign@deptname{Transportation Engineering}}{}
2171 \ifthenelse{\equal{#1}{PENT}}
2172 {\global\def\local@deptname{Engenharia de Nanotecnologia}
2173 \global\def\meta@deptname{Engenharia de Nanotecnologia}
2174 \global\def\foreign@deptname{Nanotechnology Engineering}}{}
2175 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

2176 \renewcommand\title[1]{%
2177 \global\def\local@title{#1}%
2178 \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
2179 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

2180 \newcommand\foreigntitle[1]{%
2181 \global\def\foreign@title{#1}%
2182 \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
2183 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

2184 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}
2185 % Subtitle, mirroring the title machinery slot for slot: \subtitle feeds the
2186 % brazilian slot, \foreignsubtitle the english one, \subtitlein any other.
2187 % 3.1.1 and 3.1.2.1.1(c) require the subtitle on the capa and on the folha de
2188 % rosto, subordinated to the title by a colon.
2189 \newcommand\subtitle[1]{%
2190 \expandafter\gdef\csname coppe@subtitle@brazilian\endcsname{#1}}
2191 \newcommand\foreignsubtitle[1]{%
2192 \expandafter\gdef\csname coppe@subtitle@english\endcsname{#1}}
2193 \newcommand\subtitlein[2]{%
2194 \expandafter\gdef\csname coppe@subtitle@#1\endcsname{#2}}
2195 \newcommand\coppe@selectsubtitle{\@nameuse{coppe@subtitle@\coppe@mainlang}}
2196 % Title as the capa and the folha de rosto print it. Annexes A and B show the
2197 % title in caps and the subtitle in ordinary case after a colon, so the case
2198 % change is applied to the title alone.
2199 \newcommand\coppe@titleblock{%
2200 \MakeUppercase{\coppe@selecttitle}%
2201 \@ifundefined{coppe@subtitle@\coppe@mainlang}{}{\coppe@selectsubtitle}}
2202 % Number of volumes (3.1.1 and 3.1.2.1.1(d)): printed only when there is more
2203 % than one, since a single-volume work says nothing about volumes.
2204 % Coorientador (3.1.2.1.1(f)). Same shape as \advisor, including the optional
2205 % treatment; the label switches to the plural on the second call, as the
2206 % advisor label already did.

```

```

2207 \newcount\@coadvisor\@coadvisor0
2208 \newcommand\coadvisor[5][\%
2209   \global\@namedef{CoppeCoadvisorTitle:\expandafter\the\@coadvisor}{#1}
2210   \global\@namedef{CoppeCoadvisorName:\expandafter\the\@coadvisor}{#2}
2211   \global\@namedef{CoppeCoadvisorSurname:\expandafter\the\@coadvisor}{#3}
2212   \global\@namedef{CoppeCoadvisorDegree:\expandafter\the\@coadvisor}{#4}
2213   \global\@namedef{CoppeCoadvisorInst:\expandafter\the\@coadvisor}{#5}
2214   \global\advance\@coadvisor by 1
2215 }
2216 \newcommand\local@coadvisorstring{%
2217   \ifnum\@coadvisor>1
2218     \coppemainstring{coadvisors}%
2219   \else
2220     \coppemainstring{coadvisor}%
2221   \fi}
2222 % Mesma coisa no idioma estrangeiro, para a pagina de abstract.
2223 \newcommand\foreign@coadvisorstring{%
2224   \ifnum\@coadvisor>1
2225     \coppeforeignstring{coadvisors}%
2226   \else
2227     \coppeforeignstring{coadvisor}%
2228   \fi}
2229 \newcommand\coppe@nvolumes{1}
2230 \newcommand\coppe@thisvolume{1}
2231 \newcommand\volumes[1]{\gdef\coppe@nvolumes{#1}}
2232 \newcommand\volume[1]{\gdef\coppe@thisvolume{#1}}
2233 \newcommand\coppe@volumeline{%
2234   \ifnum\coppe@nvolumes>1
2235     \par\vspace*{4mm}%
2236     \coppemainstring{volumename}\space\coppe@thisvolume\space
2237     \coppemainstring{ofname}\space\coppe@nvolumes
2238 % The closing \par matters: on the capa the next thing is \vspace{\stretch{3}},
2239 % which does not end a paragraph, so without it the city ran on into the volume
2240 % line ("Volume 2 de 3 RIO DE JANEIRO").
2241   \par
2242   \fi}

```

\advisor Defines globally the title, name and academic degree of the advisors.

```

2243 \newcount\@advisor\@advisor0
2244 % Argument order since v4.1: the TREATMENT ("Prof.", "Prof.a") became the
2245 % optional slot, and comes empty by default -- no norm asks for it and it only
2246 % clutters the folha de aprovacao. The INSTITUTION, required by 3.1.2.1.3(e),
2247 % became the last mandatory argument; it may be left empty ({} without error,
2248 % in which case nothing is printed for it. A pre-4.1 call has to be rewritten:
2249 % \advisor[UFRJ]{Prof.}{Nome}{Sobrenome}{D.Sc.} becomes
2250 % \advisor{Nome}{Sobrenome}{D.Sc.}{UFRJ}.
2251 \newcommand\advisor[5][\%
2252   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
2253   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
2254   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
2255   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
2256   \global\@namedef{CoppeAdvisorInst:\expandafter\the\@advisor}{#5}
2257   \global\advance\@advisor by 1
2258   \ifnum\@advisor>1

```

```

2259 \renewcommand\local@advisorstring{Orientadores}
2260 \renewcommand\foreign@advisorstring{Advisors}
2261 \fi
2262 }

```

\examiner

```

2263 \newcount\@examiner\@examiner0
2264 % Until v4.1 this stored only "#1 #2" and DISCARDED the third argument: the
2265 % titulacao every document passed was silently thrown away and never printed,
2266 % though 3.1.2.1.3(e) requires it. Each part is now kept separately, in the same
2267 % order \advisor uses: the optional slot is the TREATMENT, and the institution
2268 % is the last mandatory argument, which may be left empty.
2269 % CoppeExaminer: is retained with its old meaning for anything reading it.
2270 \newcommand\examiner[4][\%
2271 \global\@namedef{CoppeExaminerTitle:\expandafter\the\@examiner}{#1}
2272 \global\@namedef{CoppeExaminerName:\expandafter\the\@examiner}{#2}
2273 \global\@namedef{CoppeExaminerDegree:\expandafter\the\@examiner}{#3}
2274 \global\@namedef{CoppeExaminerInst:\expandafter\the\@examiner}{#4}
2275 \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
2276 \global\advance\@examiner by 1
2277 }

```

\author It was redefined to allow the identification of the author's first names and surname.

```

2278 \renewcommand\author[2]{%
2279 \global\def\@authname{#1}
2280 \global\def\@authsurn{#2}
2281 }

```

\date Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (\coppe@docmonth / \coppe@docyear) rather than mutating TeX's \month / \year primitives, which would clobber \today.

```

2282 \renewcommand\date[2]{%
2283 \def\coppe@docmonth{#1}%
2284 \def\coppe@docyear{#2}}

```

\local@monthname

```

2285 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
2286 Janeiro\or Fevereiro\or Mar{c c}o\or Abril\or Maio\or Junho\or
2287 Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}

```

\foreign@monthname

```

2288 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
2289 January\or February\or March\or April\or May\or June\or
2290 July\or August\or September\or October\or November\or December\fi}

```

\keyword

```

2291 \newcounter{keywords}
2292 \newcommand\keyword[1]{%
2293 \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
2294 \global\addtocounter{keywords}{1}
2295 }

```

```

2296 % Keywords for the foreign abstract. Annex F of the manual shows the English
2297 % abstract closing with English keywords, and \keyword alone cannot serve both
2298 % pages. When no \foreignkeyword is given the foreign abstract falls back to
2299 % the \keyword list, which is what a document written before v4.1 expects.
2300 \newcounter{foreignkeywords}
2301 \newcommand\foreignkeyword[1]{%
2302   \global\@namedef{CoppeForeignKeyword:\expandafter\the\c@foreignkeywords}{#1}
2303   \global\addtocounter{foreignkeywords}{1}
2304 }
2305 % Keywords for the optional third abstract (Portuguese, for the banca). Same
2306 % shape and same fallback as \foreignkeyword: without it the Portuguese page
2307 % would close with the main-language keywords under a Portuguese label, which
2308 % is what a Spanish-, French- or Italian-main work would produce.
2309 \newcounter{braziliankeywords}
2310 \newcommand\braziliankeyword[1]{%
2311   \global\@namedef{CoppeBrazilianKeyword:\expandafter\the\c@braziliankeywords}{#1}
2312   \global\addtocounter{braziliankeywords}{1}
2313 }
2314 % 3.1.2.1.4: the keywords close the resumo, preceded by the term, a colon, and
2315 % separated from one another by semicolons.
2316 % #1 label, #2 csname prefix, #3 how many.
2317 \newcommand\coppe@printkeywords[3]{%
2318   \ifnum#3>0
2319     \par\vspace*{7mm}%
2320     \noindent{\singlespacing\textbf{#1:}}\enspace
2321     \count1=0
2322     \@whilenum\count1<#3\do{%
2323       \ifnum\count1>0 ;\space\fi
2324       \csname #2\the\count1\endcsname
2325       \advance\count1 by 1}%
2326     .\par}%
2327   \fi}

\areaconcentracao Data for the folha adicional (Coleta CAPES) and for the natureza block of the
\linhapesquisa folha de rosto and folha de aprovacao. Every one of them defaults to empty;
\tipoproducao \makefolhaadicional prints an empty rule where a value is missing, so the pro-
\projetovinculado gramme can complete the sheet by hand if need be.

\nomeprojeto 2328 \newcommand\coppe@areaconc{}
\agenciafomento 2329 \newcommand\areaconcentracao[1]{\gdef\coppe@areaconc{#1}}
2330 \newcommand\coppe@dataaprov{}
2331 \newcommand\dataaprovacao[1]{\gdef\coppe@dataaprov{#1}}
2332 \newcommand\coppe@linhapesq{}
2333 \newcommand\linhapesquisa[1]{\gdef\coppe@linhapesq{#1}}
2334 \newcommand\coppe@tipoproducao{}
2335 \newcommand\tipoproducao[1]{\gdef\coppe@tipoproducao{#1}}
2336 \newcommand\coppe@projetovinc{}
2337 \newcommand\projetovinculado[1]{\gdef\coppe@projetovinc{#1}}
2338 \newcommand\coppe@nomeprojeto{}
2339 \newcommand\nomeprojeto[1]{\gdef\coppe@nomeprojeto{#1}}
2340 \newcounter{coppeagencias}
2341 \newcommand\agenciafomento[2]{%
2342   \global\@namedef{CoppeAgencia:\expandafter\the\c@coppeagencias}{#1}%
2343   \global\@namedef{CoppeAgenciaSigla:\expandafter\the\c@coppeagencias}{#2}%
2344   \global\addtocounter{coppeagencias}{1}}

```

`\freeconfig` This command allows easy changing of core class parameters.

```

2345 \newcommand\freeconfig[6]{%
2346 \providecommand\@degree{}
2347 \renewcommand\@degree{#1}
2348 \providecommand\@degreename{}
2349 \renewcommand\@degreename{#2}
2350 \providecommand\local@degname{}
2351 \renewcommand\local@degname{#3}
2352 \providecommand\foreign@degname{}
2353 \renewcommand\foreign@degname{#4}
2354 \providecommand\local@doctype{}
2355 \renewcommand\local@doctype{#5}
2356 \providecommand\foreign@doctype{}
2357 \renewcommand\foreign@doctype{#6}%
2358 }
2359 % \end{macrocode}
2360 % \end{macro}
2361 %
2362 % \begin{macro}{\frontmatter}
2363 % The number of pages for both frontmatter and mainmatter printed
2364 % in the cataloging details page is computed by means of simple
2365 % \LaTeX\ labels.
2366 % \begin{macrocode}
2367 \renewcommand\frontmatter{%
2368 \cleardoublepage
2369 \@mainmatterfalse
2370 % The count already started at the folha de rosto (see \maketitle) and simply
2371 % continues here: 2.7 requires every folha from the folha de rosto onwards to
2372 % be counted sequentially, printing arabic numerals only from the first textual
2373 % folha. Resetting the counter here -- as this class did until v4.1 -- made the
2374 % Introducao come out as page 1, which the norm does not allow.
2375 %
2376 % The numeraisromanos option keeps the legacy behaviour (lowercase roman in the
2377 % pre-textual part, arabic restarting at 1 in the textual part). That departs
2378 % from 2.7, which has the pre-textual folhas counted but NOT numbered, so the
2379 % option warns.
2380 \if@copperoman
2381 \ClassWarningNoLine{coppe}{Option ‘numeraisromanos’ prints roman numerals
2382 on the pre-textual folhas and restarts the arabic count at 1 in the
2383 textual part. Section 2.7 of the SiBI manual (9th rev. ed., 2026) has
2384 those folhas counted but not numbered, and the arabic numbering
2385 continuing the count. Drop the option to comply}%
2386 \pagenumbering{roman}%
2387 \fi
2388 % \thispagestyle vale por UMA pagina. Se a folha de aprovacao transbordar --
2389 % uma banca grande com titulo longo consegue --, a pagina seguinte pegava o
2390 % estilo corrente e saia com folio, na parte pre-textual, onde a 2.7 nao
2391 % admite. O grupo com \pagestyle{empty} cobre quantas paginas forem, e o
2392 % \endgroup devolve o estilo anterior.
2393 \begingroup
2394 \pagestyle{empty}\thispagestyle{empty}
2395 \singlespacing
2396 \makefrontpage
2397 \clearpage

```

```

2398 \endgroup
2399 \if@copperoman
2400   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2401     \renewcommand{\headrulewidth}{0pt}}%
2402   \pagestyle{coppe}%
2403 \else
2404   \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
2405   \pagestyle{empty}%
2406 \fi
2407 }

\mainmatter
2408 \renewcommand\mainmatter{%
2409   \coppe@mainBegin
2410   \cleardoublepage
2411   \@mainmattertrue
2412   \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\coppe@folio}%
2413     \renewcommand{\headrulewidth}{0pt}}%
2414   \pagestyle{coppe}%
2415   % No \pagenumbering{arabic} in the default path: it would reset the counter to
2416   % 1 and make the Introducao folha 1, contrary to 2.7. The counter has been
2417   % arabic and running since the folha de rosto; all that changes here is that
2418   % the folio starts being PRINTED. Only the legacy numeraisromanos path, which
2419   % switched to roman in the front matter, has to switch back -- and there the
2420   % restart at 1 is intrinsic to that (non-conforming) scheme.
2421   \if@copperoman\pagenumbering{arabic}\fi

\backmatter
2422 \renewcommand\backmatter{%
2423   \if@openright
2424     \cleardoublepage
2425   \else
2426     \clearpage
2427   \fi}
2428 %

```

\coppetexfinalpage Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with \vfill; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: \coppefinalengine (engine name; auto-detected via iftex), \coppefinalfont (text body font; defaults to Computer Modern, the L^AT_EX default), and \coppefinalmanual (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before \coppetexfinalpage with \renewcommand if the document uses a non-default font or follows an earlier edition of the norm.

```

2429 \providecommand{\coppefinalengine}{%
2430   \ifPDFTeX\hologo{pdfLaTeX}\fi
2431   \ifLuaTeX\hologo{LuaLaTeX}\fi
2432   \ifXeTeX\hologo{XeLaTeX}\fi}
2433 % A classe carrega lmodern (as bitmap da Computer Modern saiam em Type 3 e

```



```

2434 % inviabilizavam o PDF/A), entao o colofao diz Latin Modern -- quem trocar a
2435 % fonte redefine este comando.
2436 \providecommand{\coppefinalfont}{%
2437   \if@coppesansserif
2438     Latin Modern Sans (\texttt{lmodern})%
2439   \else
2440     Latin Modern (\texttt{lmodern})%
2441   \fi}
2442 \providecommand{\coppefinalmanual}{%
2443   \emph{Norma para a Elaborac~ao Gr'afica de Teses e Disserta~oes
2444     da COPPE/UFRJ}}
2445 \newcommand{\coppetexfinalpage}{%
2446   \clearpage
2447   % Force the colophon onto a verso (even) page so it faces the rear cover.
2448   % If the current page is odd (recto), fill it with an empty verso first.
2449   \if@twoside
2450     \ifodd\c@page
2451       \hbox{}\thispagestyle{empty}\newpage
2452     \fi
2453   \fi
2454   \thispagestyle{empty}%
2455   \null\vfill
2456   \begin{otherlanguage}{brazilian}%
2457     \begin{center}
2458       {\footnotesize\setstretch{1.0}%
2459         \textsc{Colof~ao}\par
2460         \vspace{0.7em}
2461         \begin{minipage}{0.75\textwidth}\centering
2462           Este documento foi composto em \LaTeX{} com \coppefinalengine{}
2463           em \today, na fonte \coppefinalfont. A bibliografia foi processada
2464           por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
2465           vers~ao~\coppe@version, dispon~i vel no reposit~orio GitHub
2466           \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
2467           conformidade com a \coppefinalmanual.
2468         \end{minipage}\par
2469       \end{center}
2470     \end{otherlanguage}%
2471     \vspace*{2cm}%
2472   \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

2473 \newcommand\makecover{%
2474   \begin{titlepage}%
2475   % A capa e um modelo institucional, e sai em espaco simples qualquer que seja o
2476   % espacamento do corpo. Sob a opcao doublespacing ela transbordava para uma
2477   % segunda e ate uma terceira folha -- que ainda por cima levavam folio, porque
2478   % o \thispagestyle do titlepage vale por uma pagina so. 0 \pagestyle{empty} e
2479   % local ao ambiente e cobre quantas paginas forem.

```

```

2480 \pagestyle{empty}%
2481 \singlespacing
2482 \centering
2483 \makebox[\linewidth]{%
2484   \includegraphics[height=2cm]{ufrj-logo}\hfill
2485   \includegraphics[height=2cm]{coppe-logo}}\par
2486 \vspace{1.5cm}
2487 {\singlespacing
2488 \MakeUppercase{\local@universityname}\par
2489 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de \
2490   P{\ ' o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia}}\par
2491 \MakeUppercase{Programa de \local@deptname}\par}
2492 \vspace{\stretch{2}}
2493 \textbf{\MakeUppercase{\textbf{\@authname\ \@authsur}}}\par
2494 \vspace{\stretch{2}}
2495 \nohyphens{\coppe@titleblock}\par
2496 \coppe@volumeline
2497 \vspace{\stretch{3}}
2498 \MakeUppercase{\local@cityname}\par
2499 \coppe@docyear
2500 \end{titlepage}%
2501 }

```

\maketitle

```

2502 \renewcommand\maketitle{%
2503 % The capa is neither counted nor numbered (2.7), so it lives in its own
2504 % throwaway numbering scheme; \pagenumbering{arabic} right after it restarts
2505 % the counter, making the FOLHA DE ROSTO page 1 -- which is where 2.7 says the
2506 % count begins. Nothing is printed here: \frontmatter keeps \pagestyle{empty}
2507 % until the textual part starts.
2508 \pagenumbering{alph}
2509 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
2510 \makecover
2511 \pagenumbering{arabic}
2512 \begin{titlepage}
2513 % Mesma razao da capa: modelo institucional, espaco simples, e sem folio ainda
2514 % que transborde.
2515 \pagestyle{empty}%
2516 \singlespacing
2517 \begin{flushleft}
2518 \vspace*{1.5mm}
2519 \setlength\baselineskip{0pt}
2520 \setlength\parskip{1mm}
2521 \makebox[\linewidth]{%
2522   \includegraphics[height=2cm]{ufrj-logo}\hfill
2523   \includegraphics[height=2cm]{coppe-logo}}
2524 \end{flushleft}
2525 \vspace{1.05cm}
2526 \begin{center}
2527 \nohyphens{\@authname\ \@authsur}\par
2528 \vspace*{3cm}
2529 \nohyphens{\coppe@titleblock}\par
2530 \coppe@volumeline
2531 \end{center}

```

```

2532 \vspace*{2.1cm}
2533 \begin{flushright}
2534 % 2.4: on the folha de rosto the natureza/objetivo/instituicao block is set
2535 % SINGLE spaced, and aligned from the middle of the mancha grafica to the right
2536 % margin. 0.5\textwidth tracks the mancha, so the block follows automatically if
2537 % the margins ever change -- the old fixed 8.45cm did not.
2538 \begin{minipage}{0.5\textwidth}
2539 \singlespacing\frontcover@maintext
2540 \end{minipage}\par
2541 \end{flushright}
2542 % Only the natureza block is indented to the right (2.4). The linha de pesquisa
2543 % and the orientadores return to the LEFT MARGIN, as the CAPES model of the
2544 % folha de rosto and Annex B of the manual show. Until v4.1 they sat inside the
2545 % right-hand block, which is what a recent review flagged.
2546 \vspace*{12mm}
2547 {\singlespacing
2548 \ifthenelse{\equal{\coppe@linhapesq}{}}{\%}
2549 \noindent\coppemainstring{researchline}: \coppe@linhapesq\par
2550 \vspace*{5mm}}%
2551 \nohyphens{%
2552 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2553 \local@advisorstring: &
2554 \count1=0
2555 \toks@={}
2556 \@whilenum \count1<\@advisor \do{%
2557 \ifcase\count1 % same as \ifnum0=\count1
2558 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2559 \expandafter\endcsname\expandafter\space%
2560 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2561 \else
2562 \toks@=\expandafter\expandafter\expandafter{%
2563 \expandafter\the\expandafter\toks@%
2564 \expandafter&\expandafter\space%
2565 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2566 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2567 }%
2568 \fi
2569 \advance\count1 by 1}
2570 \the\toks@
2571 \end{tabularx}}%
2572 \ifnum\@coadvisor>0
2573 \par\vspace*{2mm}%
2574 \nohyphens{%
2575 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2576 \local@coadvisorstring: &
2577 \count1=0
2578 \toks@={}
2579 \@whilenum \count1<\@coadvisor \do{%
2580 \ifcase\count1
2581 \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2582 \expandafter\endcsname\expandafter\space%
2583 \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2584 \else
2585 \toks@=\expandafter\expandafter\expandafter{%

```

```

2586         \expandafter\the\expandafter\toks@%
2587         \expandafter&\expandafter\space%
2588         \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2589         \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2590     }%
2591     \fi
2592     \advance\count1 by 1}
2593     \the\toks@
2594 \end{tabularx}}%
2595 \fi
2596 \par}
2597 \vspace*{\fill}
2598 % 3.1.2.1.1(g): the folha de rosto carries the ANO DE DEPOSITO alone -- the year
2599 % the revised final version is handed to the Programa. The month was never asked
2600 % for here (the capa, Annex A, also shows only the year), and printing it
2601 % invited confusion with the defence date. \date still takes month and year:
2602 % the month is what the abstract pages and the colophon use.
2603 \begin{center}
2604 \local@cityname\par
2605 \coppe@docyear
2606 \end{center}
2607 \end{titlepage}
2608 % Folha adicional (3.1.2.1.2): a sheet of its own right after the folha de
2609 % rosto, ahead of the approval sheet produced by \frontmatter. Until the 2026
2610 % edition this content lived on the VERSO of the folha de rosto (NBR 14724,
2611 % R57); the verso is gone with the printed copy.
2612 % LaTeX's titlepage environment ends with \setcounter{page}{1} whenever the
2613 % document is one-sided (see \endtitlepage in classes.dtx). That silently ate
2614 % the folha de rosto from the count, putting the folha de aprovacao at 1 instead
2615 % of 2. The count is therefore restated here: the folha de rosto WAS folha 1, so
2616 % whatever follows it is folha 2.
2617 \setcounter{page}{2}%
2618 % The page carrying the ficha catalografica is explicitly excluded from the
2619 % count -- 2.7 of the 2026 manual ("a folha adicional que contem a ficha
2620 % catalografica nao pode ser contada ou numerada"), and the 2025 edition said
2621 % the same of the verso of the folha de rosto. Hence the counter is stepped
2622 % back after it.
2623 % An exame de qualificacao is not deposited with the library: no ficha
2624 % catalografica, and no Coleta CAPES sheet. Everything else gets both.
2625 \if@coppeexame\else
2626     \makefolhaadicional\clearpage\addtocounter{page}{-1}%
2627 \fi
2628 \global\let\maketitle\relax%
2629 \global\let\and\relax}

2630 % 3.1.2.1.3 lists the elements of the folha de aprovacao in order: (a) the
2631 % AUTHOR, (b) the title in full, (c) the natureza/objetivo/instituicao/area de
2632 % concentracao, (d) the date of approval, (e) the examining board. Until v4.1
2633 % this page opened with the title in caps and the author under it, and set the
2634 % natureza as a wall of uppercase (\frontpage@maintext). Annex D shows neither.
2635 % One board member. \coppe@membroid is the identification line required by
2636 % 3.1.2.1.3(e): name, titulacao and institution. Since v4.1 the TREATMENT
2637 % ("Prof.", "Prof.a") is the optional slot and comes empty by default -- no norm
2638 % asks for it -- while titulacao and institution are what actually identify the

```

2639 % member and are shown whenever they are given. Each of the three is dropped
2640 % silently when empty, so nothing breaks if the author leaves one blank.

2641 \newcommand\coppe@membroid[4]{%
2642 \ifthenelse{\equal{#1}{}}{ }\{#1\ }#2%
2643 \ifthenelse{\equal{#3}{}}{ }\{, #3}%
2644 \ifthenelse{\equal{#4}{}}{ }\{, #4}%
2645 % Until v4.1 the class could draw a rule above each name, for a hand signature,
2646 % and \coppe@membrogap held the gap above that rule, shrinking as the board grew
2647 % so seven members still fitted on one sheet. The deposit has been digital only
2648 % since Resolucao CEPG n. 246/2023, so there is nothing left to sign by hand and
2649 % the rules are gone. The length stays declared because the seven-member sizing
2650 % below still uses it as the gap between entries.

2651 \newlength\coppe@membrogap
2652 \setlength\coppe@membrogap{9mm minus 2mm}
2653 \newcount\coppe@bancasize
2654 \newcommand\coppe@membro[4]{%
2655 \noindent\hspace*{8mm}%
2656 \begin{minipage}{\dimexpr\textwidth-8mm\relax}
2657 \coppe@membroid{#1}{#2}{#3}{#4}
2658 \end{minipage}\par\vspace{2mm}}
2659 \newcommand\makefrontpage{%
2660 \begin{center}
2661 \nohyphens{\@authname\ \@authsurn}\par
2662 \vspace*{7mm}
2663 \sloppy\nohyphens{\coppe@selecttitle
2664 \@ifundefined{coppe@subtitle@\coppe@mainlang}{:}
2665 \coppe@selectsubtitle}}\par
2666 \end{center}\par
2667 \vspace*{8mm}

2668 % Same block, same indentation and same single spacing as the folha de rosto:
2669 % 2.4 names the two pages together.

2670 \begin{flushright}
2671 \begin{minipage}{0.5\textwidth}
2672 \singlespacing\frontcover@maintext
2673 \end{minipage}
2674 \end{flushright}
2675 \vspace*{12mm}

2676 % (d) date of approval. Annex D prints "Aprovada em:" followed by a blank to be
2677 % filled in; \dataaprovacao supplies it when the date is already known.

2678 \noindent\local@approvedonname:\enspace
2679 \ifthenelse{\equal{\coppe@dataaprov}{}}{\hrulefill}{\coppe@dataaprov}\par
2680 \vspace*{12mm}

2681 % (e) the examining board, with the ORIENTADOR FIRST, as president of the board
2682 % (3.1.2.1.3(e)). Until v4.1 the advisors were printed in a block of their own
2683 % above a second list headed "Aprovada por:", which split the board in two and
2684 % did not mark the advisor as its president. They are now one list.

2685 %
2686 % The list is assembled into \coppe@bancalist with \g@addto@macro instead of
2687 % being built inside a tabularx: each entry now carries four fields (treatment,
2688 % name, titulacao, institution) and the old \toks@/\expandafter accumulation
2689 % could not carry them without becoming unreadable. \protected@edef keeps
2690 % accents and other protected commands in the names intact.
2691 % Size of the board decides the spacing above. The thresholds are measured, not
2692 % guessed: with the 9mm gap a seven-member board pushed three members onto a

2693 % second page, 7mm brought six back onto one sheet, 5mm did the same for seven,
 2694 % and eight needed 3.5mm -- eight came up in the adversarial suite, in a
 2695 % Spanish thesis whose longer natureza block leaves less room than the
 2696 % Portuguese one. Boards of five or fewer keep the original 9mm, so every
 2697 % document written before this change renders exactly as it did.
 2698 \coppe@bancasize=\@advisor
 2699 \advance\coppe@bancasize by \@coadvisor
 2700 \advance\coppe@bancasize by \@examiner
 2701 \ifnum\coppe@bancasize>7
 2702 \setlength\coppe@membrogap{3.5mm minus 1.5mm}%
 2703 \else\ifnum\coppe@bancasize>6
 2704 \setlength\coppe@membrogap{5mm minus 2mm}%
 2705 \else\ifnum\coppe@bancasize>5
 2706 \setlength\coppe@membrogap{7mm minus 2mm}%
 2707 \fi\fi
 2708 \gdef\coppe@bancalist{}%
 2709 \count1=0
 2710 \@whilenum\count1<\@advisor\do{%
 2711 \protected@edef\coppe@tmp{\noexpand\coppe@membro
 2712 {\csname CoppeAdvisorTitle:\the\count1\endcsname}%
 2713 {\csname CoppeAdvisorName:\the\count1\endcsname\space
 2714 \csname CoppeAdvisorSurname:\the\count1\endcsname}%
 2715 {\csname CoppeAdvisorDegree:\the\count1\endcsname}%
 2716 {\csname CoppeAdvisorInst:\the\count1\endcsname}}%
 2717 \expandafter\g@addto@macro\expandafter\coppe@bancalist
 2718 \expandafter{\coppe@tmp}%
 2719 \advance\count1 by 1}%
 2720 % Coorientadores come right after the orientadores and before the rest of the
 2721 % board, keeping the order 3.1.2.1.3(e) prescribes.
 2722 \count1=0
 2723 \@whilenum\count1<\@coadvisor\do{%
 2724 \protected@edef\coppe@tmp{\noexpand\coppe@membro
 2725 {\csname CoppeCoadvisorTitle:\the\count1\endcsname}%
 2726 {\csname CoppeCoadvisorName:\the\count1\endcsname\space
 2727 \csname CoppeCoadvisorSurname:\the\count1\endcsname}%
 2728 {\csname CoppeCoadvisorDegree:\the\count1\endcsname}%
 2729 {\csname CoppeCoadvisorInst:\the\count1\endcsname}}%
 2730 \expandafter\g@addto@macro\expandafter\coppe@bancalist
 2731 \expandafter{\coppe@tmp}%
 2732 \advance\count1 by 1}%
 2733 \count1=0
 2734 \@whilenum\count1<\@examiner\do{%
 2735 \protected@edef\coppe@tmp{\noexpand\coppe@membro
 2736 {\csname CoppeExaminerTitle:\the\count1\endcsname}%
 2737 {\csname CoppeExaminerName:\the\count1\endcsname}%
 2738 {\csname CoppeExaminerDegree:\the\count1\endcsname}%
 2739 {\csname CoppeExaminerInst:\the\count1\endcsname}}%
 2740 \expandafter\g@addto@macro\expandafter\coppe@bancalist
 2741 \expandafter{\coppe@tmp}%
 2742 \advance\count1 by 1}%
 2743 \noindent\local@approvedname:\par
 2744 \nohyphens{\singlespacing\coppe@bancalist}\par
 2745 % No city/date block at the foot: 3.1.2.1.3 ends the folha de aprovacao at the
 2746 % examining board, and Annex D shows nothing below the signature lines. The date

```

2747 % that belongs on this page is the date of APPROVAL, printed above.
2748 % \frontpage@bottomtext stays defined for anyone calling it directly.
2749 \vspace*{\fill}}

2750 \newcommand\coppe@hypersetup{%
2751 \begin{group}
2752 % changes to \toks@ and \count@ are kept local;
2753 % it's not necessary for them, but it is usually the case
2754 % for \count1, because the first ten counters are written
2755 % to the DVI file, thus you got lucky because of PDF output
2756 \toks@={}% in this special case not necessary
2757 \count@=0 %
2758 \@whilenum\count@<\value{keywords}\do{%
2759 % * a keyword separator is not necessary,
2760 % if there is just one keyword
2761 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
2762 % at least once, to get rid of the loop depended \count@
2763 \ifcase\count@ % same as \ifnum0=\count@
2764 \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
2765 \else
2766 \toks@=\expandafter\expandafter\expandafter{%
2767 \expandafter\the\expandafter\toks@
2768 \expandafter;\expandafter\space
2769 \csname CoppeKeyword:\the\count@\endcsname
2770 }%
2771 \fi
2772 \advance\count@ by 1 %
2773 }%
2774 \edef\x{\endgroup
2775 \noexpand\hypersetup{%
2776 pdfkeywords={\the\toks@}%
2777 }%
2778 }%
2779 \x
2780 \hypersetup{%
2781 pdfauthor={\@authname\ \@authsur},
2782 pdftitle={\local@title},
2783 pdfsubject={\local@doctype\ de \@degree\ em \local@deptname\ da COPPE/UFRJ},
2784 pdfcreator={LaTeX with CoppeTeX toolkit},
2785 breaklinks={true},
2786 raiselinks={true},
2787 pageanchor={true},
2788 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\isundefined` (cf. the T_EX FAQ).

```

2789 \newcommand\makecatalog{%
2790 \vspace*{\fill}
2791 \begin{center}
2792 \setlength{\fboxsep}{5mm}
2793 \framebox[120mm][c]{\makebox[5mm][c]{}}%
2794 \begin{minipage}[c]{105mm}

```

```

2795 \setlength{\parindent}{5mm}
2796 \noindent\sloppy\nohyphens\@authsurn,
2797 \nohyphens\@authname\par
2798 \nohyphens{%
2799 \coppe@selecttitle/\@authname\ \@authsurn. -- \local@cityname:
2800 UFRJ/COPPE, \coppe@docyear.}\par
2801 \pageref{front:pageno},
2802 \pageref{LastPage}
2803 p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}} $29,7$cm.\par
2804 % There is an issue here. When the last entry must be split between lines,
2805 % the spacing between it and the next paragraph becomes smaller.
2806 % Should we manually introduce a fixed space? But how could we know that
2807 % a name was split? Is this happening yet?
2808 \nohyphens{%
2809 \begin{tabularx}{100mm}[b]{@{}l@{ }}>\raggedright\arraybackslashX@{}
2810 \local@advisorstring: &
2811 \count1=0
2812 \toks@={}
2813 \@whilenum \count1<\@advisor \do{%
2814 \ifcase\count1 % same as \ifnum0=\count1
2815 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2816 \expandafter\endcsname\expandafter\space%
2817 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2818 \else
2819 \toks@=\expandafter\expandafter\expandafter{%
2820 \expandafter\the\expandafter\toks@
2821 \expandafter&\expandafter\space
2822 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2823 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2824 }%
2825 \fi
2826 \advance\count1 by 1}
2827 \the\toks@
2828 \end{tabularx}}\par
2829 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
2830 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
2831 Refer{\^ e}ncias Bibliogr{\` a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
2832 \count1=0
2833 \count2=1
2834 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
2835 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
2836 \advance\count1 by 1
2837 \advance\count2 by 1}
2838 I. \csname CoppeAdvisorSurname:0\endcsname,%
2839 \ \csname CoppeAdvisorName:0\endcsname%
2840 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
2841 II. \local@universityname, COPPE, Programa de \local@deptname.
2842 III. T{\` i}tulo.}
2843 \end{minipage}}
2844 \end{center}
2845 \vspace*{\fill}}

```

\fichacatalografica The folha adicional required from August 2026 (SiBI manual, 9th rev. ed.,
\makefolhaadicional 3.1.2.1.2 and Annex H): the CAPES collection fields plus the ficha cata-

lografica, sitting immediately after the folha de rosto and counted by nobody. \fichacatalografica takes the PDF produced by the SiBI generator at <<http://fichacatalografica.sibi.ufrj.br/>>, optionally with a page number.

The sheet is institutional, so it is always in Portuguese whatever the main language of the work – and deliberately without \selectlanguage, since the brazilian babel language need not be loaded in, say, a Spanish-main document.

```

2846 \newcommand\coppe@fichafile{}
2847 \newcommand\coppe@fichapage{1}
2848 \newcommand\fichacatalografica[2][1]{%
2849   \gdef\coppe@fichapage{#1}\gdef\coppe@fichafile{#2}}
2850 \newcommand\coppe@marcatipo[1]{%
2851   \ifthenelse{\equal{\coppe@tipoproducao}{#1}}{$\times$}{\phantom{$\times$}}}
2852 \newcommand\coppe@marcavinc[1]{%
2853   \ifthenelse{\equal{\coppe@projetovinc}{#1}}{$\times$}{\phantom{$\times$}}}
2854 % Annex H prints the five fields inside a CLOSED FRAME, one cell each, with a
2855 % horizontal rule between cells and NOTHING to write on: the space to fill in is
2856 % the blank space of the cell. Until v4.1 the class laid the fields out as loose
2857 % paragraphs and drew a dotted \hrulefill to write on, which is not what the
2858 % model shows. \coppe@campo now prints the value when there is one and leaves
2859 % empty vertical space when there is not.
2860 \newlength\coppe@capesgap
2861 \setlength\coppe@capesgap{11mm}
2862 \newcommand\coppe@campo[1]{%
2863   \ifthenelse{\equal{#1}}{\vspace*{\coppe@capesgap}}{\par#1}}
2864 % One cell of the frame: the label, then whatever the field holds.
2865 \newcommand\coppe@capescelula[2]{%
2866   \begin{minipage}[t]{\dimexpr\linewidth-2\tabcolsep\relax}
2867     \vspace{1mm}#1\par#2\vspace{1.5mm}
2868   \end{minipage}\\hline}
2869 \newcommand\coppe@checativo{%
2870   \ifthenelse{\equal{\coppe@tipoproducao}{}}{}{%
2871     \ifthenelse{\equal{\coppe@tipoproducao}{bibliografica}\OR
2872       \equal{\coppe@tipoproducao}{artistica}\OR
2873       \equal{\coppe@tipoproducao}{tecnologica}\OR
2874       \equal{\coppe@tipoproducao}{tecnica}}{}{%
2875       \ClassWarning{coppe}{Unknown \string\coppe@tipoproducao{} value
2876         '\coppe@tipoproducao'. Annex H of the SiBI manual allows only
2877         bibliografica, artistica, tecnologica or tecnica; no box will be
2878         ticked}}}}
2879 \newcommand\makefolhaadicional{%
2880   \clearpage
2881   \thispagestyle{empty}%
2882   \coppe@checativo
2883   {\singlespacing
2884     \begin{center}
2885       {\large\bfseries Informa\c c\~oes Coleta CAPES}
2886     \end{center}
2887     \vspace{7mm}
2888     \noindent\begin{tabular}{|@{\hspace{\tabcolsep}}p{\dimexpr\textwidth-2\tabcolsep-2\arrayru
2889       \hline
2890       \coppe@capescelula{Tipo de produ\c c\~ao intelectual:}{%
2891         \vspace{1mm}
2892         \hspace*{10mm}(\coppe@marcatipo{bibliografica})\enspace

```

```

2893 Bibliogr\'afica\par
2894 \hspace*{10mm}(\coppe@marcatipo{artística})\enspace Art\'{i}stica\par
2895 \hspace*{10mm}(\coppe@marcatipo{tecnológica})\enspace Tecnol\'ogica\par
2896 \hspace*{10mm}(\coppe@marcatipo{técnica})\enspace T\'ecnica}
2897 \coppe@capescelula{Projeto de pesquisa vinculado \'a produo
2898 intelectual?}{%
2899 \vspace{1mm}
2900 \hspace*{10mm}(\coppe@marcavinc{sim})\enspace Sim\par
2901 \hspace*{10mm}(\coppe@marcavinc{nao})\enspace No}
2902 \coppe@capescelula{Nome do projeto de pesquisa:}{%
2903 \coppe@campo{\coppe@nomeprojeto}}
2904 \coppe@capescelula{\'Area de concentrao da produo
2905 intelectual:}{\coppe@campo{\coppe@areaconc}}
2906 \coppe@capescelula{Agncia(s) de fomento financiadora(s) da produo
2907 intelectual:}{%
2908 \ifthenelse{\value{coppeagencias}=0}{\vspace*{\coppe@capesgap}}{%
2909 \count1=0
2910 \@whilenum\count1<\value{coppeagencias}\do{%
2911 \hspace*{10mm}\csname CoppeAgencia:\the\count1\endcsname
2912 \ (\csname CoppeAgenciaSigla:\the\count1\endcsname)\par
2913 \advance\count1 by 1}}
2914 \end{tabular}\par
2915 \vspace{9mm}
2916 \begin{center}
2917 {\large\bfseries Ficha catalogr\'afica}\par
2918 \vspace{4mm}
2919 \ifthenelse{\equal{\coppe@fichafile}{}}{%
2920 \if@copperascunho
2921 \makecatalog
2922 \else
2923 \framebox[125mm][c]{\parbox[c][62mm][c]{115mm}{\centering\footnotesize
2924 Espao reservado \'a ficha catalogr\'afica.\\[1ex]
2925 Gere-a em \texttt{fichacatalografica.sibi.ufrj.br} e inclua o
2926 arquivo com \texttt{\string\fichacatalografica\{arquivo.pdf\}},
2927 ou solicite-a \'a biblioteca do seu Programa.}}%
2928 \fi}{%
2929 \includegraphics[page=\coppe@fichapage,width=125mm]{\coppe@fichafile}}%
2930 \end{center}
2931 \par}}

```

\dedication

```

2932 \newcommand\dedication[1]{
2933 \gdef\@dedic{#1}
2934 \cleardoublepage
2935 \vspace*{\fill}
2936 \begin{flushright}
2937 \begin{minipage}{60mm}
2938 \raggedleft \it \normalsize \@dedic
2939 \end{minipage}
2940 \end{flushright}}

```

\coppe@coadvisorrows The coadvisor block of the abstract pages, in the shape the folha de rosto already uses. It prints nothing at all unless the `coorientador` class option is given AND the document actually declared a \coadvisor, so the default output is byte for

byte what it was. The argument is the label macro, which differs per page: the main-language pages pass `\local@coadvisorstring` and the foreign one passes `\foreign@coadvisorstring`.

```

2941 \newcommand\coppe@coadvisorrows[1]{%
2942   \if@coppecoadvresumo
2943     \ifnum\@coadvisor>0
2944       \par\vspace*{2mm}%
2945       \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2946         #1: &
2947         \count1=0
2948         \toks@={}
2949         \@whilenum \count1<\@coadvisor \do{%
2950           \ifcase\count1
2951             \toks@=\expandafter{\csname CoppeCoadvisorName:\the\count1%
2952               \expandafter\endcsname\expandafter\space%
2953               \csname CoppeCoadvisorSurname:\the\count1\endcsname\\}
2954           \else
2955             \toks@=\expandafter\expandafter\expandafter{%
2956               \expandafter\the\expandafter\toks@%
2957               \expandafter&\expandafter\space%
2958               \csname CoppeCoadvisorName:\the\count1\expandafter\endcsname%
2959               \expandafter\space\csname CoppeCoadvisorSurname:\the\count1\endcsname\\
2960             }%
2961           \fi
2962           \advance\count1 by 1}
2963         \the\toks@
2964       \end{tabularx}%
2965     \fi
2966   \fi}

```

`abstract (env.)` This is a specialization of the abstract in the article standard class.

```

2967 \newenvironment{abstract}{%
2968   \cleardoublepage
2969   \thispagestyle{plain}
2970   \abstract@toptext\par
2971   \vspace*{8.6mm}
2972   \begin{center}
2973     \sloppy\nohyphens{\MakeUppercase\local@title}\par
2974     \vspace*{13.2mm}
2975     \@authname\ \@authsurn \par
2976     \vspace*{7mm}
2977     \local@monthname/\coppe@docyear
2978   \end{center}\par
2979   \vspace*{1.5cm}
2980   \noindent%
2981   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
2982     \local@advisorstring: &
2983     \count1=0
2984     \toks@={}
2985     \@whilenum \count1<\@advisor \do{%
2986       \ifcase\count1 % same as \ifnum0=\count1
2987         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2988           \expandafter\endcsname\expandafter\space%
2989           \csname CoppeAdvisorSurname:\the\count1\endcsname\\}

```

```

2990 \else
2991 \toks@=\expandafter\expandafter\expandafter{%
2992 \expandafter\the\expandafter\toks@
2993 \expandafter&\expandafter\space
2994 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2995 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2996 }%
2997 \fi
2998 \advance\count1 by 1}
2999 \the\toks@
3000 \end{tabularx}%
3001 \coppe@coadvisorrows{\local@coadvisorstring}\par
3002 \vspace*{2mm}
3003 \noindent\local@deptstring: \local@deptname\par
3004 \vspace*{7mm}}}%
3005 \coppe@printkeywords{\coppemainstring{keywordsname}}%
3006 {CoppeKeyword:}\value{keywords}}}%
3007 \vspace*{\fill}}

```

foreignabstract (*env.*)

```

3008 \newenvironment{foreignabstract}{%
3009 \cleardoublepage
3010 \thispagestyle{plain}
3011 \begin{otherlanguage}{\coppe@foreignlang}
3012 \foreignabstract@toptext\par
3013 \vspace*{8.6mm}
3014 \begin{center}
3015 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
3016 \vspace*{13.2mm}
3017 \@authname\ \@authsurn \par
3018 \vspace*{7mm}
3019 \foreign@monthname/\coppe@docyear
3020 \end{center}\par
3021 \vspace*{1.5cm}
3022 \noindent%
3023 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
3024 \foreign@advisorstring: &
3025 \count1=0
3026 \toks@={}
3027 \@whilenum \count1<\@advisor \do{%
3028 \ifcase\count1 % same as \ifnum0=\count1
3029 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
3030 \expandafter\endcsname\expandafter\space%
3031 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3032 \else
3033 \toks@=\expandafter\expandafter\expandafter{%
3034 \expandafter\the\expandafter\toks@
3035 \expandafter&\expandafter\space
3036 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3037 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3038 }%
3039 \fi
3040 \advance\count1 by 1}
3041 \the\toks@

```

```

3042 \end{tabularx}%
3043 \coppe@coadvisorrows{\foreign@coadvisorstring}\par
3044 \vspace*{2mm}
3045 \noindent\foreign@deptstring: \foreign@deptname\par
3046 \vspace*{7mm}}{%
3047 \ifnum\value{foreignkeywords}>0
3048   \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3049   {CoppeForeignKeyword:}{\value{foreignkeywords}}%
3050 \else
3051   \coppe@printkeywords{\coppeforeignstring{keywordsname}}%
3052   {CoppeKeyword:}{\value{keywords}}%
3053 \fi
3054 \end{otherlanguage}
3055 \vspace*{\fill}
3056 \global\let\@author\@empty
3057 \global\let\@date\@empty
3058 \global\let\foreign@title\@empty
3059 \global\let\foreign@title\relax
3060 \global\let\local@title\@empty
3061 \global\let\local@title\relax
3062 \global\let\author\relax
3063 \global\let\author\relax
3064 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

3065 \newenvironment{brazilianabstract}{%
3066   \cleardoublepage
3067   \thispagestyle{plain}
3068   \begin{otherlanguage}{brazilian}
3069   \abstract@toptext\par
3070   \vspace*{8.6mm}
3071   \begin{center}
3072     \sloppy\nohyphens{\MakeUppercase\local@title}\par
3073     \vspace*{13.2mm}
3074     \@authname\ \@authsurv \par
3075     \vspace*{7mm}
3076     \local@monthname/\coppe@docyear
3077   \end{center}\par
3078   \vspace*{1.5cm}
3079   \noindent%
3080   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}
3081     \local@advisorstring: &
3082     \count1=0
3083     \toks@={}
3084     \@whilenum \count1<\@advisor \do{%
3085       \ifcase\count1
3086         \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%

```

```

3087     \expandafter\endcsname\expandafter\space%
3088     \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
3089   \else
3090     \toks@=\expandafter\expandafter\expandafter{%
3091       \expandafter\the\expandafter\toks@
3092       \expandafter&\expandafter\space
3093       \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
3094       \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
3095     }%
3096   \fi
3097   \advance\count1 by 1}
3098   \the\toks@
3099 \end{tabularx}%
3100 \coppe@coadvisorrows{\local@coadvisorstring}\par
3101 \vspace*{2mm}
3102 \noindent\local@deptstring: \local@deptname\par
3103 \vspace*{7mm}}}%
3104 % \coppestring takes ONE argument and resolves against the current language;
3105 % this runs before \end{otherlanguage}, so \language is already brazilian.
3106 \ifnum\value{braziliankeywords}>0
3107   \coppe@printkeywords{\coppestring{keywordsname}}%
3108   {CoppeBrazilianKeyword:}{\value{braziliankeywords}}%
3109 \else
3110   \coppe@printkeywords{\coppestring{keywordsname}}%
3111   {CoppeKeyword:}{\value{keywords}}%
3112 \fi
3113 \end{otherlanguage}
3114 \vspace*{\fill}}

```

\listoffigures

```

3115 \renewcommand\listoffigures{%
3116   \coppe@hasLof
3117   \if@twocolumn
3118     \@restonecoltrue\onecolumn
3119   \else
3120     \@restonecolfalse
3121   \fi
3122   \chapter*{\listfigurename}%
3123   \coppe@listtoc{\listfigurename}%
3124   \@mkboth{\MakeUppercase\listfigurename}%
3125           {\MakeUppercase\listfigurename}%
3126   \@starttoc{lof}%
3127   \if@restonecol\twocolumn\fi
3128 }

```

\listoftables

```

3129 \renewcommand\listoftables{%
3130   \if@twocolumn
3131     \@restonecoltrue\onecolumn
3132   \else
3133     \@restonecolfalse
3134   \fi
3135   \chapter*{\listtablename}%
3136   \coppe@listtoc{\listtablename}%

```

```

3137      \@mkboth{%
3138          \MakeUppercase\listtablename}%
3139          {\MakeUppercase\listtablename}%
3140      \@starttoc{lot}%
3141      \if@restonecol\twocolumn\fi
3142      }

```

\printlosymbols

```

3143 % 0 \renewcommand ficava FORA do grupo e vazava: depois de imprimir a lista de
3144 % simbolos, \glossaryname continuava valendo "Lista de Simbolos" pelo resto do
3145 % documento, e o Glossario pos-textual da 3.1.4.2 saia com o titulo errado --
3146 % no sumario tambem. Dentro do grupo, a troca dura o que tem de durar.
3147 \newcommand\printlosymbols{%
3148 \begingroup\@coppeglosspretrue
3149 \renewcommand\glossaryname{\listsymbolname}%
3150 \@input@{\jobname.los}\endgroup}

```

\makelosymbols

```

3151 \def\makelosymbols{%
3152   \newwrite\@losfile
3153   \immediate\openout\@losfile=\jobname.syx
3154   \newcommand\syml[3][\@bsphack\begingroup
3155     \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}}%
3156     \@sanitize%
3157     \@wrtlos{\@tempsymb1}{##2}{##3}\typeout%
3158     {Writing index of symbols file \jobname.syx}}%
3159   \let\makelosymbols\empty%
3160 }%
3161 \@onlypreamble\makelosymbols

3162 \AtBeginDocument{%
3163 \@ifpackageloaded{hyperref}{%
3164   \newcommand\@wrtlos[3]{%
3165     \protected@write\@losfile{%
3166       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3167     }%
3168     \@esphack}}%
3169   \newcommand\@wrtlos[3]{%
3170     \protected@write\@losfile{%
3171       {\string\indexentry{#1[#2] #3}{\thepage}}%
3172     }%
3173     \@esphack}}}%

```

\printloabbreviations

```

3174 \newcommand\printloabbreviations{%
3175 \begingroup\@coppeglosspretrue
3176 \renewcommand\glossaryname{\listabbreviationname}%
3177 \@input@{\jobname.lab}\endgroup}

```

\makeloabbreviations

```

3178 \def\makeloabbreviations{%
3179   \newwrite\@labfile
3180   \immediate\openout\@labfile=\jobname.abx
3181   \newcommand\abbrev[3][\@bsphack\begingroup

```

```

3182             \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}
3183             \@sanitize
3184             \@wrlab{\@tempsymb1}{##2}{##3}\typeout
3185 {Writing index of abbreviations file \jobname.abx}%
3186 \let\makeloabbreviations\@empty
3187 }
3188 \@onlypreamble\makeloabbreviations

3189 \AtBeginDocument{%
3190 \ifpackageloaded{hyperref}{%
3191   \newcommand\@wrlab[3]{%
3192     \protected@write\@labfile{%
3193       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
3194     \endgroup%
3195     \@esphack}}{%
3196   \newcommand\@wrlab[3]{%
3197     \protected@write\@labfile{%
3198       {\string\indexentry{#1[#2] #3}{\thepage}}%
3199     \endgroup%
3200     \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

3201 \newcommand\newsigla[3]{%
3202   \@namedef{coppe@sigla@s@#1}{#2}%
3203   \@namedef{coppe@sigla@l@#1}{#3}}
3204 \newcommand\coppe@sigla@reg[1]{%
3205   \ifundefined{abbrev}{%
3206     {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}}
3207 \newcommand\coppe@sigla@full[1]{%
3208   \@nameuse{coppe@sigla@l@#1}\space(\@nameuse{coppe@sigla@s@#1})}
3209 \newcommand\coppe@sigla@undef[1]{%
3210   \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
3211   \textbf{???#1???}}
3212 \newcommand\coppe@sigla@first[1]{%
3213   \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}
3214 \newcommand\coppe@sigla@norm[1]{%
3215   \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3216   {\ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
3217     {\@nameuse{coppe@sigla@s@#1}}}}
3218 \newcommand\coppe@sigla@star[1]{%
3219   \ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
3220   {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}}
3221 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

3222 %%% \AtBeginDocument{%
3223 %%% \ifpackageloaded{hyperref}{%
3224 %%%   \def\@wrlab#1#2{%
3225 %%%     \protected@write\@labfile{%
3226 %%%       {\string\indexentry{[#1] #2|hyperpage}{\thepage}}%
3227 %%%     \endgroup

```



```

3228 %%%% \esphack}}{%
3229 %%%% \def\@wrlab#1#2{%
3230 %%%% \protected@write\@labfile{%
3231 %%%% {\string\indexentry{[#1] #2}{\arabic{page}}}%
3232 %%%% \endgroup
3233 %%%% \esphack}}}

3234 % Catalog-label writers. Two language-related traps must be avoided here:
3235 % 1. The label NAMES contain ':' and '.', which babel makes active in
3236 % some languages (e.g. Brazilian's "shorthand" set). If the write
3237 % argument is tokenised while one of those is active, pdfTeX aborts
3238 % with "<inserted text> }\endwrite" because the shorthand expansion
3239 % injects unbalanced braces. \detokenize emits its argument as
3240 % catcode-12 characters regardless of the current activation, so the
3241 % label NAMES survive whatever language is current at writer time.
3242 % 2. Spanish babel redefines \roman (and therefore \roman{...}) to
3243 % produce small-caps roman numerals via \es@scroman, whose expansion
3244 % contains an \uppercase group that is unbalanced inside \write. The
3245 % same crash signature appears. The fix is to bypass LaTeX's
3246 % \roman/\arabic counter wrappers entirely and use the raw e-TeX
3247 % primitives \romannumeral and \number directly on the \c@page count
3248 % register --- both are fully expandable, language-agnostic, and
3249 % always produce plain ASCII digits or lowercase roman letters.
3250 % \Roman{page} is left in place because babel-spanish only redefines
3251 % \@Roman under the (non-default) ucroman option.
3252 \AtBeginDocument{%
3253 \ltx@ifpackageloaded{hyperref}{%
3254 \def\coppe@bibEnd{%
3255 \immediate\write\@auxout{%
3256 \string\newlabel{\detokenize{bib:end}}%
3257 {{}\{\number\c@page\}\{\detokenize{page.}\number\c@page\}}}%
3258 \def\coppe@bibBegin{%
3259 \immediate\write\@auxout{%
3260 \string\newlabel{\detokenize{bib:begin}}%
3261 {{}\{\number\c@page\}\{\detokenize{page.}\number\c@page\}}}%
3262 \def\coppe@mainBegin{%
3263 \immediate\write\@auxout{%
3264 \string\newlabel{\detokenize{front:pageno}}%
3265 {{}\{\Roman{page}\}\{\detokenize{page.}\romannumeral\c@page\}}}%
3266 \def\coppe@hasLof{%
3267 \immediate\write\@auxout{%
3268 \string\newlabel{\detokenize{cat:lofflag}}%
3269 {{}\{\detokenize{:~il.}\}\{\detokenize{page.}\romannumeral\c@page\}}}%
3270 }{%
3271 \def\coppe@bibEnd{%
3272 \immediate\write\@auxout{%
3273 \string\newlabel{\detokenize{bib:end}}{{}\{\number\c@page\}}}%
3274 \def\coppe@bibBegin{%
3275 \immediate\write\@auxout{%
3276 \string\newlabel{\detokenize{bib:begin}}{{}\{\number\c@page\}}}%
3277 \def\coppe@mainBegin{%
3278 \immediate\write\@auxout{%
3279 \string\newlabel{\detokenize{front:pageno}}{{}\{\Roman{page}\}}}%
3280 \def\coppe@hasLof{%
3281 \immediate\write\@auxout{%

```

```

3282      \string\newlabel{\detokenize{cat:lofflag}}{\detokenize{:~il.;}}{}}}%
3283    }%
3284  }
3285  % The reference list is now produced by biblatex's \printbibliography (see the
3286  % bibliography setup after babel, above). The former thebibliography
3287  % redefinition and its \bibindent are no longer used and have been removed; the
3288  % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
3289  % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

3290 \newlength{\recuolongquote}
3291 \setlength{\recuolongquote}{4cm}
3292
3293 \newenvironment*{longquote}[1][default]{%
3294   \begin{list}{}%
3295     \setlength{\leftmargin}{\recuolongquote}%
3296     \setlength{\rightmargin}{0pt}%
3297     \setlength{\itemindent}{0pt}%
3298     \setlength{\listparindent}{0pt}%
3299     \setlength{\parsep}{0pt}%
3300     \setlength{\topsep}{\baselineskip}%
3301   }%
3302   \item\relax
3303   \footnotesize
3304   \singlespacing
3305   \ifthenelse{\not\equal{#1}{default}}{%
3306     {\itshape\selectlanguage{#1}}%
3307   }%
3308   \ignorespaces
3309 }{%
3310 \end{list}%
3311 \ignorespacesafterend
3312 }

3313 \newenvironment{theglossary}{%
3314   \if@twocolumn%
3315     \@restonecoltrue\onecolumn%
3316   \else%
3317     \@restonecolfalse%
3318   \fi%
3319   \@mkboth{\MakeUppercase\glossaryname}%
3320   {\MakeUppercase\glossaryname}%
3321   \chapter*{\glossaryname}%
3322   \if@coppeglosspre
3323     \coppe@listtoc{\glossaryname}%
3324   \else
3325     \addcontentsline{toc}{chapter}{\glossaryname}%
3326   \fi
3327   \list{}
3328   {\setlength{\listparindent}{0in}%
3329     \setlength{\labelwidth}{1.0in}%
3330     \setlength{\leftmargin}{1.5in}%
3331     \setlength{\labelsep}{0.5in}%
3332     \setlength{\itemindent}{0in}}%

```

```

3333 \sloppy}%
3334 {\if@restonecol\twocolumn\fi%
3335 \endlist}
3336 %
3337 \renewenvironment{theindex}{%
3338 \if@twocolumn
3339 \@restonecolfalse
3340 \else
3341 \@restonecoltrue
3342 \fi
3343 \twocolumn[\@makeschapterhead{\indexname}]}%
3344 \@mkboth{\MakeUppercase\indexname}%
3345 {\MakeUppercase\indexname}%
3346 \thispagestyle{plain}\parindent\z@
3347 \addcontentsline{toc}{chapter}{\indexname}
3348 \parskip\z@ \@plus .3\p@\relax
3349 \columnseprule \z@
3350 \columnsep 35\p@
3351 \let\item\@idxitem}
3352 {\if@restonecol\onecolumn\else\clearpage\fi}
3353 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
3354 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
3355 \newcommand\glossaryname{\coppemainstring{glossary}}
3356 %
3357 \newcommand\local@advisorstring{Orientador}
3358 \newcommand\foreign@advisorstring{Advisor}
3359 % Fed by the string dispatcher instead of a hard-coded Portuguese literal.
3360 % Expansion is lazy (it happens inside \makefrontpage), so \coppe@mainlang is
3361 % already set by the time this is used.
3362 \ifthenelse{\boolean{maledoc}}{%
3363 \newcommand\local@approvedname{\coppemainstring{approvedmale}}}%
3364 }{%
3365 \newcommand\local@approvedname{\coppemainstring{approvedfemale}}}%
3366 }
3367 \ifthenelse{\boolean{maledoc}}{%
3368 \newcommand\local@approvedonname{\coppemainstring{approvedonmale}}}%
3369 }{%
3370 \newcommand\local@approvedonname{\coppemainstring{approvedonfemale}}}%
3371 }
3372 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
3373 \newcommand\local@deptstring{Programa}
3374 \newcommand\foreign@deptstring{Department}
3375 \newcommand\local@cityname{Rio de Janeiro}
3376 \newcommand\local@statename{RJ}
3377 \newcommand\local@countryname{Brasil}
3378 %
3379 % 3.1.2.1.1(e) of the 2026 manual requires the natureza, the objetivo, the
3380 % name of the institution AND the area de concentracao in this block. The area
3381 % was missing; it is appended as its own sentence when \areaconcentracao was
3382 % given, and simply omitted otherwise.
3383 \newcommand\frontcover@maintext{
3384 \sloppy\nohyphens{\local@doctype\ de \@degreename\
3385 \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3386 ao Programa de P{\` o}s-gradua{\c c}{\~ a}o em \local@deptname,

```

```

3387 COPPE, da \local@universityname, como parte dos requisitos
3388 necess{\' a}rios {\' a} obten{\c c}{\~ a}o do t{\' i}tulo de
3389 \local@degname\ em \local@deptname.}%
3390 \ifthenelse{\equal{\coppe@areaconc}{}}{\}%
3391 \par\vspace{1ex}%
3392 \sloppy\nohyphens{\'Area de concentra{\c c}{\~ a}o: \coppe@areaconc.}}%
3393 }
3394 %
3395 \newcommand\frontpage@maintext{
3396 \noindent {\MakeUppercase\local@doctype}
3397 \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
3398 \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
3399 DE P{\' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
3400 UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
3401 NECESS{\' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
3402 {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
3403 {\MakeUppercase\local@deptname.\par}}%
3404 }
3405 %
3406 \newcommand\frontpage@bottomtext{%
3407 \begin{center}
3408 {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
3409 {\MakeUppercase\local@monthname\ DE \coppe@docyear}
3410 \end{center}%
3411 }
3412 %
3413 \newcommand\abstract@toptext{%
3414 \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
3415 \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
3416 \sloppy\nohyphens{\' a} COPPE/UFRJ como parte dos requisitos
3417 necess{\' a}rios para a obten{\c c}{\~ a}o do grau de
3418 \local@degname\ em Ci{\^e}ncias (\@degree)}
3419 }
3420 \newcommand\foreignabstract@toptext{%
3421 \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
3422 COPPE/UFRJ as a partial fulfillment of the requirements for the
3423 degree of \foreign@degname\ of Science (\@degree)}
3424 }
3425 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- **setspace** – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before **hyperref** patches all the cross-reference and TOC plumbing.
- **hyperref** – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, list-of-* commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the **coppe** class redefines (e.g. `\@chapter`, `\listofquadros`).

- `biblatex` – must come after `hyperref` so its citation links and back-references hook into `hyperref` correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

3426 \RequirePackage{setspace}
3427 % Apply the default line spacing now that setspace is available. The flag was
3428 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
3429 \if@coppepdoublespace\doublespacing\else\onehalfspacing\fi
3430 % pdfx MUST precede hyperref -- it loads hyperref itself, with the options
3431 % PDF/A needs, and cannot fix up an already-loaded one. Level a-2b rather than
3432 % a-1b (the "ISO 19005-1" of Annex I): a-1b forbids transparency, which
3433 % tcolorbox and several graphics packages produce, so a-1b would fail on
3434 % documents that are otherwise perfectly fine. a-2b is ISO 19005-2, accepted by
3435 % Pantheon and by the SiBI's own instructions, which also list PDF/A-2b.
3436 \if@coppepdfa
3437   \RequirePackage[a-2b]{pdfx}
3438 % pdfTeX stamps /PTEX.Fullbanner into the document information dictionary.
3439 % It is not one of the keys ISO 19005-2 6.6.2.3.2 maps to an XMP property, and
3440 % no extension schema declares it, so it is a DocInfo entry with no XMP
3441 % counterpart -- a finding a PDF/A checker has every reason to raise, and one
3442 % that costs nothing to remove. -1 sets every suppression bit.
3443 %
3444 % The two engines spell this differently, and not just in the name. pdfTeX has
3445 % \pdfsuppressptexinfo, whose bits only ever govern PTEX.* entries, so -1 (all
3446 % of them) is safe. LuaTeX moved the pdfTeX integer parameters behind
3447 % \pdfvariable in 0.85 AND renamed this one to suppressoptionalinfo, whose bits
3448 % reach further: besides the four PTEX.* entries (1, 2, 4, 8) it can also
3449 % suppress Creator, the dates, Producer, Trapped and -- the dangerous one --
3450 % the trailer /ID, which PDF/A requires. So -1 is exactly what must NOT be
3451 % written there; 15 is the four PTEX.* bits and nothing else. Checked on a real
3452 % LuaTeX run: /ID survives.
3453 %
3454 % Getting the name wrong is not a quiet failure. \pdfvariable with an unknown
3455 % variable name makes LuaTeX typeset the name, and the document stops with
3456 % "Missing \begin{document}" -- which is how the first attempt at this was
3457 % caught.
3458 \ifdefined\pdfsuppressptexinfo
3459   \pdfsuppressptexinfo=-1
3460 \else
3461   \ifdefined\pdfvariable
3462     \pdfvariable suppressoptionalinfo=15
3463   \fi
3464 \fi
3465 % pdfx takes the XMP metadata from \jobname.xmpdata, which it reads when it is
3466 % LOADED -- before the preamble has run, so before \title, \author and
3467 % \keyword exist. The file is therefore WRITTEN by the class and picked up on
3468 % the NEXT run. The build already runs pdflatex three times for cross-
3469 % references, so the deposited PDF always carries current metadata; only a
3470 % first-ever compilation of a brand-new file emits pdfx's "no \jobname.xmpdata"
3471 % warning and lacks dc:title.
3472 %
3473 % The write happens at the END of the document, not at \begin{document}. This
3474 % class has always documented -- and every example follows it -- that \title,
3475 % \author, \department and \keyword go AFTER \begin{document}, next to

```

```

3476 % \maketitle. A write at \begin{document} therefore ran before any of them
3477 % existed and died with "Undefined control sequence" on \@authname, so the
3478 % pdfa option was unusable in exactly the style the manual teaches. It only
3479 % ever worked for a document that set its metadata in the preamble. At the end
3480 % everything is set whichever style the author chose, and \immediate still puts
3481 % the file on disk in time for the next run.
3482 %
3483 % The hook is enddocument/afteraux, and the stream is \@mainaux rather than one
3484 % of our own. TeX has sixteen output streams and example.tex was already using
3485 % all of them (toc, lof, lot, loa, lol, loq, los, idx, glo, aux, bcf, ...), so
3486 % \newwrite here failed with "No room for a new \write" on precisely the
3487 % document that most needs validating. enddocument/afteraux fires after LaTeX
3488 % has closed the .aux, which frees that stream for the one \immediate\openout
3489 % we need and costs the document nothing. Kernels older than 2020-10-01 have no
3490 % such hook; there the class falls back to \AtEndDocument with a stream of its
3491 % own, which is the pre-existing behaviour.
3492 %
3493 % Every field is guarded, and the guards are not cosmetic. \coppe@selecttitle
3494 % is \@nameuse{coppe@title@<lang>}, and \csname on a name that does not exist
3495 % yet MAKES it \relax: unexpandable, so \write puts the control sequence NAME
3496 % into the file instead of the title. The .xmpdata then carried the literal
3497 % \coppe@title@brazilian, pdfx read it back on the following run, and the
3498 % document stopped compiling at all -- a single bad run bricked the file until
3499 % someone deleted the .xmpdata by hand. A field that is not set is now simply
3500 % omitted, which pdfx handles, instead of poisoning the next run.
3501 %
3502 % The writes must be \immediate. \protected@write defers to the shipout, and
3503 % the stream is closed here and now, so deferred writes would land nowhere --
3504 % which is exactly what happened first time round: a 0-byte .xmpdata.
3505 % \let\protect\string inside the group gives the protection that
3506 % \protected@write would have given: accents and other robust commands are
3507 % written literally instead of being expanded. pdfx reads the .xmpdata as
3508 % LaTeX, so \'a arrives there intact.
3509 \ifdefined\AddToHook
3510   \let\coppe@xmpout\@mainaux
3511   \def\coppe@xmphook{\AddToHook{enddocument/afteraux}}%
3512 \else
3513   \newwrite\coppe@xmpout
3514   \let\coppe@xmphook\AtEndDocument
3515 \fi
3516 \coppe@xmphook{%
3517   \gdef\coppe@xmpkeys{%
3518     \count@=0
3519     \@whilenum\count@<\value{keywords}\do{%
3520       \ifnum\count@>0
3521         \g@addto@macro\coppe@xmpkeys{\string\sep\space}%
3522       \fi
3523       \protected@edef\coppe@tmp{\csname CoppeKeyword:\the\count@\endcsname}%
3524       \expandafter\g@addto@macro\expandafter\coppe@xmpkeys
3525       \expandafter{\coppe@tmp}%
3526       \advance\count@ by 1}%
3527   \immediate\openout\coppe@xmpout=\jobname.xmpdata
3528   \begingroup
3529     \let\protect\string

```

```

3530 \ifundefined{coppe@title@\coppe@mainlang}{\%
3531 \immediate\write\coppe@xmpout{\string\Title{\coppe@selecttitle}}}%
3532 \ifundefined{@authname}{\%
3533 \immediate\write\coppe@xmpout{\%
3534 \string\Author{\@authname\space\@authsurn}}}%
3535 \immediate\write\coppe@xmpout{\string\Keywords{\coppe@xmpkeys}}}%
3536 \ifthenelse{equal{\coppe@deptcode}{}}{\%
3537 \immediate\write\coppe@xmpout{\string\Subject{\local@doctype\space de
3538 \@degree\name. Programa de \meta@deptname\space(\coppe@deptcode),
3539 COPPE/UFRJ}}}%
3540 \immediate\write\coppe@xmpout{\string\Publisher{\local@universityname}}}%
3541 \ifundefined{coppe@docyear}{\%
3542 \immediate\write\coppe@xmpout{\string\Date{\coppe@docyear}}}%
3543 \endgroup
3544 \immediate\closeout\coppe@xmpout}%
3545 \fi
3546 \RequirePackage{hyperref}
3547 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
3548 % former natbib+bibtex setup. The numbers class option selects the numeric
3549 % citation style; otherwise the author-date default is used. natbib=true keeps
3550 % \citet/\citep available for users migrating from natbib.
3551 \if@coppenumeric
3552 \RequirePackage[backend=biber,datamodel=coppe,%
3553 bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
3554 \else
3555 \RequirePackage[backend=biber,datamodel=coppe,%
3556 bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
3557 \fi
3558 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
3559 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
3560 % "Bibliografia" (babel's default \bibname). The title is rendered through
3561 % \coppestring so it follows the current main language in BOTH the chapter
3562 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
3563 % the .toc write, and babel's \bibname caption override was equally fragile).
3564 \defbibheading{bibliography}{%
3565 \chapter*{\coppestring{references}}%
3566 \addcontentsline{toc}{chapter}{\coppestring{references}}}
3567 % The catalog page range (bib:begin/bib:end) was written by the old
3568 % thebibliography redefinition, which biblatex's \printbibliography does not
3569 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
3570 % begin label uses that hook and the end label is emitted by wrapping
3571 % \printbibliography to write it just after the reference list.
3572 \AtBeginBibliography{\coppe@bibBegin}
3573 \let\coppe@orig@printbibliography\printbibliography
3574 \renewcommand\printbibliography[1][\%
3575 \coppe@orig@printbibliography[#1]]%
3576 \coppe@bibEnd}
3577 </class>
3578 <*glossary>
3579 actual '='
3580 quote '!'
3581 level '>'
3582 %%% delim_0 ", p. "

```

```

3583 delim_0 "\\dotfill "
3584 lethead_flag 0
3585 headings_flag 0
3586 preamble
3587 "\\n\\begin{theglossary}\\n \\makeatletter"
3588 postamble
3589 "\\n \\end{theglossary}\\n"
3590 </glossary>

3591 <*class>
3592 \\newcommand{\\annex}{
3593     \\renewcommand{\\appendixname}{\\coppestring{annex}}%
3594     \\appendix
3595     \\@coppeannextrue
3596 % \\appendix has already re-issued the centred chapter format, but it baked in
3597 % the \\appendixname of that moment. Re-issue it now that the name is "Anexo".
3598     \\coppe@appendixheadings
3599 }
3600 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

3601 <*dbx>
3602 \\ProvidesFile{coppe.dbx}[2026/09/09 v4.1 CoppeTeX ABNT datamodel]
3603 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
3604 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
3605 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
3606 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
3607 % performance) are already in biblatex's standard datamodel.
3608 \\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
3609 % Custom fields.
3610 \\DeclareDatamodelFields[type=field,datatype=literal]{course} % "(... em <course>)"
3611 \\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
3612 \\DeclareDatamodelFields[type=field,datatype=literal]{platform} % videogame platform
3613 \\DeclareDatamodelFields[type=field,datatype=literal]{scale} % map scale (escala)
3614 \\DeclareDatamodelFields[type=list,datatype=literal]{artist} % game/illustration art
3615 \\DeclareDatamodelFields[type=list,datatype=literal]{director} % movie/tvshow/video
3616 \\DeclareDatamodelFields[type=list,datatype=literal]{producer}
3617 \\DeclareDatamodelFields[type=field,datatype=literal]{caderno} % newspaper section
3618 \\DeclareDatamodelFields[type=field,datatype=literal]{relator} % jurisprudence rapporteur
3619 \\DeclareDatamodelFields[type=field,datatype=literal]{filingdate} % patente: deposito
3620 \\DeclareDatamodelFields[type=field,datatype=literal]{grantdate} % patente: concessao
3621 \\DeclareDatamodelEntryfields{course,coppedegree}
3622 \\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
3623 {artist,platform,director,producer,version}
3624 \\DeclareDatamodelEntryfields[map]{scale}
3625 \\DeclareDatamodelEntryfields[article]{caderno}
3626 \\DeclareDatamodelEntryfields[jurisdiction]{relator}
3627 \\DeclareDatamodelEntryfields[patent]{filingdate,grantdate}
3628 </dbx>
3629 <*bbx>
3630 \\ProvidesFile{coppe.bbx}[2026/09/09 v4.1 CoppeTeX ABNT bibliography style]
3631 % The reference list is single-spaced (ABNT 2.4) via \\singlespacing in

```



```

3632 % \AtBeginBibliography. coppe.cls already loads setspace, but the style must work
3633 % standalone too (e.g. \usepackage[bibstyle=coppe]{biblatex} in a plain article),
3634 % so load setspace here as well (idempotent).
3635 \RequirePackage{setspace}
3636 % ABNT reference list keeps the date at the end (imprenta), same for author-date
3637 % and numeric modes, so we derive from the standard style, not authoryear.
3638 \RequireBibliographyStyle{standard}
3639
3640 % Note: standard style never dashes repeated authors, which already matches the
3641 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
3642 \ExecuteBibliographyOptions{%
3643   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
3644   giveninits=false,
3645   uniquename=false,
3646   uniquelist=false,
3647   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
3648   sorting=nyt,
3649 }
3650
3651 % Language strings (Disponível em / Acesso em / In, etc.)
3652 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
3653 \DeclareLanguageMapping{english}{english-coppe}
3654
3655 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
3656 \DeclareNameAlias{author}{family-given}
3657 \DeclareNameAlias{editor}{family-given}
3658 \DeclareNameAlias{translator}{family-given}
3659 \DeclareNameAlias{default}{family-given}
3660 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
3661 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
3662
3663 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
3664 % organizer prints the kind of responsibility in parentheses after the
3665 % name, abbreviated, lowercase and in the singular even for several
3666 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
3667 % standard style prints ", Org." (comma, no parentheses) when the editor
3668 % takes the author slot, so override editor+others to wrap the role string
3669 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
3670 % the role via coppe:bookeditor below. The role word itself is the
3671 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
3672 % packs (brazilian-coppe.lbx et al.).
3673 \renewbibmacro*{editor+others}{%
3674   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}}} }
3675     {\printnames{editor}%
3676      \setunit*\addspace}%
3677     \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
3678     \clearname{editor}}
3679   {}
3680
3681 % Same parenthesised role for drivers that take the editor through the plain
3682 % editor macro instead of editor+others (e.g. @manual).
3683 \renewbibmacro*{editor}{%
3684   \ifbool{expr{ test \ifuseeditor and not test {\ifnameundef{editor}}} }
3685     {\printnames{editor}%
3686      \setunit*\addspace}%

```

```

3686 \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3687 \clearname{editor}}
3688 {}
3689
3690 % Uppercase the surname (family + prefix) only inside the bibliography list;
3691 % citations keep normal caps (post-2020 rule).
3692 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
3693 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
3694 % field names; the month stays abbreviated via \mkbibmonth.
3695 % Autor-entidade com subordinacao: a 4.3.2.13 poe em caixa alta a jurisdicao ou
3696 % a entidade superior, e deixa em caixa normal o que vem depois -- o orgao
3697 % subordinado ("BRASIL. Supremo Tribunal Federal.") e o qualificador entre
3698 % parenteses ("SAO PAULO (Estado)."). \mkbibnamefamily so conhecia nome de
3699 % pessoa e punha tudo em caixa alta. Corta-se no primeiro " (" e, dentro do que
3700 % sobra a esquerda, no primeiro ". ": o que estiver antes vai para caixa alta,
3701 % o resto sai como foi digitado.
3702 % O biber entrega o nome literal ENTRE CHAVES ("{Brasil. Supremo Tribunal
3703 % Federal}"), e as chaves escondem o ponto do casamento de padrao abaixo: sem
3704 % desembrulhar, o nome inteiro caia no \MakeUppercase. \coppe@ucstrip tira a
3705 % chave externa -- um argumento delimitado que seja um grupo unico perde as
3706 % chaves na captura.
3707 % Autor-entidade: a 4.3.2.13 poe em caixa alta a jurisdicao ou a entidade
3708 % superior e deixa o orgao subordinado em caixa normal -- "BRASIL. Supremo
3709 % Tribunal Federal." Qual das duas coisas o nome e, so quem escreve sabe:
3710 % "Associacao Brasileira de Normas Tecnicas" e uma entidade so, e vai inteira
3711 % para versal; "Brasil. Supremo Tribunal Federal" tem subordinacao, e a parte
3712 % depois do ponto fica como esta.
3713 %
3714 % A regra, entao, e mecanica e previsivel: um nome que TENHA ponto sai
3715 % exatamente como o autor digitou; um nome sem ponto vai para caixa alta. O
3716 % teste e feito sobre a forma detokenizada, porque entre as palavras de um
3717 % nome o biblatex nao poe espaco -- poe \bibnamedelim e \bibnamedelimi --, e
3718 % procurar ". " no fluxo de tokens nunca encontra nada.
3719 % O parentese conta junto com o ponto: "Sao Paulo (Estado)" leva o
3720 % qualificador em caixa normal, e so quem escreve sabe disso.
3721 % O argumento chega como \namepartfamily -- uma sequencia de controle, nao o
3722 % texto. Detokenizar sem expandir antes devolve a string "\namepartfamily",
3723 % que nao tem ponto nenhum, e por isso TODO nome caia no \MakeUppercase.
3724 \protected\def\coppe@ucfamily#1{%
3725 \protected@edef\coppe@uctmp{#1}%
3726 \edef\coppe@ucstr{\detokenize\expandafter{\coppe@uctmp}}%
3727 \expandafter\coppe@ucscan\coppe@ucstr.\coppe@ucstop{#1}}
3728 \protected\def\coppe@ucscan#1.#2\coppe@ucstop#3{%
3729 \ifblank{#2}
3730 {\expandafter\coppe@ucscanp\coppe@ucstr(\coppe@ucstop{#3}}
3731 {#3}}
3732 \protected\def\coppe@ucscanp#1(#2\coppe@ucstop#3{%
3733 \ifblank{#2}{\MakeUppercase{#3}}{#3}}
3734 % A 4.3 separa os meses de um intervalo por barra -- "jan./jun. 1991" --, e o
3735 % biblatex usa travessao.
3736 \renewcommand*{\bibdaterangesep}{/}
3737 \protected\def\coppe@abntdate#1#2#3{%
3738 \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
3739 \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%

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3740 \iffielddundef{#1}{\stripzeros{\thefield{#1}}}}
3741 \AtBeginBibliography{%
3742 \singlespacing % entries single-spaced internally (2.4)
3743 \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
3744 \renewcommand*{\mkbibnamefamily}[1]{\coppe@ucfamily{#1}}%
3745 \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
3746 \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
3747 \let\mkbibdateshort\coppe@abntdate
3748 }
3749
3750 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
3751 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
3752 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
3753 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
3754 % article AND the next word are uppercased; otherwise just the first word. The
3755 % delimited macros split off words; \coppeUCkeep strips the trailing-space
3756 % sentinel so no spurious space is left before the following punctuation.
3757 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
3758 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
3759 \coppeUCifart{#1}
3760 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
3761 {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
3762 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
3763 \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
3764 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
3765 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
3766 \protected\def\coppeUCifart#1{%
3767 \begingroup
3768 \lowercase{\def\coppeUC@w{#1}}%
3769 \edef\coppeUC@w{\coppeUC@w}%
3770 \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,}%
3771 \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
3772 \endgroup}
3773 \DeclareFieldFormat{title}{%
3774 \ifboolexpr{ test {\ifnameundef{labelname}} }
3775 {#1}%
3776 {\mkbibbold{#1}}}
3777 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
3778 % format receives the RAW title text (the title format above only sees the
3779 % field-printing commands), so \coppeUCtitle can split off and uppercase the
3780 % first word here.
3781 \DeclareFieldFormat{titlecase}{%
3782 \ifboolexpr{ test {\ifnameundef{labelname}} }
3783 {\coppeUCtitle{#1}}%
3784 {#1}}
3785 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%
3786 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
3787 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
3788 % the title of a whole document, so override those back to bold.
3789 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
3790 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
3791 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
3792 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
3793 \DeclareFieldFormat[legislation,jurisdiction,legal]{volume}{v\adddot\addnbspace#1}

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3794 \DeclareFieldFormat[legislation,jurisdiction,legal]{number}{n\addot\addnbspace#1}
3795 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
3796 \DeclareFieldFormat{booktitle}{\mkbibbold{#1}}
3797 \DeclareFieldFormat{maintitle}{\mkbibbold{#1}}
3798
3799 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
3800 \DeclareFieldFormat[pagetotal]{#1\addnbspace p\addot}
3801 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\addot}
3802 \DeclareFieldFormat{pages}{p\addot\addnbspace#1}
3803
3804 % --- Periodical volume / number prefixes ---
3805 \DeclareFieldFormat[article]{volume}{v\addot\addnbspace#1}
3806 \DeclareFieldFormat[article]{number}{n\addot\addnbspace#1}
3807
3808 % --- "In:" before host title (incollection/inproceedings/inbook) ---
3809 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
3810
3811 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
3812 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
3813 % O 4.2 do manual fecha toda entrada eletrônica assim, sem exceção:
3814 %   Disponível em: <url>. Acesso em: <data>.
3815 % O biblatex, por padrão, põe a data de acesso entre parênteses e sem
3816 % dois-pontos -- "<url> (Acesso em <data>)" --, que é a convenção dele, não a
3817 % da norma.
3818 \DeclareFieldFormat[urldate]{\bibstring{urlseen}\addcolon\space#1}
3819 % A descrição física ("1 carta", "3 fotografias", "1 partitura") vem ANTES do
3820 % bloco eletrônico em todos os exemplos da 4.2. Os drivers padrão do biblatex
3821 % a imprimem DEPOIS, então ela é impressa aqui e o campo é limpo, para não
3822 % sair duas vezes.
3823 \renewbibmacro*{url+urldate}{%
3824   \iffielddundef{addendum}
3825   {}
3826   {\printfield{addendum}\clearfield{addendum}%
3827     \setunit{\addot\space}}%
3828   \iffielddundef{url}
3829   {}
3830   {\printfield{url}%
3831     \iffielddundef{urlyear}{\setunit{\addot\space}\printurldate}}
3832
3833 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
3834 %   Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
3835 %   differ: the newspaper name (journaltitle) and city, then either
3836 %       "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
3837 %       "p. <pages>, <date>" (page before date, when none),
3838 %   per NBR 6023 §4.2.3.6. ---
3839 \DeclareBibliographyDriver{article}{%
3840   \usebibmacro{bibindex}%
3841   \usebibmacro{begentry}%
3842   \usebibmacro{author/translator+others}%
3843   \setunit{\labelnamepunct}\newblock
3844   \usebibmacro{title}%
3845   \newunit\newblock
3846   \printfield{journaltitle}%
3847 % "J. Manag. Stud., Oxford": a vírgula depois de um título abreviado terminado

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3848 % em ponto e engolida pelo rastreador de pontuacao, e a 4.2.3.4 a exige.
3849 \iflistundef{location}{\printtext{,}\setunit{\space}\printlist{location}}%
3850 \iffieldequalstr{entrysubtype}{newspaper}
3851   {\iffieldundef{caderno}
3852     {\setunit{\addcomma\space}\printfield{pages}%
3853       \setunit{\addcomma\space}\printdate}%
3854     {\setunit{\addcomma\space}\printdate%
3855       \setunit{\adddot\space}\printfield{caderno}%
3856       \setunit{\addcomma\space}\printfield{pages}}}%
3857   {\setunit{\addcomma\space}\printfield{volume}%
3858     \setunit{\addcomma\space}\printfield{number}%
3859     \setunit{\addcomma\space}\printfield{pages}%
3860     \setunit{\addcomma\space}\printdate}%
3861 \newunit\newblock
3862 \printfield{doi}%
3863 \newunit\newblock
3864 \usebibmacro{url+urldate}%
3865 \usebibmacro{finentry}}
3866
3867 % --- incollection: ABNT puts the host organizer before the booktitle, as
3868 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
3869 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
3870 \newbibmacro*{coppe:bookeditor}{%
3871   \ifnameundef{editor}
3872     {}
3873     {\printnames{editor}%
3874       \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
3875       \clearname{editor}%
3876       \setunit{\adddot\space}}
3877 \newbibmacro*{coppe:partdriver}{%
3878   \usebibmacro{bibindex}\usebibmacro{begentry}%
3879   \usebibmacro{author/translator+others}%
3880   \setunit{\labelnamepunct}\newblock
3881   \usebibmacro{title}%
3882   \newunit\newblock
3883   \usebibmacro{in:}%
3884   \usebibmacro{coppe:bookeditor}%
3885   \printfield{booktitle}%
3886   \newunit\newblock
3887   \printfield{edition}%
3888   \newunit\newblock
3889   \printlist{location}%
3890   \setunit*{\addcolon\space}\printlist{publisher}%
3891   \setunit*{\addcomma\space}\printdate%
3892   \newunit\newblock
3893   \printfield{pages}%
3894   \setunit{\addcomma\space}\printfield{chapter}%
3895   \newunit\newblock
3896   \usebibmacro{url+urldate}%
3897   \usebibmacro{finentry}}
3898 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
3899 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
3900
3901 % --- type field: resolve a known bibliography string (babel-aware), else literal ---

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3902 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
3903
3904 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
3905 % "(Mestrado em Memoria Social e Documento)": a 4.2.1.1 poe o grau dentro dos
3906 % parenteses, antes do curso, e deixa o substantivo do documento fora deles --
3907 % "Dissertacao (Mestrado em ...)". O grau vem do campo coppedegree, que o
3908 % sourcemap preenche junto com o tipo.
3909 % O grau sai do proprio tipo do trabalho, que ja e uma chave de string; passar
3910 % a chave por um campo e depois resolve-la com \ifbibstring nao funciona,
3911 % porque \thefield nao se expande onde \ifbibstring precisa. Um campo
3912 % coppedegree preenchido a mao pelo autor tem precedencia.
3913 \newcommand*\coppe@degword{%
3914   \iffieldequalstr{coppedegree}
3915     {\iffieldequalstr{type}{mscdiss}{\bibstring{coppemscdeg}}}%
3916     {\iffieldequalstr{type}{dscsthesis}{\bibstring{coppedscdeg}}}%
3917     {\iffieldequalstr{type}{phdthesis}{\bibstring{coppephddeg}}}%
3918     {\iffieldequalstr{type}{tcc}{\bibstring{coppetccdeg}}}{}}
3919   {\printfield{coppedegree}}
3920 \DeclareFieldFormat{course}{%
3921   \mkbibparens{\coppe@degword\addspace\coppestring{in:}\addspace#1}}
3922 % A ordem da 4.2.1.1 para tese e dissertacao:
3923 %   AUTOR. Titulo. ano. NNN f. Tipo (Grau em Curso) -- Instituicao, Local, ano.
3924 % A data aparece duas vezes de proposito, como nos exemplos do manual: uma
3925 % depois do titulo, outra fechando a nota de tese.
3926 \DeclareBibliographyDriver{thesis}{%
3927   \usebibmacro{bibindex}\usebibmacro{begentry}%
3928   \usebibmacro{author}%
3929   \setunit{\labelnamepunct}\newblock
3930   \usebibmacro{title}%
3931   \newunit\newblock
3932   \printdate%
3933   \newunit\newblock
3934   \printfield{pagetotal}%
3935   \newunit\newblock
3936   \printfield{type}\setunit*{\addspace}\printfield{course}%
3937   \iflistundef{institution}
3938     {}
3939     {\setunit*{\addspace\textendash\addspace}\printlist{institution}}%
3940   \setunit*{\addcomma\space}\printlist{location}%
3941   \setunit*{\addcomma\space}\printdate%
3942   \newunit\newblock
3943   \printfield{note}%
3944   \newunit\newblock
3945   \usebibmacro{url+urldate}%
3946   \usebibmacro{finentry}}
3947 % --- Games / audiovisual: render the custom fields (artist, platform,
3948 %   director, producer, version) the generic misc driver ignores. ---
3949 \newbibmacro*{coppe:media}{%
3950   \iflistundef{artist}{ }\setunit{\addperiod\space}%
3951   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
3952   \iffieldequalstr{platform}{ }\setunit{\addperiod\space}%
3953   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}}%
3954   \iflistundef{director}{ }\setunit{\addperiod\space}%
3955   \printtext{\coppestring{directedby}\addspace}\printlist{director}}%

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3956 \iflistundef{producer}{\setunit{\addperiod\space}%
3957 \printtext{\coppestring{producedby}\addspace}\printlist{producer}}
3958 \newbibmacro*{coppe:mediadriv}{%
3959 \usebibmacro{bibindex}\usebibmacro{begentry}%
3960 \usebibmacro{author/translator+others}%
3961 \setunit{\labelnamepunct}\newblock
3962 \usebibmacro{title}%
3963 \newunit\newblock
3964 \usebibmacro{coppe:media}%
3965 \newunit\newblock
3966 \usebibmacro{coppe:impreta}%
3967 \newunit\newblock
3968 \printfield{version}%
3969 \newunit\newblock
3970 \printfield{note}%
3971 \newunit\newblock
3972 \usebibmacro{url+urldate}%
3973 \usebibmacro{finentry}}
3974 % --- Publicacao periodica (4.2.3.1 a 4.2.3.3) -----
3975 % O titulo do periodico vai INTEIRO em caixa alta, e nao so a primeira palavra
3976 % como numa obra sem autor: e a forma dos exemplos do manual. Sem driver
3977 % proprio saia o default do biblatex -- "SAO Paulo Medical Journal (1941)." --,
3978 % sem local, sem editora e com o ano entre parenteses.
3979 \DeclareFieldFormat[periodical,suppperiodical]{title}{\MakeUppercase{#1}}
3980 \DeclareFieldFormat[periodical,suppperiodical]{volume}{v\adddot\addnbspace#1}
3981 \DeclareFieldFormat[periodical,suppperiodical]{number}{n\adddot\addnbspace#1}
3982 \DeclareBibliographyDriver{periodical}{%
3983 \usebibmacro{bibindex}\usebibmacro{begentry}%
3984 % Pelo campo, e nao pelo bibmacro: o bibmacro 'title' passa pelo formato
3985 % 'titlecase', e num periodico -- que nao tem autor -- ele nao chegava a
3986 % imprimir coisa nenhuma. A entrada saia comecando por um ponto solto.
3987 \printfield{title}%
3988 \newunit\newblock
3989 \printlist{location}%
3990 \iflistundef{publisher}{\setunit*{\addcolon\space}\printlist{publisher}}%
3991 \setunit*{\addcomma\space}\printfield{volume}%
3992 \setunit*{\addcomma\space}\printfield{number}%
3993 \setunit*{\addcomma\space}\printdate%
3994 \newunit\newblock
3995 \printfield{note}%
3996 \newunit\newblock
3997 \usebibmacro{url+urldate}%
3998 \usebibmacro{finentry}}
3999 \DeclareBibliographyDriver{suppperiodical}{%
4000 \usedriver{}{periodical}\usebibmacro{finentry}}
4001 % --- Eventos (4.2.4) -----
4002 % O nome do evento vai em caixa alta, seguido do numero da edicao, do ano e da
4003 % cidade -- "ENCONTRO ANUAL DA ANPAD, 14., 1982, Florianopolis." --, e so
4004 % depois vem o titulo dos anais, que e o elemento em destaque. Por isso o nome
4005 % do evento tem campo proprio (|eventtitle|, sinonimo |evento|) em vez de vir
4006 % empacotado no titulo: nada mais na entrada vai para caixa alta.
4007 \DeclareFieldFormat{eventtitle}{\MakeUppercase{#1}}
4008 % "Anais [...]" nao e uma obra entrada pelo titulo: quem faz as vezes de
4009 % cabeca da entrada e o nome do evento, ja em caixa alta. Sem esta declaracao

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4010 % o titulo dos anais saia "ANAIS [...]" e ainda perdia o ponto que o separa
4011 % da impressa.
4012 \DeclareFieldFormat[proceedings]{title}{#1}
4013 \DeclareFieldFormat[proceedings,inproceedings]{number}{#1\adddot}
4014 \DeclareFieldFormat[proceedings,inproceedings]{volume}{v\adddot\addnbspace#1}
4015 \newbibmacro*{coppe:evento}{%
4016   \printfield{eventtitle}%
4017   \setunit*{\addcomma\space}\printfield{number}%
4018   \setunit*{\addcomma\space}\printeventdate%
4019   \setunit*{\addcomma\space}\printfield{venue}}
4020 \DeclareBibliographyDriver{proceedings}{%
4021   \usebibmacro{bibindex}\usebibmacro{begentry}%
4022   \usebibmacro{coppe:evento}%
4023   \newunit\newblock
4024 % O ponto que separa o titulo dos anais da impressa e literal: o rastreador de
4025 % pontuacao do biblatex nao reconhece o fim de um titulo terminado em "]", e
4026 % nem \newunit nem \adddot produzia alguma coisa ali.
4027   \iffielddundef{title}{\printfield{title}\printtext{.}}%
4028   \setunit{\space}\newblock
4029   \usebibmacro{coppe:impressa}%
4030   \newunit\newblock
4031   \usebibmacro{url+urldate}%
4032   \usebibmacro{finentry}}
4033 \DeclareBibliographyDriver{inproceedings}{%
4034   \usebibmacro{bibindex}\usebibmacro{begentry}%
4035   \usebibmacro{author/translator+others}%
4036   \setunit{\labelnamepunct}\newblock
4037   \usebibmacro{title}%
4038   \newunit\newblock
4039   \usebibmacro{in:}%
4040   \usebibmacro{coppe:evento}%
4041   \newunit\newblock
4042   \iffielddundef{booktitle}{\printfield{booktitle}\printtext{.}}%
4043   \setunit{\space}\newblock
4044   \usebibmacro{coppe:impressa}%
4045   \newunit\newblock
4046   \printfield{volume}%
4047   \setunit*{\addcomma\space}\printfield{pages}%
4048   \newunit\newblock
4049   \usebibmacro{url+urldate}%
4050   \usebibmacro{finentry}}
4051 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
4052 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
4053 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
4054 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
4055 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
4056 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
4057 % impressa) plus the mandatory scale ("Escala 1:N").
4058 \DeclareFieldFormat{scale}{\bibstring{coppescale}\addnbspace#1}
4059 % O dois-pontos separa local e editora; sem editora ele vazava para a data
4060 % ("Recife: 1976"), e a 4.2.12 quer "Recife, 1976". Por isso a guarda.
4061 \newbibmacro*{coppe:impressa}{%
4062   \printlist{location}%
4063   \iflistundef{publisher}

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4064 {}
4065 {\setunit*{\addcolon\space}\printlist{publisher}}%
4066 \setunit*{\addcomma\space}\printdate}
4067 \DeclareBibliographyDriver{map}{%
4068 \usebibmacro{bibindex}\usebibmacro{begentry}%
4069 \usebibmacro{author/translator+others}%
4070 \setunit{\labelnamepunct}\newblock
4071 \usebibmacro{title}%
4072 \newunit\newblock
4073 \usebibmacro{coppe:imprenta}%
4074 \newunit\newblock
4075 % "Recife, 1976. 71 fotografias. Escala 1: 20.000." -- a descricao fisica vem
4076 % ANTES da escala, e por isso ela nao pode esperar pelo bloco eletronico.
4077 \iffielddundef{addendum}{\printfield{addendum}\clearfield{addendum}}%
4078 \newunit\newblock
4079 \printfield{scale}%
4080 \newunit\newblock
4081 \printfield{note}%
4082 \newunit\newblock
4083 \usebibmacro{url+urldate}%
4084 \usebibmacro{finentry}}
4085 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
4086 % number. date. The holder (depositante) is printed in normal case, not the
4087 % ALL-CAPS family treatment the bibliography gives the leading author.
4088 \newbibmacro*{coppe:depositor}{%
4089 \ifnameundef{holder}
4090 {}
4091 {\printtext{\bibstring{depositor}\addcolon\space}%
4092 \begingroup
4093 \def\mkbibnamefamily##1{##1}%
4094 \def\mkbibnameprefix##1{##1}%
4095 \printnames{holder}%
4096 \endgroup}}
4097 \DeclareBibliographyDriver{patent}{%
4098 \usebibmacro{bibindex}%
4099 \usebibmacro{begentry}%
4100 \usebibmacro{author/translator+others}%
4101 \setunit{\labelnamepunct}\newblock
4102 \usebibmacro{title}%
4103 \newunit\newblock
4104 \usebibmacro{coppe:depositor}%
4105 \newunit\newblock
4106 \printfield{type}%
4107 \setunit*{\addspace}\printfield{number}%
4108 \newunit\newblock
4109 % A 4.2.5 fecha a patente com as duas datas, nomeadas: "Deposito: 27 jan. 2006.
4110 % Concessao: 25 mar. 2008." Sem elas so saia o ano.
4111 \iffielddundef{filingdate}
4112 {}
4113 {\printtext{\bibstring{coppefiling}\addcolon\space}\printfield{filingdate}%
4114 \setunit{\addot\space}}%
4115 \iffielddundef{grantdate}
4116 {}
4117 {\printtext{\bibstring{coppegrant}\addcolon\space}\printfield{grantdate}}%

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4118 \ifboolexpr{ test {\iffieldundef{filingdate}} and test {\iffieldundef{grantdate}} }
4119   {\printdate}
4120   }%
4121 \newunit\newblock
4122 \usebibmacro{url+urldate}%
4123 \usebibmacro{finentry}}
4124 % Exotic ABNT types that biblatex's standard style leaves without a driver:
4125 % give each a baseline (book-like for published docs: norma/partitura; misc for
4126 % the rest) so they print. Per-type ABNT refinements can follow later.
4127 % \usedriver desliga o \finentry do driver chamado, porque quem chama e que
4128 % deve fechar a entrada -- e aqui ninguem fechava. Resultado: norma, partitura,
4129 % audiovisual, iconografico, obra de arte, tridimensional e conjunto de dados
4130 % saiam sem o ponto final, que a 4.2 poe em toda referencia.
4131 \DeclareBibliographyDriver{standard}{\usedriver{}{book}\usebibmacro{finentry}}
4132 \DeclareBibliographyDriver{music}{\usedriver{}{book}\usebibmacro{finentry}}
4133 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
4134 % the law id + ementa (plain, not bold), then location and date.
4135 \newbibmacro*{coppe:legaldriver}{%
4136   \usebibmacro{bibindex}\usebibmacro{begentry}%
4137   \usebibmacro{author/translator+others}%
4138   \setunit{\labelnamepunct}\newblock
4139   \usebibmacro{title}%
4140   \newunit\newblock
4141   % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
4142   \iffieldundef{relator}{ }\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
4143   \printfield{edition}%
4144   \newunit\newblock
4145   \printlist{location}%
4146 % A 4.2.6 traz o veiculo de publicacao com volume, numero e pagina -- "Sao
4147 % Paulo, v. 62, n. 3, p. 217-220, 1998." --, que se perdiam por completo.
4148 \iflistundef{publisher}{ }\setunit*{\addcolon\space}\printlist{publisher}}%
4149 \setunit*{\addcomma\space}\printfield{volume}%
4150 \setunit*{\addcomma\space}\printfield{number}%
4151 \setunit*{\addcomma\space}\printfield{pages}%
4152 \setunit{\addcomma\space}\printdate%
4153 \newunit\newblock
4154 \printfield{note}%
4155 \newunit\newblock
4156 \usebibmacro{url+urldate}%
4157 \usebibmacro{finentry}}
4158 % 4.2.14: "UNIVERSIDADE ... Rio de Janeiro, 1998. Disponivel em: ..." -- o
4159 % driver padrao do biblatex para @online nao imprime o local.
4160 \DeclareBibliographyDriver{online}{%
4161   \usebibmacro{bibindex}\usebibmacro{begentry}%
4162   \usebibmacro{author/translator+others}%
4163   \setunit{\labelnamepunct}\newblock
4164   \usebibmacro{title}%
4165   \newunit\newblock
4166   \usebibmacro{coppe:imprenta}%
4167   \newunit\newblock
4168   \printfield{note}%
4169   \newunit\newblock
4170   \usebibmacro{url+urldate}%
4171   \usebibmacro{finentry}}

```

```

4172 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
4173 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}
4174 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
4175 % Audio e video vao pelo driver de midia, que imprime diretor e produtor --
4176 % "Producao de Jorge Ramos de Andrade." da 4.2.9. O driver misc os ignorava.
4177 \DeclareBibliographyDriver{audio}{\usebibmacro{coppe:mediadriver}}
4178 \DeclareBibliographyDriver{video}{\usebibmacro{coppe:mediadriver}}
4179 \DeclareBibliographyDriver{software}{\usedriver{}{misc}\usebibmacro{finentry}}
4180 \DeclareBibliographyDriver{image}{\usedriver{}{misc}\usebibmacro{finentry}}
4181 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}\usebibmacro{finentry}}
4182 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}\usebibmacro{finentry}}
4183 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}\usebibmacro{finentry}}
4184
4185 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
4186 \DeclareSourceMap{%
4187   \maps[datatype=bibtex]{%
4188     % entry-type PT synonyms -> canonical English
4189     \map{step[typesource=livro,typetarget=book]}
4190     \map{step[typesource=artigo,typetarget=article]}
4191     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
4192     \map{step[typesource=partedelivro,typetarget=incollection]}
4193     \map{step[typesource=emlivro,typetarget=inbook]}
4194     \map{step[typesource=verbete,typetarget=inreference]}
4195     \map{step[typesource=anais,typetarget=proceedings]}
4196     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
4197     \map{step[typesource=periodico,typetarget=periodical]}
4198     \map{step[typesource=relatorio,typetarget=report]}
4199     \map{step[typesource=relatoriotecnico,typetarget=report]}
4200     \map{step[typesource>manual,typetarget>manual]}
4201     \map{step[typesource=correspondencia,typetarget=letter]}
4202     \map{step[typesource=patente,typetarget=patent]}
4203     \map{step[typesource=diverso,typetarget=misc]}
4204     \map{step[typesource=jogo,typetarget=game]}
4205     \map{step[typesource=jogoeletronico,typetarget=videogame]}
4206     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
4207     \map{step[typesource=filme,typetarget=movie]}
4208     \map{step[typesource=serietv,typetarget=tvshow]}
4209     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
4210     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
4211     % except map (declared in coppe.dbx).
4212     \map{step[typesource=legislacao,typetarget=legislation]}
4213     \map{step[typesource=atonomativo,typetarget=legislation]}
4214     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
4215     \map{step[typesource=documentojuridico,typetarget=legal]}
4216     \map{step[typesource=documentocivil,typetarget=legal]}
4217     \map{step[typesource=norma,typetarget=standard]}
4218     \map{step[typesource=partitura,typetarget=music]}
4219     \map{step[typesource=iconografico,typetarget=image]}
4220     \map{step[typesource=obradearte,typetarget=artwork]}
4221     \map{step[typesource=tridimensional,typetarget=artwork]}
4222     \map{step[typesource=cartografico,typetarget=map]}
4223     \map{step[typesource=mapa,typetarget=map]}
4224     \map{step[typesource=audiovisual,typetarget=video]}
4225     \map{step[typesource=conjuntodedados,typetarget=dataset]}

```

```

4226 \map{\step[tources=fasciculo,typetarget=supperiodical]}
4227 \map{\step[tources=trabalhoapresentado,typetarget=unpublished]}
4228 % thesis family: retype to thesis and preset the (babel-aware) type word
4229 \map{\step[tources=tese,typetarget=thesis]}
4230 \map{\step[tources=dissertacao,typetarget=thesis]}
4231 \map{\step[tources=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
4232 \map{\step[tources=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
4233 \map{\step[tources=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
4234 \map{\step[tources=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
4235 \map{\step[tources=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
4236 \map{\step[tources=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
4237 \map{\step[tources=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
4238 \map{\step[tources=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
4239 % field PT synonyms (unaccented) -> canonical English
4240 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
4241 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
4242 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
4243 \map{\step[fieldsource=titulo,fieldtarget=title]}
4244 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
4245 \map{\step[fieldsource=autor,fieldtarget=author]}
4246 \map{\step[fieldsource=editora,fieldtarget=publisher]}
4247 \map{\step[fieldsource=local,fieldtarget=location]}
4248 \map{\step[fieldsource=ano,fieldtarget=year]}
4249 \map{\step[fieldsource=data,fieldtarget=date]}
4250 \map{\step[fieldsource=edicao,fieldtarget=edition]}
4251 \map{\step[fieldsource=paginas,fieldtarget=pages]}
4252 \map{\step[fieldsource=numero,fieldtarget=number]}
4253 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
4254 \map{\step[fieldsource=organizador,fieldtarget=editor]}
4255 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
4256 \map{\step[fieldsource=tipo,fieldtarget=type]}
4257 \map{\step[fieldsource=nota,fieldtarget=note]}
4258 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
4259 \map{\step[fieldsource=curso,fieldtarget=course]}
4260 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
4261 \map{\step[fieldsource=artista,fieldtarget=artist]}
4262 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
4263 \map{\step[fieldsource=diretor,fieldtarget=director]}
4264 \map{\step[fieldsource=produtor,fieldtarget=producer]}
4265 \map{\step[fieldsource=versao,fieldtarget=version]}
4266 \map{\step[fieldsource=escala,fieldtarget=scale]}
4267 \map{\step[fieldsource=evento,fieldtarget=eventtitle]}
4268 \map{\step[fieldsource=dataevento,fieldtarget=eventdate]}
4269 \map{\step[fieldsource=localevento,fieldtarget=venue]}
4270 \map{\step[fieldsource=deposito,fieldtarget=filingdate]}
4271 \map{\step[fieldsource=concessao,fieldtarget=grantdate]}
4272 }%
4273 }
4274
4275 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
4276 \setlength{\bibhang}{0pt}
4277 \setlength{\bibitemsep}{\baselineskip}
4278
4279 </bbx>

```

```

4280 <*cbx>
4281 \ProvidesFile{coppe.cbx}[2026/09/09 v4.1 CoppeTeX ABNT author-date citations]
4282 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
4283 % author-year cites with normal-caps names (post-2020 rule) out of the box.
4284 \RequireCitationStyle{authoryear-comp}
4285
4286 % --- NBR 10520:2023 author separator -----
4287 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
4288 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->
4289 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
4290 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
4291 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
4292 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
4293 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
4294 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
4295
4296 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
4297 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
4298 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
4299 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
4300 % so it is never added to the bibliography.
4301 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4302 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
4303 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
4304 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
4305 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
4306 \DeclareRobustCommand{\citetapud}[4][]{%
4307   #2\space(#3)\space\mkbibemph{apud}\space\textcite[#1]{#4}}
4308 \DeclareRobustCommand{\citepapud}[4][]{%
4309   \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
4310     \citeauthor{#4}\addcomma\space\citeyear{#4}%
4311     \ifblank{#1}{ }\{\addcomma\space\mkpageprefix[pagination]{#1}\}}
4312 </cbx>
4313 <*lbr>
4314 \ProvidesFile{brazilian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX brazilian strings]
4315 \InheritBibliographyExtras{brazilian}
4316 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale,
4317   coppemscdeg,coppedscdeg,coppesphddeg,coppetccdeg,coppefiling,coppegrant}
4318 \DeclareBibliographyStrings{%
4319   inherit = {brazilian},
4320   availablefrom = {{Dispon'\i vel em}{Dispon'\i vel em}},
4321   coppemscdeg = {{Mestrado}{Mestrado}},
4322   coppedscdeg = {{Doutorado}{Doutorado}},
4323   coppesphddeg = {{Doutorado}{Doutorado}},
4324   coppetccdeg = {{Gradua\c{c}\~{a}o}{Gradua\c{c}\~{a}o}},
4325   coppefiling = {{Dep\osito}{Dep\osito}},
4326   coppegrant = {{Concess\~ao}{Concess\~ao}},
4327   urlseen = {{Acesso em}{Acesso em}},
4328 % A 4.3 abrevia os meses nas tres primeiras letras seguidas de ponto -- menos
4329 % maio, que se escreve por extenso. O biblalex-brazilian abrevia "mai.".
4330   may = {{maio}{maio}},
4331   in = {{In}{In}},
4332   editor = {{organizador}{org\addot}},
4333   editors = {{organizadores}{org\addot}},

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4334 depositor      = {{Depositante}{Depositante}},
4335 coppescale      = {{Escala}{Escala}},
4336 msdiss          = {{Disserta\c{c}\~{a}o}{Disserta\c{c}\~{a}o}},
4337 dscthesi        = {{Tese}{Tese}},
4338 phdthesis       = {{Tese}{Tese}},
4339 tcc             = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
4340 }
4341 </lbr>
4342 <*lbr>
4343 \ProvidesFile{english-coppe.lbx}[2026/09/09 v4.1 CoppeTeX english strings]
4344 \InheritBibliographyExtras{english}
4345 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale,
4346   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4347 \DeclareBibliographyStrings{%
4348   inherit        = {english},
4349   availablefrom = {{Available from}{Available from}},
4350   coppemscdeg   = {{M\addot Sc\addot}{M\addot Sc\addot}},
4351   coppedscdeg   = {{D\addot Sc\addot}{D\addot Sc\addot}},
4352   coppephddeg   = {{Ph\addot D\addot}{Ph\addot D\addot}},
4353   coppetccdeg   = {{B\addot Sc\addot}{B\addot Sc\addot}},
4354   coppefiling   = {{Filed}{Filed}},
4355   coppegrant    = {{Granted}{Granted}},
4356   urlseen       = {{Access on}{Access on}},
4357   in            = {{In}{In}},
4358   editor        = {{editor}{ed\addot}},
4359   editors       = {{editors}{ed\addot}},
4360   depositor     = {{Depositor}{Depositor}},
4361   coppescale    = {{Scale}{Scale}},
4362   msdiss        = {{M\addot Sc\addot\ Dissertation}{M\addot Sc\addot\ Diss\addot}},
4363   dscthesi      = {{D\addot Sc\addot\ Thesis}{D\addot Sc\addot\ Thesis}},
4364   phdthesis     = {{Ph\addot D\addot\ Thesis}{Ph\addot D\addot\ Thesis}},
4365   tcc           = {{Undergraduate Thesis}{Undergraduate Thesis}},
4366 }
4367 </lbr>
4368 <*lbr>
4369 \ProvidesFile{spanish-coppe.lbx}[2026/09/09 v4.1 CoppeTeX spanish strings]
4370 \InheritBibliographyExtras{spanish}
4371 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale,
4372   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4373 \DeclareBibliographyStrings{%
4374   inherit        = {spanish},
4375   availablefrom = {{Disponible en}{Disponible en}},
4376   coppemscdeg   = {{Maestr'\{i}a}{Maestr'\{i}a}},
4377   coppedscdeg   = {{Doctorado}{Doctorado}},
4378   coppephddeg   = {{Doctorado}{Doctorado}},
4379   coppetccdeg   = {{Grado}{Grado}},
4380   coppefiling   = {{Dep\ 'osito}{Dep\ 'osito}},
4381   coppegrant    = {{Concesi\ 'on}{Concesi\ 'on}},
4382   urlseen       = {{Consultado el}{Consultado el}},
4383   in            = {{En}{En}},
4384   editor        = {{editor}{ed\addot}},
4385   editors       = {{editores}{ed\addot}},
4386   depositor     = {{Solicitante}{Solicitante}},
4387   coppescale    = {{Escala}{Escala}},

```

```

4388 msdiss      = {{Tesis de Maestr'\{i\}a}{Tesis de Maestr'\{i\}a}},
4389 dscthesiis   = {{Tesis Doctoral}{Tesis Doctoral}},
4390 phdthesis     = {{Tesis Doctoral}{Tesis Doctoral}},
4391 tcc           = {{Trabajo de Fin de Carrera}{Trabajo de Fin de Carrera}},
4392 }
4393 </lbxes>
4394 <*lbxfr>
4395 \ProvidesFile{french-coppe.lbx}[2026/09/09 v4.1 CoppeTeX french strings]
4396 \InheritBibliographyExtras{french}
4397 \NewBibliographyString{availablefrom,msdiss,dscthesiis,tcc,depositor,coppescale,
4398   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4399 \DeclareBibliographyStrings{%
4400   inherit      = {french},
4401   availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
4402   coppemscdeg   = {{Master}{Master}},
4403   coppedscdeg   = {{Doctorat}{Doctorat}},
4404   coppephddeg   = {{Doctorat}{Doctorat}},
4405   coppetccdeg   = {{Licence}{Licence}},
4406   coppefiling   = {{D\'ep\'ot}{D\'ep\'ot}},
4407   coppegrant    = {{D\'elivrance}{D\'elivrance}},
4408   urlseen       = {{Consult\'e le}{Consult\'e le}},
4409   in            = {{Dans}{Dans}},
4410   editor        = {{\'editeur}{\'ed\adddot}},
4411   editors       = {{\'editeurs}{\'ed\adddot}},
4412   depositor     = {{D\'eposant}{D\'eposant}},
4413   coppescale    = {{\'Echelle}{\'Echelle}},
4414   msdiss        = {{M\'emoire de Master}{M\'emoire de Master}},
4415   dscthesiis    = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
4416   phdthesis     = {{Th\'ese de Doctorat}{Th\'ese de Doctorat}},
4417   tcc           = {{M\'emoire de fin d\'\'etudes}{M\'emoire de fin d\'\'etudes}},
4418 }
4419 </lbxfr>
4420 <*lbxit>
4421 \ProvidesFile{italian-coppe.lbx}[2026/09/09 v4.1 CoppeTeX italian strings]
4422 \InheritBibliographyExtras{italian}
4423 \NewBibliographyString{availablefrom,msdiss,dscthesiis,tcc,depositor,coppescale,
4424   coppemscdeg,coppedscdeg,coppephddeg,coppetccdeg,coppefiling,coppegrant}
4425 \DeclareBibliographyStrings{%
4426   inherit      = {italian},
4427   availablefrom = {{Disponibile da}{Disponibile da}},
4428   urlseen       = {{Consultato il}{Consultato il}},
4429   in            = {{In}{In}},
4430   editor        = {{curatore}{cur\adddot}},
4431   editors       = {{curatori}{cur\adddot}},
4432   depositor     = {{Depositante}{Depositante}},
4433   coppescale    = {{Scala}{Scala}},
4434   msdiss        = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
4435   dscthesiis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
4436   phdthesis     = {{Tesi di Dottorato}{Tesi di Dottorato}},
4437   tcc           = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
4438 }
4439 </lbxit>
4440 <*langes>
4441 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.

```

```

4442 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
4443 %% spanish slot) is active. Populates the dispatcher table for the
4444 %% spanish language and registers the biblatex language mapping.
4445 %% Institutional names (universityname/cityname/statename/countryname)
4446 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
4447 %% rosto are a Brazilian institutional template that does not translate.
4448 \ProvidesFile{coppe-lang-spanish.def}[2026/09/09 v4.1 CoppeTeX spanish class strings]
4449 \copperdefstring{spanish}{appendix}      {Ap'endice}
4450 \copperdefstring{spanish}{annex}         {Anexo}
4451 \copperdefstring{spanish}{frame}         {Cuadro}
4452 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
4453 \copperdefstring{spanish}{source}        {Fuente}
4454 \copperdefstring{spanish}{program}       {Programa}
4455 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
4456 \copperdefstring{spanish}{references}    {Referencias}
4457 \copperdefstring{spanish}{in:}          {en}
4458 \copperdefstring{spanish}{art}          {Arte}
4459 \copperdefstring{spanish}{platform}     {Plataforma}
4460 \copperdefstring{spanish}{directedby}   {Direcci'on}
4461 \copperdefstring{spanish}{producedby}   {Producci'on}
4462 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
4463 \copperdefstring{spanish}{listsymbol}   {Lista de S'imbolos}
4464 \copperdefstring{spanish}{glossary}     {Glosario}
4465 \copperdefstring{spanish}{advisor}      {Director}
4466 \copperdefstring{spanish}{advisors}     {Directores}
4467 \copperdefstring{spanish}{dept}         {Programa}
4468 \copperdefstring{spanish}{approvedmale} {Aprobado por}
4469 \copperdefstring{spanish}{approvedfemale} {Aprobada por}
4470 \copperdefstring{spanish}{approvedonmale} {Aprobado el}
4471 \copperdefstring{spanish}{approvedonfemale} {Aprobada el}
4472 \copperdefstring{spanish}{keywordsname} {Palabras clave}
4473 \copperdefstring{spanish}{volumename} {Volumen}
4474 \copperdefstring{spanish}{coadvisor}   {Codirector}
4475 \copperdefstring{spanish}{coadvisors} {Codirectores}
4476 \copperdefstring{spanish}{ofname}     {de}
4477 \copperdefstring{spanish}{researchline}{L'\i nea de investigaci'on}
4478 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
4479 \copperdefstring{spanish}{cityname}    {Rio de Janeiro}
4480 \copperdefstring{spanish}{statename}   {RJ}
4481 \copperdefstring{spanish}{countryname} {Brasil}
4482 \copperdefstring{spanish}{monthname1} {enero}
4483 \copperdefstring{spanish}{monthname2} {febrero}
4484 \copperdefstring{spanish}{monthname3} {marzo}
4485 \copperdefstring{spanish}{monthname4} {abril}
4486 \copperdefstring{spanish}{monthname5} {mayo}
4487 \copperdefstring{spanish}{monthname6} {junio}
4488 \copperdefstring{spanish}{monthname7} {julio}
4489 \copperdefstring{spanish}{monthname8} {agosto}
4490 \copperdefstring{spanish}{monthname9} {septiembre}
4491 \copperdefstring{spanish}{monthname10}{octubre}
4492 \copperdefstring{spanish}{monthname11}{noviembre}
4493 \copperdefstring{spanish}{monthname12}{diciembre}
4494 \copperdefstring{spanish}{algorithm2eopt}{spanish}
4495 \copperdefstring{spanish}{babelname}   {spanish}

```



```

4496 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
4497 %% language-mapping registration until \begin{document} so biblatex's
4498 %% \DeclareLanguageMapping is available.
4499 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
4500 \
```

```

4550 \copperdefstring{french}{monthname12}{d\'ecembre}
4551 \copperdefstring{french}{algorithm2eopt}{french}
4552 \copperdefstring{french}{babelname} {french}
4553 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
4554 \
```

```

4604 \copperdefstring{italian}{monthname10}{ottobre}
4605 \copperdefstring{italian}{monthname11}{novembre}
4606 \copperdefstring{italian}{monthname12}{dicembre}
4607 \copperdefstring{italian}{algorithm2eopt}{italiano}
4608 \copperdefstring{italian}{babelname} {italian}
4609 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
4610 </langit>
4611 <*numbbx>
4612 \ProvidesFile{coppe-numeric.bbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric bibliography]
4613 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
4614 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,
4615 % the PT->EN source map and the language mappings) and only adds the [n] list
4616 % label and the numeric sorting on top.
4617 \RequireBibliographyStyle{coppe}
4618
4619 % ABNT numeric system lists references in the order they are first cited
4620 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
4621 % numbered list. 'labelnumber' generates the [n] field.
4622 \ExecuteBibliographyOptions{labelnumber,sorting=none}
4623
4624 % Bracketed labels, both in the list and for any shorthand entries.
4625 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
4626 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
4627
4628 % Reference list driven by the [n] label (copied from biblalex's numeric.bbx),
4629 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
4630 \defbibenvironment{bibliography}
4631 { \list
4632   {\printtext[labelnumberwidth]{%
4633     \printfield{labelprefix}%
4634     \printfield{labelnumber}}}
4635   {\setlength{\labelwidth}{\labelnumberwidth}%
4636     \setlength{\leftmargin}{\labelwidth}%
4637     \setlength{\labelsep}{\biblabelsep}%
4638     \addtolength{\leftmargin}{\labelsep}%
4639     \setlength{\itemsep}{\bibitemsep}%
4640     \setlength{\parsep}{\bibparsep}}%
4641     \renewcommand*{\makelabel}[1]{\hss##1}}
4642 {\endlist}
4643 {\item}
4644
4645 </numbbx>
4646 <*numcbx>
4647 \ProvidesFile{coppe-numeric.cbx}[2026/09/09 v4.1 CoppeTeX ABNT numeric citations]
4648 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
4649 % numbers [1], compressed and sorted within a single cite group. This is the
4650 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
4651 % bibstyle=coppe-numeric. With the biblalex natbib option, \citep -> [1] and
4652 % \citet -> Author [1].
4653 \RequireCitationStyle{numeric-comp}
4654 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

10.3 The generated companion files

Everything the distribution ships is extracted from this file: the class, the BibLaTeX styles, the language packs, the databases, the `.ist`, the demonstration documents and the `latexmkrc`. One `pdflatex coppe.ins` rebuilds all of it and there is no second place to edit. Each one sits in an `\iffalse` block: the documentation driver never typesets them, and only `coppe.ins` extracts them.

The line is drawn at what is DISTRIBUTED. What exists only to prove the class works stays outside, as ordinary files under version control:

`tests/` the regression suite. Nobody installing CoppeTeX needs it, and generating it from here would put proof and product in the same box.

`adversativa/` twelve documents – four kinds of work by three languages – each firing everything the class offers at once.

`tools/*.ps1` the build harness itself.

`NORMA_COPPE_2026.tex`, `futuremanual2026.tex` documents about the norm that live in this repository with their own life cycle.

Both suites are run by one command, `tools\prova.ps1`, and that is what has to come out clean before a version is tagged.

10.3.1 `example_pt.tex` -- pt-main demo

`example_pt.tex` – CoppeTeX 4.0 multilingual demo, main language: brazilian. Short demonstration document (cover + folha + abstracts + 1 chapter) used by the CPGP submission package. The full pt-main example with bibliography and lists is the historical `example.tex`.

10.3.2 `example_en.tex` -- en-main demo

`example_en.tex` – CoppeTeX 4.0 multilingual demo, main language: english.

10.3.3 `example_es.tex` -- es-main demo

`example_es.tex` – CoppeTeX 4.0 multilingual demo, main language: spanish. Requires the language pack files `coppe-lang-spanish.def` and `spanish-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.4 `example_fr.tex` -- fr-main demo

`example_fr.tex` – CoppeTeX 4.0 multilingual demo, main language: french. Requires the language pack files `coppe-lang-french.def` and `french-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.5 `example_it.tex` -- it-main demo

`example_it.tex` – CoppeTeX 4.0 multilingual demo, main language: italian. Requires the language pack files `coppe-lang-italian.def` and `italian-coppe.lbx` to be installed alongside `coppe.cls`.

10.3.6 `example_pdfa.tex` -- the full example as PDF/A

`example_pdfa.tex` – o `example.tex` completo, compilado com a opção ‘pdfa’.

Existe para dar ao veraPDF o documento mais exigente que a classe produz: o exemplo inteiro, com bibliografia, listas, figuras, tabelas, algoritmos e as caixas do `tcolorbox` – ou seja, transparência, que é justamente o que o `a-1b` proibiria e o `a-2b` admite. Validar só um documento curto não diria nada sobre o que vai de fato para o depósito.

`\PassOptionsToClass` tem de vir ANTES do `\documentclass` que o `example.tex` traz dentro de si; é o único jeito de acrescentar uma opção de classe a um arquivo gerado pelo `docstrip` sem duplicá-lo aqui e deixar as duas cópias divergirem com o tempo.

10.3.7 `covers_5languages.tex` -- the five covers side by side

`covers_5languages.tex` – One-page visual demonstration of the CoppeTeX 4.0 multilingual model: the five built-in main languages side by side, all sharing the same Portuguese institutional template, differing only in the title (and the body language, not visible on the cover).

This file uses the article class deliberately – it is a meta-document about the `coppe` class, not a thesis. It expects the five cover PNGs `cover_pt.png`, `cover_en.png`, `cover_es.png`, `cover_fr.png`, `cover_it.png` to live in the same directory (generated by `pdftoppm` from the `example_<lang>.pdf` demos).

10.3.8 `latexmkrc` -- latexmk/Overleaf build configuration

`latexmkrc` – CoppeTeX build configuration (works locally and on Overleaf).

Run with: `latexmk example` (or `latexmk futuremanual2026`, your thesis...) `latexmk` reads this file automatically when it is in the build directory. On Overleaf, place this file at the project root; Overleaf runs `latexmk`.

Engine: pdfLaTeX by default (fast). For LuaLaTeX (recommended, full Unicode) set `$pdf_mode = 4` below, or use the `"% !TeX program = lualatex"` magic comment / the Overleaf "Menu > Compiler" setting.

Acknowledgments

Thanks to all COPPETEX users who have reported their experience with this class. We also thank to professor Fernando Lizarralde and Heiko Oberdiek for their

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Change History

v0.0		v3.1	
General: Creation Date.	1	<code>\department</code> : Included a sort key in <code>sybl</code>	59
v0.1		v3.2	
General: Documentation:		General: Fixed version problem between <code>cls</code> and <code>dtx</code>	1
bibliography fixed, title translation.	1	v3.3	
Sourceforge submission.	1	General: Extend <code>abrev</code> to work like <code>sybl</code> and accept a sorting key .	1
v0.2		v3.4	
General: Unification of the code for the list of symbols and abbreviations.	1	General: Some examples for figures, tables, longtables, etc. .	1
v0.3		v3.4.1	
General: Added 'draft' option. . . .	1	General: Fix english option to consider brazilian a second language, describe it in document	57
<code>\maketitle</code> : Added number of examiners test.	68	v3.5	
Generalization.	68	General: New command <code>annex</code> . .	109
v0.4		v3.5.1	
General: Beta documentation. . . .	1	General: New command <code>annex</code> now supports brazilian or english	109
v0.5		v4.0	
<code>abstract</code> : Changed from macro to environment.	76	General: Catalog-label writers (<code>coppe@mainBegin</code> , <code>coppe@bibBegin</code> , <code>coppe@bibEnd</code> , <code>coppe@hasLof</code>) made language-agnostic: <code>detokenize</code> wrap on label names protects ':' and '.' from babel shorthand activation, and the raw <code>etex</code> primitives <code>romannumeral/number</code> replace LaTeX's <code>roman/arabic</code> to bypass <code>spanish-babel</code> 's <code>small-caps @roman</code> redefinition that would otherwise crash <code>pdftex</code> with "Unbalanced write command" on every <code>es-main</code> document.	1
<code>\backmatter</code> : Added <code>mainmatter</code> pages counter.	64	Class-wide string dispatcher (<code>copperdefstring</code> , <code>coppestring</code> , <code>coppemainstring</code> , <code>coppeforeignstring</code>) replaces scattered <code>iflanguage</code> and if english branches; <code>pt/en</code> outputs remain byte-identical. .	1
<code>foreignabstract</code> : Changed from macro to environment.	77	Multilingual model: three fixed slots (<code>main</code> , <code>foreign</code> , <code>optional</code> <code>third</code>); five built-in language options (<code>brazilian</code> , <code>english</code> ,	
v1.0			
General: First COPPETeX release. .	1		
v2.0			
General: COPPETeX release 2.0. . .	1		
v2.1			
General: COPPETeX release 2.1:			
Matching the new rules.	1		
<code>\advisor</code> : Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. .	61		
v2.2			
General: Matching new guidelines, including new logo.	1		
v2.2.2			
General: Fixed some text constants in <code>.bib</code> and documented it here. .	1		
v3.0			
General: Added support for monographs in English.	57		
New approval page layout. . . .	71		
<code>\department</code> : Added new course on Nanotechnology.	59		
<code>\examiner</code> : Examiners expansion without degree.	61		

spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.	1	sumario indicative.	1
New user API: titlein, usecoppelanguage, brazilianabstract environment. .	1	New options: pdfa (PDF/A-2b via pdfx, XMP built from the document), assinaturas, coorientador, rascunhoficha, listasnosumario, morewrites/semmorewrites. New command coadvisor. Under pdfTeX the class loads morewrites by itself when installed, because a work that asks for every list needs seventeen of TeX's sixteen output streams.	1
<code>\coppetexfinalpage</code> : Initial coppetexfinalpage colophon page.	65	newcoppfloat fixed twice: newfloat was wrapped in a group, and since newfloat defines the environment locally the new float vanished when the group closed; and its list came out unnumbered and without leaders because l@figure was not inherited. Author-declared floats are now numbered within the chapter like every other illustration. . . .	1
v4.1 General: Breaking change in how the board is declared: advisor, coadvisor and examiner now take the TREATMENT as the optional argument, empty by default, and the INSTITUTION as the last mandatory one, which may be left empty. 3.1.2.1.3(e) asks for name, titulacao and institution and asks for no treatment, so that is what the folha de aprovacao prints.	1	No signature rules on the folha de aprovacao: the deposit has been digital only since Resolucao CEPG n. 246/2023, so there is nothing to sign by hand. The assinaturas option is still accepted and only warns. .	1
Conformance with the revised 9th edition (2026) of the UFRJ/SiBI manual: continuous pagination from the folha de rosto, folio in 10pt at 2cm from the top and right edges, one-sided layout with a 3cm left margin, additional sheet with the catalog card and the CAPES collection fields, approval sheet of 3.1.2.1.3, keywords closing the three abstracts, pre-textual lists out of the sumario, Apendice and Anexo headings centred, fifth heading level, and Latin Modern instead of the bitmap fonts.	1	Single source: pdfatex coppe.ins generates everything distributed – class, biblatex styles, language packs, .bib bases, .ist, the five per-language examples, example_pdfa, the cover montage and latexmkrc. The regression suite, the adversarial documents and the harness prove the class and are NOT distributed.	1
Fixes found by the adversarial review: cover and folha de rosto compose in single spacing so they do not overflow under doublespacing, no folio is printed on a pre-textual sheet when something does overflow, the approval sheet holds a board of eight, and es-nosectiondot removes the period babel-spanish put in the		The class sets the document in the sans-serif family by DEFAULT; comserifa restores the serif one. Neither the manual nor the COPPE norm prescribes a family, and the SiBI manual that carries the rule is itself set in Arial. semserifa is still accepted and	

now does nothing.	1	with dotted fill rules.	1
The folha adicional draws the		The research-line label on the	
Coleta CAPES fields as the		folha de rosto follows the main	
closed frame of Annex H – one		language of the work instead of	
cell per field, a rule between		being hard-coded in	
cells, and no rules to write on –		Portuguese.	1
instead of loose paragraphs			